SEMESTER 3

FOOD TECHNOLOGY

SEMESTER S3

MATHEMATICS FOR LIFE SCIENCE – 3

(Group D)

Course Code	**GDMAT301**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus	**Course Type**	Theory

Course Objectives:

To provide a solid understanding of probability theory and its practical applications to address uncertainty and variability in engineering systems effectively.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Summarizing data sets, Sample mean, Median and mode, Sample variance and standard deviation, Normal data sets, Paired data sets and sample correlation coefficient. **(Text1: Relevant topics from sections 2.3.1, 2.3.2, 2.5, 2.6)**	**9**
2	Introduction to probability, Sample space and events, Venn diagrams, Axioms of probability, Sample spaces having equally likely outcomes, Conditional probability, Baye's formula, Independent events. **(Text1: Relevant topics from sections 3.1, 3.2, 3.3,3.4, 3.5, 3.6, 3.7, 3.8)**	**9**
3	Random variables, Discrete and continuous random variables, Probability mass function and Probability density function, Cumulative distribution function, Expectation, Expectation of a function of a random variable and its properties, Variance and standard deviation. **(Text1: Relevant topics from sections 4.1, 4.2, 4.4, 4.5, 4.6)**	**9**
4	The Bernoulli and binomial random variables and its distribution, The Poisson random variable and its distribution, Uniform distribution, Normal and standard normal distribution. **(Text1: Relevant topics from sections 5.1, 5.2, 5.4, 5.5)**	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the methods of descriptive statistics to summarize and analyze data systematically.	**K2**
CO2	Understand the basics of probability theory and to apply the fundamental concepts and methods to analyze uncertainty and make predictions.	**K3**
CO3	Apply the concept of random variables and their associated probability distributions to study random phenomena.	**K3**
CO4	Understand the fundamental probability distributions and to apply statistical techniques effectively to make informed decisions based on probabilistic analysis.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Probability and Statistics for Engineers and Scientists	Sheldon M Ross	Academic Press	6th edition, 2021

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Probability, Statistics and Random processes	T Veerarajan	McGraw Hill	3rd edition, 2008.
2	Probability and Statistics for Engineering and the Sciences	Devore Jay L	Cengage	9th edition, 2016.
3	Statistical Methods	S.P. Gupta	Sultan Chand & Sons	46th edition, 2021.

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_mg31/preview
2	https://onlinecourses.nptel.ac.in/noc22_mg31/preview
3	https://onlinecourses.nptel.ac.in/noc22_mg31/preview
4	https://onlinecourses.nptel.ac.in/noc22_mg31/preview

SEMESTER S3

FOOD THERMODYNAMICS AND PROCESS CALCULATIONS

Course Code	**PCFTT302**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1: 0 :0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs 30 Mins
Prerequisites (if any)	Knowledge of fundamental Physics, Chemistry and Mathematics	**Course Type**	Theory

Course Objectives:

1. Introduction of key thermodynamics concepts in food technology
2. Familiarization with important concepts such as heat and mass balance in industrial food processing
3. Basic concepts of reaction engineering.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Ideal and real gas laws:** – Gas constant - calculations of pressure, volume and temperature using ideal gas law, Humidity, Calculation of absolute humidity, molal humidity, relative humidity and percentage humidity - Use of humidity in condensation and drying - Humidity chart, dew point.	**11**
2	M**aterial balance in unit operations**:- such as evaporation, crystallization, mixing, leaching, extraction, absorption, distillation and drying.Heat capacity of solids, liquids, gases, and solutions, use of mean heat capacity in heat calculations, problems involving sensible heat and latent heat.	**11**

3	**Introduction: Fundamental concepts and definitions** - closed, open, and isolated system – intensive and extensive properties - path and state functions - reversible and irreversible process -Zeroth law of thermodynamics - First law of thermodynamics - internal energy- enthalpy – heat capacity - first law for cyclic, non-flow and flow processes - applications - Second law of thermodynamics - limitations of first law - general statements of second law - concept of entropy - calculation of entropy changes - Carnot's principle -absolute scale of temperature - Clausius inequality - entropy and irreversibility - Third law of thermodynamics.	**11**
4	O**verview of chemical reaction engineering:** classification of chemical reactions, the definition of reaction rate, concentration and temperature dependant term of rate equation, order, and molecularity, elementary and non-elementary reactions, Reactors: Batch reactor, continuous stirred tank reactor (CSTR), Plug flow reactor (PFR), Enzymatic reaction fundamentals, Michaelis - Menten kinetics, Batch reactor calculations forenzymatic reactions. Bioreactors- cell growth kinetics- Monod equation.	**11**

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Develop and solve basic material balance equations for the unit operations employed in food processing industries.	**K3**
CO2	Develop and solve basic energy balance equations for the unit process employed in food processing industries	**K3**
CO3	Apply the laws of thermodynamics to analyse various processes .	**K2**
CO4	State and prove the equivalence of two statements of the second law of thermodynamics and evaluate the limitations of the first law of thermodynamics	**K2**
CO5	The concept of reaction rates and different reactors used in food industries.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1									3
CO2	2	3	2									3
CO3	3	1										3
CO4	3	1										3
CO5	3	2	1									3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Chemical Engineering Thermodynamics	J.M. Smith, Hendrick Van Ness & Michael Abbott	McGraw-Hill Higher Education	7th Edition, 2005
2	A Textbook of Chemical Engineering Thermodynamics	K V Narayanan	PHI Learning Pvt. Ltd	2011
3	Unit Operations in Chemical Engineering.	Warren McCabe, Julian Smith & Peter Harriot	Mc Graw Hill Book Co.	5th Edition, 1993
4	Stoichiometry and process calculations	K V Narayanan & B Lakshmikutty	PHI Learning Pvt. Ltd	2nd Edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Engineering Thermodynamics	Y.V.C. Rao	Universities Press	1997
2	Chemical and Process Thermodynamics	Kyle B.G.	Prentice Hall International	3rd edition 1999
3	Basic Principles & Calculations in Chemical Engineering	David M. Himmelblau& James B. Riggs	PHI Learning Pvt. Ltd	7th edition, 2006
4	Stoichiometry	B. I. Bhatt & S. M. Vora.	Tata McGraw-Hill	1976

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/103/103/103103144/
2	https://archive.nptel.ac.in/courses/103/101/103101004/
3	https://archive.nptel.ac.in/courses/103/103/103103165/
4	https://onlinecourses.nptel.ac.in/noc22_ch02/preview

SEMESTER S3

FOOD MICROBIOLOGY

Course Code	**PCFTT303**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Understanding the morphology and structure of microbes
2. To study the microbiological techniques for quality evaluation of food
3. To acquire knowledge on basic concepts of microbial contamination and spoilage
4. To study the various foodborne illness of microbial origin and their control measures

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Morphology, structure, microbiology and reproduction of Bacteria, Yeast, mold,actinomycetes and algae. **Microbial growth and death kinetics:** Definition, Growth curves (different phases), synchronous growth, doubling/generation time, intrinsic and extrinsic factors, relationship between number of generations and total number of microbes. **Microscopic Techniques:** Light Microscopy, Fluorescence Microscopy, Electron Microscopy, Scanning Probe Microscopy	**11**
2	**Enumeration of microorganisms** – quantitative and qualitative methods, microscopic and culture-dependent methods- Direct microscopic observation, culture, enumeration and isolation methods; Chemical (LAL, ATPase, radiometry), Physical (Impedance, microcalorimetry, flow cytometry) and immunological methods (ELISA, RIA, Immunomagnetic); Biosensors, Test for bacterial toxins in foods, Rapid methods and automation (FACS, PCR).	**11**

3	**Microflora associated with various food groups**:- Food spoilage: Important food spoilage bacteria – psychrotrophic, thermoduric, thermophilic, aciduric; characteristic features and significance of spoilage of different groups of foods – Cereals, fruits & vegetables, meat, sea foods, eggs, milk and milk products. Sources of contamination. Fermented food products- prebiotics, probiotics; Spoilage of fermented foods.	**11**
4	**Food poisoning and microbial foodborne diseases:** Foodborne infections – Salmonellosis, listeriosis, pathogenic E. coli, shigellosis, campylobacteriosis, enteric viruses, brucellosis, Q fever; Foodborne Intoxications – staphylococcal, botulism, mycotoxicosis; Foodborne toxicoinfections – gastroenteritis, cholera. Foodborne parasites. Control of microorganisms in foods.	**11**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the basics of food microbiology, microscopy, major microorganisms in food, their morphology, isolation, growth, quantification and culture.	**K1**
CO2	Explain the qualitative and quantitative methods to detect microbes and microbial toxins in food.	**K2**
CO3	Identify the factors and microbes involved in food spoilage, food-borne pathogens and fermented foods.	**K1**
CO4	Explore different illnesses produced by microorganisms in food	**K3**
CO5	Elaborate the strategies to control the growth of microorganisms in food	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping od Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	**2**											**3**
CO2	**3**		**2**			**3**						**2**
CO3	**3**					**2**						**2**
CO4	**3**		**1**			**2**						**2**
CO5	**3**		**1**			**3**	**1**					**3**

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Modern Food Microbiology	James MJ	CBS Publishers	5^{th}Edition, 2000
2	Basic Food Microbiology	Barnatt J	CBS Publishers	1997

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Microbiology	William C Frazier & Dennis C Westoff	McGraw Hill Education	5th Edition, 2017
2	Fundamental of Food Microbiology	Bibek Ray	CRC Press	5th Edition, 2013

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/102106096
2	https://nptel.ac.in/courses/102103015
3	https://onlinecourses.swayam2.ac.in/cec19_ag03/preview
4	https://onlinecourses.swayam2.ac.in/cec22_ag01/preview

SEMESTER S3

FOOD PROCESS ENGINEERING

Course Code	**PBFTT304**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	**ESE Marks**	40
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	PBL

Course Objectives:

1. Understand the principles and techniques of food processing
2. Apply theoretical knowledge to analyse and solve practical problems in food processing

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Food Processing-** Primary, Secondary & Tertiary Processing, Introduction to different types of food manufacturing industries and the products manufactured - Raw material properties - physical, functional – Preparation and Cleaning of raw materials - wet and dry techniques - Peeling, Sorting and grading methods of raw materials, Handling of raw materials using equipment like conveyers and elevators.	**9**
2	**Size reduction of solids** – Principle and laws of size reduction – Kicks, Bond, Rittinger's laws – Energy required for grinding- Universal law of grinding, Equipment used in size reduction- roller mill, impact mill, attrition mill, tumbling mills, crushers and pulveriser, Importance and principles of Homogenization and emulsification, Basic concepts of Mixing, Forming, Filtration.	**9**

3	**Thermal & Low Temperature Processing of Foods-** Blanching, Pasteurization, and Shelf life of Thermally processed foods using microbial death kinetics, Drying- Drying rate curve, Drying time prediction - Driers - Tray, tunnel, puff, fluidized bed, spray, Rotary drier etc - Freeze drying. VA, VC refrigeration systems- components - compressor, condenser, evaporator, refrigerant, COP - Chilling and freezing, effect of low temperature on food spoilage, Freezing curve - Frozen food storage	**9**
4	**Non-Thermal Processing-** Extrusion - operating characteristics, single screw, twin screw extruders, application, effect on foods. Principle, Process and applications of Ohmic heating, RF heating, Pulsed electric field heating, high pressure processing, Food irradiation and ultrasound techniques. Hurdle technology - Principle and application **Packaging techniques-** CAP, MAP, Vacuum packaging, Retort packaging, Aseptic Packaging.	**9**

Suggestion on Project Topics

1. **Select a Focus Area**: Choose a specific aspect of food processing (e.g., thermal processing methods, size reduction techniques, non-thermal processing innovations) as the focus of your project.
2. **Literature Review**: Conduct a comprehensive literature review on your chosen topic. Gather information on theories, principles, laws (e.g., Kicks, Bond, Rittinger's), and case studies related to food processing techniques.
3. **Define Objectives**: Clearly define the objectives of your project. State what you aim to achieve or investigate through your study.
4. **Plan Experimental Design**: Design experiments or simulations to explore your objectives. Consider factors like equipment selection (e.g., roller mill, impact mill), processing conditions (e.g., temperature, pressure), and methods (e.g., drying techniques, extrusion).
5. **Data Collection**: Conduct experiments or gather data using appropriate methods (e.g., trials with different equipment settings, measurements of temperature profiles during processing).

6. **Data Analysis**:

    - Organize your collected data systematically.
    - Perform statistical analysis or use appropriate analytical methods (e.g., regression analysis for grinding laws, comparison of drying rates) to interpret your data.
    - Present your data analysis results using tables, graphs, or charts as necessary to illustrate trends and relationships.

7. **Compare and Contrast Techniques**: Compare different techniques within your chosen area (e.g., compare the effectiveness of different drying methods, analyse the energy requirements of various size reduction techniques).
8. **Evaluate Results**: Evaluate the results of your experiments or simulations. Discuss how well your findings align with existing theories and whether they suggest improvements or modifications to current practices.
9. **Drafting the Report**:
    - **S**tructure your report according to academic standards (e.g., introduction, literature review, methodology, results, discussion, conclusions, recommendations).
    - Clearly articulate your research question, objectives, and methodology used.
    - Include detailed descriptions of experiments conducted and data collected.
    - Present and discuss your findings, linking them back to relevant theories and literature.

10. **Draw Conclusions and Recommendations**: Summarize your findings and draw conclusions based on the results of your project. Provide recommendations for future research or practical applications based on your insights.

11. **Review and Finalize**: Review your draft report for clarity, coherence, and completeness. Ensure all sections are well-organized and effectively communicate your research findings and analysis.

12. **Submit and Present**: Submit your final report according to the specified guidelines. Prepare to present your findings in a clear and engaging manner, addressing any questions or feedback from your audience.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Identify the principles and techniques of food processing.	K3
CO2	Apply principles of size reduction and homogenization in food processing	K2
CO3	Analyse thermal and low-temperature processing methods in food preservation	K2
CO4	Evaluate non-thermal processing techniques and their application	K2
CO5	Develop food processing strategies integrating multiple techniques for optimal food quality and safety	K3

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3		2					2		2	3
CO2	3	3		2					2		2	3
CO3	3	3		2					2		2	3
CO4	3	3		2					2		2	3
CO5	3	3		2					2		2	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Food Processing Technology: Principles and Practice	PJ Fellows	CRC	Second Edition, 2000
2	Fundamentals of Food Process Engineering	Romeo T. Toledo, Fanbin Kong, Rakesh K. Singh	Springer	2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Food Process Engineering and Technology	Zeki Berk	Elsevier	Second Edition, 2013
2	Introduction to Food Engineering	R Paul Singh and Dennis R Heldmann	Academic Press	Fifth Edition, 2014

Video Links (NPTEL, SWAYAM…)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_ag01/preview
2	
3	
4	

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

1. **Project Planning and Proposal (5 Marks)**

   - Clarity and feasibility of the project plan
   - Research and background understanding
   - Defined objectives and methodology

2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**

   - Individual contribution to the presentation
   - Effectiveness in answering questions and handling feedback

3. **Involvement in the Project Work and Team Work (3 Marks)**

- Active participation and individual contribution
- Teamwork and collaboration

4. **Execution and Implementation (10 Marks)**

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. **Final Presentation (5 Marks)**

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. **Project Quality, Innovation, and Creativity (3 Marks)**

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to AI and Machine Learning**: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	**11**
2	**Mathematical Foundations of AI and Data science**: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	**11**

3	**Applied Probability and Statistics for AI and Data Science**: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis- linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	**11**
4	**Basics of Data Science**: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	**11**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	**K3**
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems**.**	**K3**
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	**K3**
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	**3**	**3**	**3**	**3**								
CO2	**3**	**3**	**3**	**3**								
CO3	**3**	**3**	**3**	**3**								
CO4	**3**	**3**	**3**	**3**								

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2nd edition,2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1st edition, 2016

Reference Books				
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	**UCHUT346**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	**6**
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	**6**

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	**6**
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return –- Payback – Accounting Rate of Return.	**6**

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	**15**	**12.5**	**12.5**	**50**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	**50**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	**K2**
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	**K3**
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	**K2**
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7^{TH} Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: UCHUT346

Course Name: Economics for Engineers

Max. Marks: 50		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 8 marks

Module 1

9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

		Module 2		
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$.. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
		Module 3		
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNPfc from GDPmp.	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDPmp and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
		Module 4		
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

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SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	**UCHUT347**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Fundamentals of ethics -** Personal vs. professional ethics, Civic Virtue, Respect for others, **Profession and Professionalism,** Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, **Technology and digital revolution-**Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, **High technologies: connecting people and places-**accessibility and social impacts, **Managing conflict**, Collective bargaining, **Confidentiality**, Role of confidentiality in moral integrity, **Codes of Ethics**. **Basic concepts in Gender Studies -** sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, **Gender disparity and discrimination in education**, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations**, Ethical values and practices in**	**6**

	connection with gender - equity, diversity & gender justice, **Gender policy and women/transgender empowerment initiatives.**	
2	**Introduction to Environmental Ethics:** Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). **Sustainable Engineering Principles:** Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. **Ecosystems and Biodiversity:** Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. **Landscape and Urban Ecology:** Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	**6**
3	**Hydrology and Water Management:** Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. **Zero Waste Concepts and Practices:** Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. **Circular Economy and Degrowth:** Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. **Mobility and Sustainable Transportation:** Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	**6**
4	**Renewable Energy and Sustainable Technologies:** Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. **Climate Change and Engineering Solutions:** Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. **Environmental Policies and Regulations:** Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. **Case Studies and Future Directions:** Analysis of real-	**6**

	world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	

Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
		Total Marks		**50**

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis**: Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts**: Ability to apply course concepts to real-world problems and local contexts.
- **Creativity**: Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills**: Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	**K3**
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	**K4**
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	**K5**
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	**K4**
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessmen	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements -

calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio

- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

FOOD MICROBIOLOGY LAB

Course Code	PCFTL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0-0-3-0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. Goal of this course is to develop the knowledge of students in the basic area of Food Microbiology. This is essential for the better understanding of various basic microbiological technologies and acquires knowledge about different types microorganisms related to food and techniques of microbial analysis

Expt. No.	Experiments
1	Introduction to Basic Microbiological Practices and Equipment
2	Sterilization techniques and disinfection
3	Preparation of culture media – solid and liquid
4	Cultivation and Subculturing of microbes
5	Isolation techniques for microorganisms – spread, pour & streak plate
6	Simple staining techniques
7	Gram staining method for differentiation of bacteria
8	Motility of microorganism – hanging drop method
9	Enumeration of microorganisms in air
10	Evaluation of microbiological quality of potable water
11	Evaluation of microbiological quality of milk (MBRT)
12	Quantitative analysis of milk by Standard plate count (SPC) method
13	Evaluation of microbiological quality of food products
14	Microbial Examination of Fruit sample-surface washing and internal tissue

15	Methods for identification of Escherichia coli
16	Methods for identification of Staphylococcus aureus
17	Counting for yeasts and moulds
18	Demonstration of microbial production of curd

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	**25**	**20**	**50**

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	**15**	**10**	**10**	**5**	**50**

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Demonstrate safety practices inside a microbiology laboratory and familiarise with different equipments	**K2**
CO2	Preparation and sterilisation of common culture media for subculturing microbes	**K3**
CO3	Identify suitable methods for qualitative and quantitative evaluation of food	**K3**
CO4	Evaluate the total microbial load in the food or in the environment	**K3**
CO5	Demonstrate the procedure for identification of specific microorganism in food and usage of beneficial microbes for food production	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Creat*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3								3			
CO2	3					1			3			
CO3	3		2			2			3			
CO4	3		2			2			3			
CO5	3		2			2			3			

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Practical Microbiology	R C Dubey and D K Maheshwary	S. Chand & Co. Ltd.	2006
2	Microbiology: A Laboratory Manual	James Cappuccino		10th edition
3	Fundamental Food Microbiology	Bibek Ray	CRC press	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.classcentral.com/course/youtube-how-to-pass-your-microbiology-lab-practical-skills-97364
2	https://onlinecourses.swayam2.ac.in/cec24_ag01/preview
3	https://www.classcentral.com/course/youtube-how-to-pass-your-microbiology-lab-practical-skills-97364
4	https://www.classcentral.com/course/youtube-food-microbiology-97376

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students’ preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

    - Accuracy of Results: Precision and correctness of the obtained results.
    - Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

FOOD CHEMISTRY LAB

Course Code	PCFTL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs 30 Mins
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. Provides an experimental experience with the basic scientific principles of food systems and practical applications to Food Chemistry.
2. Demonstrate the chemical/biochemical reactions of carbohydrates, lipids, proteins, and other constituents in fresh and processed foods with respect to food quality.
3. Helps to examine the reaction conditions and processes that affect the color, flavor, and texture of food

Expt. No.	Experiments
1	Karl Fischer titration for determination of moisture content
2	Qualitative tests for carbohydrates a) Distinguishing reducing from non- reducing sugars b) Keto sugar from aldo sugar
3	Determination of rate/ extent of hydrolysis of sucrose/starch
4	Acid hydrolysis and action of salivary amylase on starch
5	Estimation of gluten in wheat
6	Study of gelling properties of starch- gelatinisation and retrogradation
7	Quantitative tests for proteins- Kjelhdal method
8	Effect of heat treatment on protein quality
9	Effect of pH on hydration of meat proteins
10	Water holding capacity and oil holding capacity of proteins

11	Detection and estimation of oxidative rancidity in fats/oils
12	Determination of enzymatic browning reactions
13	Determination of non enzymatic browning reactions
14	Determination of free fatty acid content in fats and oils
15	Determination of hardness of water
16	Determination of heat stability of vitamin C
17	Detection of adulteration in food products
18	Detection / Estimation of some additives in foods

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	**25**	**20**	**50**

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	**15**	**10**	**10**	**5**	**50**

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Students will be able to estimate the moisture content in food	K3
CO2	Understand the biochemical reactions in food components	K3
CO3	Demonstrate the knowledge on chemical properties of food components	K3
CO4	Evaluate and interpret food analysis data and scientifically communicate this	K3

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2					1	1	2			1
CO2	2	2					1	1	2			1
CO3	2	2					1	1	2			1
CO4		1						2		2		1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The chemical analysis of foods and food products	Morris B. Jacobs	CBS Publishers and distributors New Delhi.	III edition
2	Hand book of analysis and quality control for fruit and vegetable products	Ranganna S	Tata McGraw Hill Publishing Co. New Delhi	II edition
3	Methods in Food Analysis	Rui M. S. Cruz, Igor Khmelinskii, Margarida Vieira	CRC Publishers	2014 (1 Ed.)

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Modern Experimental Biochemistry,	Rodney and Boyer	Pearson education, India.	
2	Fundamental Laboratory Approaches forBiochemistry and Biotechnology	Alexander J. Ninfa and David P. Ballou	Fitzgerald Science Press Inc, USA.	
3	An introduction to Practical Biochemistry,	David T. Plummer	McGraw- Hill.	
4	Principles and Techniques of Practical Biochemistry,	Wilson K and Walker J	Cambridge University Press.	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/-GR8Z3UerE0
2	
3	
4	

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.

- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

   - Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

   - Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
   - Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
   - Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
   - Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

   - Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

   - Accuracy of Results: Precision and correctness of the obtained results.
   - Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

   - Ability to explain the experiment, procedure results and answer related questions
   - Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

   - Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

FOOD TECHNOLOGY

SEMESTER S4

MATHEMATICS FOR LIFE SCIENCE – 4

(D Group)

Course Code	GDMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GDMAT301	Course Type	Theory

Course Objectives:

To provide a comprehensive understanding of the concepts of population and sample distinctions, interval estimation, hypothesis testing, the method of least squares for regression, and ANOVA for variance analysis and to apply them in practical applications across various fields.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Population and sample, The sample mean, Population mean and variance, The central limit theorem – Distribution of sum of random variables, Approximate distribution of the sample mean, Sample variance and sample standard deviation. **[Text 1: Relevant topics from sections 6.1,6.2,6.3, 6.3.1,6.3.2, 6.4]**	**9**
2	Interval estimation of population mean - Two sided confidence interval, One-sided lower and upper confidence intervals, Confidence interval for the mean when the variance is unknown, Confidence intervals for the variance of a normal distribution. **[Text 1: Relevant topics from sections 7.3, 7.3.1, 7.3.3]**	**9**
3	Hypothesis testing, Significance levels, Null hypothesis, Critical region, Type I and Type II error, Testing of mean of a normal population – Case of known variance, One sided tests; Testing of mean of a normal population- Case of unknown variance, One sided *t*- test. **[Text 1: Relevant topics from sections 8.1, 8.2, 8.3.1, 8.3.1.1, 8.3.2]**	**9**

4	The method of least squares-regression line, Multiple regression (for two independent variables only), Analysis of variance, Completely randomized design (One-way ANOVA), Randomized-block design (Two-way ANOVA). **[Text 2: Relevant topics from sections 11.1, 11.4, 12.1, 12.2, 12.3]**	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand population and sample distinctions, calculate means and variances and to apply the central limit theorem to distributions of sums of random variables.	**K3**
CO2	Construct interval estimates for population means using two-sided and one-sided confidence intervals, to determine confidence intervals for the mean with unknown variance, and to calculate confidence intervals for the variance of a normal distribution.	**K3**
CO3	Understand the basic concepts of hypothesis tests and to apply them in testing means of normal populations with known and unknown variances.	**K3**
CO4	Apply the method of least squares to determine regression lines to conduct multiple regression analysis for two independent variables and to perform analysis of variance (ANOVA) including completely randomized and randomized-block designs.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Probability and Statistics for Engineers and Scientists	Sheldon M. Ross	Academic Press	6th edition, 2021
2	Miller & Freund's Probability and Statistics for Engineers	Richard A. Johnson	Pearson	9th Global edition, 2018

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Statistical Methods	S.P. Gupta	Sultan Chand & Sons	46^{th} edition, 2021
2	Probability and Statistics for Engineering and the Sciences	Jay L. Devore	Cengage	9^{th} edition, 2016
3	Statistics for engineers and scientists	William Navidi	McGraw-Hill	6^{th} edition, 2023
4	Probability and Statistics for Engineers and Scientists	Walpole, Myers, Ye	Pearson	9^{th} edition, 2010

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	NPTEL :: Mathematics - NOC:Introduction to Probability Theory and Stochastic Processes
2	NPTEL :: Mathematics - NOC:Introduction to Probability Theory and Stochastic Processes
3	NPTEL :: Mathematics - NOC:Introduction to Probability Theory and Stochastic Processes
4	NPTEL :: Mathematics - NOC:Introduction to Probability Theory and Stochastic Processes

SEMESTER S4

HEAT AND MASS TRANSFER IN FOOD PROCESSING

(Common to FT Branches D Group/Groups)

Course Code	**PCFTT402**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Identify the important modes of heat transfer and their applications.
2. Impart the knowledge of various mass transfer operation related with food processing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Mechanisms and modes of heat transfer**:- conduction and Fourier law – Thermal conductivity of food products, General Heat conduction Equation, Heat conduction in Plane wall, cylinder, sphere for single and composite resistances, Critical Thickness of Insulation – wall, cylinder, and sphere (Basic numerical problems).	**11**
2	**Basics and concepts, Types of Convection**:– Free convection – Forced Convection, heat transfer coefficient, and overall heat transfer coefficient, factors affecting heat transfer coefficients, Types of Boiling and Condensation. Importance of boilers and condensers in the food industry, dimensionless numbers used in heat transfer. Radiation heat transfer- Basic Concepts, Laws of Radiation – Stefan Boltzmann Law, Kirchoff Law, Weins displacement law, Black Body Radiation.	**11**
3	**Molecular diffusion:-** Fick's Law, concentration, mass flux, Steady State molecular diffusion of A through stagnant gas B and equimolar counter diffusion in liquids, Application; Mass Transfer coefficients: Types of mass Transfer Coefficient, Dimensionless Groups in Mass Transfer, types of	**11**

	diffusion, Theories of Mass Transfer, Numerical Problems on molar flux and diffusivity.	
4	**Application of heat and mass transfer equipment in food processing**:- Heat Exchanger equipment –classification and constructional details-Gas absorption- Packed tower and plate tower- types and properties of packing, choice of solvent, the solubility of a gas in the liquid, Basic principle and operation of Distillation, Crystallization, leaching and its equipment, supercritical fluid extraction.	**11**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize problems and develop solutions for steady state conduction in simple geometrics.	**K3**
CO2	Apply principles of heat and mass transfer to predict transfer coefficients.	**K3**
CO3	Understand the mechanism of mass transfer operation	**K2**
CO4	Analyse the working of various heat and mass transfer equipment.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2				1					
CO2	3	3	2									
CO3	3	2	2				1					
CO4	3	1	1									

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Food Process Engineering	Romeo T. Toledo	Springer	Third Edition, 2001
2	Coulson and Richardson's Chemical Engineering: Volume 1B: Heat and Mass Transfer: Fundamentals and Applications	R. P. Chhabra, V. Shankar	Elsevier Science	2017
3	Mass Transfer Operations	Treybal R. E	McGraw Hill	3rd edition. 1981
4	Heat and Mass Transfer: A practical Approach	Yunus A Cengel	McGraw Hill	3rd edition, 2006

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A text book of heat and mass transfer	R K Rajput	S Chand Publishing	2016
2	Principles of Mass Transfer and Separation Processes	Binay K Dutta	PHI Learning Private Ltd	2009
3	Mass Transfer Theory and Applications	K. V. Narayanan	CBS Publishers & Distributors	2017
4	Mass Transfer Theory and Practice	N. Anantharaman, K. M. Meera Sheriffa Begum	PHI Learning Private Ltd	2011

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/103/101/103101137/
2	https://archive.nptel.ac.in/courses/103/101/103101137/
3	https://archive.nptel.ac.in/courses/103/103/103103034/
4	https://archive.nptel.ac.in/courses/103/103/103103034/

SEMESTER S4

PROCESSING OF CEREALS, PULSES AND OILSEEDS

Course Code	**PCFTT403**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To study various properties and functions of cereals pulses and oilseeds
2. To acquire knowledge on various unit operations involved in processing of cereals, pulses and oilseeds

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Status of grain processing industries in India:** – rice- varieties- nutrient composition, Parboiling – physicochemical changes during parboiling modern methods of parboiling- Rice milling – equipment involved in the rice milling processing – Cleaner – Stoner – Husker- Separator- Whitening machine, Wheat - nutrient composition – classification of wheat - Milling of wheat - equipment used in milling – break roll – reduction roll- purifier – plan sifter – scalping - scratch system, Types of wheat.	**11**
2	**Barley**: – Nutrient composition – Commercial production of malt, Milling of corn -Dry milling and wet milling- Gluten and starch separation – enzyme hydrolysis – acid hydrolysis- Processing of oats- Processing of Millets- Processing of Pulses- antinutritional factors in pulses- Dhal mill	**11**
3	**Value added products from Cereals**:- Processing of breakfast cereals – processing of extruded products (noodles and pasta) - bread - Processing of protein isolates, flaked products-puffed snacks, HFCS - Processing of special dietary foods	**11**
4	**Milling of oilseeds:-** mechanical expression, screw press, hydraulic press, solvent extraction methods, preconditioning of oilseeds, refining of oil, stabilization of rice bran.- Processing of soya bean- Storage structure- Bag storage, Cover and plinth, CAP storage (Ceiling and Plinth Storage)- Silos and large bins - Silos flow pattern.	**11**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the processing technology of rice	**K2**
CO2	Describe the processing technology of wheat	**K2**
CO3	Illustrate the processing of value added products from cereals	**K2**
CO4	Explain the extraction of oil from oilseeds	**K2**
CO5	Evaluate the suitability of different storage structures for specific grain types	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	2										3
CO3	3	2										3
CO4	3	2										3
CO5	3	2										3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Unit Operations of Agricultural Processing	K M Sahai & K K Singh	Vikas Publishing House Pvt Ltd	Second Edition, 2009
2	Post Harvest Technology of Cereals, Pulses and Oilseeds	A Chakraverty	Oxford & IBH	Third Edition, 2008
3	Chemistry and Technology of Cereals as Food and Feed	Martz Samuel A	CBS	Second Edition, 2014

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Cereals Processing Technology	Gavin ed Owens	Bio-green Elsevier	Nil edition, 2015
2	Millets: Nutritional Value and Processing Technology	UD Chavan and JV Patil	Astrall Publications	Ist Edition, 2016

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
3	https://onlinecourses.swayam2.ac.in/cec19_ag05/preview
4	https://onlinecourses.swayam2.ac.in/cec19_ag05/preview

SEMESTER S4

FOOD ANALYSIS

Course Code	**PBFTT404**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	4	**ESE Marks**	40
Credits	3:0:0:1	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	4	**Course Type**	Theory

Course Objectives:

1. Introduce the basic concepts of Food Analysis, and explain the principles behind sampling and proximate analysis.
2. Familiarize the regulations by Government agencies based on the analysis of food.
3. Strengthen the knowledge about the different equipment and techniques employed in food analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to food analysis, significance, FSSAI regulations for food analysis, Sampling and sampling plan in relation to food analysis, AOAC, NABL Accreditation, Export Inspect Council Proximate Analysis – Moisture, Ash, Fiber	**9**
2	Proximate Analysis –Carbohydrates, Protein,Protein Quality Fat analysis – method of analysis, oxidation test for fat methods, modern analytical techniques for proximate analysis Protein separation techniques – Electrophoresis – principle. Types of electrophoresis – capillary electrophoresis, Gel electrophoresis – PAGE and SDS PAGE, 1Dimensional and 2-Dimensional electrophoresis	**9**
3	Spectroscopy – principles, Beer Lambert's law- deviation from Beer lambert's law, construction of calibration curve, UV Visible spectroscopy. Fluorescence spectroscopy. Atomic Absorption and emission spectroscopy,	**9**
4	Chromatography – Principles of chromatography column chromatography, Gel filtration chromatography, GLC, HPLC, Supercritical fluid chromatography	**9**

Suggestion on Project Topics

1. Identify any one of the food products from market and check for its label for nutritional value and health claim if any
2. Identify the official method of analysis protocol – AOAC, FSSAI Method
3. Conduct the proximate analysis of selected food – moisture, ash, protein, fat, carbohydrate, fiber
4. Evaluate the rancidity study – peroxide value, acid number of oil, saponification value
5. Conduct the shelf-life analysis and sensory analysis of food product
6. Identify the methods available for spectroscopy and chromatography for the specific analysis of food
7. Refer FSSAI standard for the food product from FSSAI regulations
8. Compare the result of your analysis with FSSAI standards and infer the result
9. Prepare a project report and present the results.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To be familiar with the Government Regulations pertaining to Food Analysis	**K1**
CO2	Identify the different methodologies employed in proper sampling of food	**K2**
CO3	Review the principles behind the analysis of various food components	**K2**
CO4	To recognise the different techniques employed for analysis of components in food	**K3**
CO5	To gain knowledge of the instrumentation and working of equipment used for Food Analysis	**K3**

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2					1		1				
CO2	3			1				1				2
CO3	3	2	1									2
CO4	2						1					
CO5	2				1							

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Food Analysis	Nielsen, Suzanne (Ed.),	Springer US	2010
2	Handbook of Food Analysis Vol I and II,	Leo M. L. Nollet	CRC Press	2004
3	Methods of Analysis of Food Components and Additives	Semih Otles	CRC Press	2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Spectroscopic Methods in Food Analysis	Adriana S. Franca and Leo M.L. Nollet,	CRC Press	2018
2	Handbook of Food Analysis Instruments	Semih Otles	CRC Press	2016
3	Methods in Food Analysis	Rui M. S. Cruz, Igor Khmelinskii, Margarida Vieira,	CRC Press	2014
4	Modern Methods of Food Analysis	Kent K. Stewart and John R. Whitaker	AVI PUBLISHING COMPANY	1984

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/104105084
2	https://nptel.ac.in/courses/104105102
3	https://nptel.ac.in/courses/104106122

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

1. **Project Planning and Proposal (5 Marks)**
   - Clarity and feasibility of the project plan
   - Research and background understanding
   - Defined objectives and methodology

2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**
   - Individual contribution to the presentation
   - Effectiveness in answering questions and handling feedback

3. **Involvement in the Project Work and Team Work (3 Marks)**
   - Active participation and individual contribution
   - Teamwork and collaboration

4. **Execution and Implementation (10 Marks)**
   - Adherence to the project timeline and milestones
   - Application of theoretical knowledge and problem-solving
   - Final Result

5. **Final Presentation (5 Marks)**
   - Quality and clarity of the overall presentation
   - Individual contribution to the presentation
   - Effectiveness in answering questions

6. **Project Quality, Innovation, and Creativity (3 Marks)**

- Overall quality and technical excellence of the project
- Innovation and originality in the project

Creativity in solutions and approaches

SEMESTER S4

FOOD PRODUCT DESIGN AND DEVELOPMENT

Course Code	PEFTT411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To acquire knowledge about food product development process
2. To analyse various concepts about food product commercialization

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Concept behind food product design and development process :-** Concept of product development, product success and failure, factors for success- Market survey process for product development, managing for product's success- Innovation strategy -possibilities for innovation, building up strategy, product development programme. Stages of product development process - product life cycle	**9**
2	**Food product commercialization**:- product launch and evaluation, Technology, knowledge and the food system, knowledge management, knowledge for the conversion of raw material properties, equipment needed. Role of consumers in product development - consumer behaviour, Food preferences, Avoiding and acceptance of consumers, Integration of consumer needs in product development.	**9**
3	**Principles of product development management:-** People in product development management, Establishing outcomes, budgets and constraints.Improving the product development process - key message, evaluating product development, innovative matrices, striving for continuous improvement, Improving success potential of new products, market exploration and acquisition, Legal aspects of new product launch.	**9**

4	**Product Stability:-** Evaluation of shelf life; changes in sensory attributes and effects of environmental conditions; accelerated shelf-life determination; developing packaging systems for maximum stability and cost effectiveness; interaction of package with food; Regulatory Aspects; whether standard product and conformation to standards; Approval for Proprietary Product.	**9**

Course Assessment Method (CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the basic concept of product development and innovation strategy	**K2**
CO2	Apply the concept of Product Development Process in real life by product commercialisation.	**K3**
CO3	Describe the role and behaviour of consumer in Product Development	**K2**
CO4	Restate the establishment out comes, budget and constraints in product development	**K1**
CO5	Analyse product development strategies to optimize packaged food stability through advanced packaging systems and environmental controls, integrating sensory attributes and regulatory compliance.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2		2			2						3
CO2	2	2	2			2						3
CO3	2		2			3						3
CO4	2		2			2						3
CO5	2		2			2						3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managing new product and process development	Steven C. Wheelwright	Harvard Business School, Free Press	1992
2	Food Product Development	Earle R & Anderson A	Woodhead Publishing	2004
3	Accelerating New product design and development	Bekley, Topp &Prinyawiwatku	IFT Press	2007

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Case studies in Food Product Development	Mary Earle & Richard Earle	CRC Press	2008
2	Creating new Foods	Mary D Earle &Richard Earle	Chadwick House Group Ltd	2001
3	Guidelines for sensory analysis in food product development and quality control.	Lyon, Francombe, Hasdell& Lawson	Chapman & Hall	1992

Video Links (NPTEL, SWAYAM…)

Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_me83/preview https://onlinecourses.nptel.ac.in/noc21_de01/preview

SEMESTER S4

BAKERY AND CONFECTIONERY PRODUCTS

Course Code	PEFTT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Develop proficiency in fundamental baking techniques such as dough preparation, fermentation processes, and baking temperatures to consistently produce high-quality bread and pastry products.
2. Acquire advanced skills in confectionery, including tempering chocolate, creating intricate sugar decorations, and mastering techniques for producing a variety of candies and chocolates with different textures and flavors.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction**:- History of bakery and confectionery-trends-Bakery products-types, specifications and composition and raw materials. Flour types-characteristics-water-salt-usage-function, yeast, production enzymes used-function role of sugar and milk-leavening agents-types functions..	9
2	**Baking Technology:-**Major baking ingredients and their functions, role of baking ingredients in improving the quality of bread. Characteristics of flour used for making bread, biscuits, and cakes. **Manufacturing of bread:-**Ingredients used for bread manufacture, methods of mixing the ingredients, dough development methods straight dough, sponge dough, moulding, proofing, baking and packaging, spoilage, bread staling, methods to reduce bread staling and spoilage. Biscuits/Cakes processing-types-ingredients Processing of Pizza and Pastry. Equipment required-Quality and standards-Regulations to be followed Packaging requirement and methods for products, Bakery unit layout	9
3	**Confectionery Science and Technology:-**Introduction, present trends in the industry. Ingredients. Sugar-boiled confectionery, crystalline and amorphous confectionery,	9

	Stages of sugar cookery:- Rockcandy, Hardcandy, soft candy, Fondants and creams, Marshmallows, toffee, lollypop, honeycomb, Caramel and fudge. processing of toffee. Quality and standards Regulations to be followed- Packaging requirements for confectionery products.	
4	**Confectionery and Chocolate processing:-** Liquorice processing-Aerated confectionery-chewing gum-bubble gum, gum tablets.Cocoa production, chocolate processing steps, chemical composition of ingredients,manufacture of chocolate beverages. Chocolates-Sweet chocolate, Milk chocolates, white chocolate, wafer-coated chocolates, cocoa butter. Fat bloom, cause, and effect on quality, ways to store chocolate and candies. Cocoa and cocoa powder, low-fat cocoa	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze the fundamental raw materials vital in bakery units	**K1**
CO2	Explain the important unit operations used in bakery and confectionery technology	**K2**
CO3	Discover an idea about the basic concepts of setting up the bakery units	**K2**
CO4	Classify the different types of confectioneries in food sector. Apply knowledge in the processing of different confectionary products and its packaging requirements	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	2					3						
CO2	3											
CO3	2		2									
CO4	2											

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Technology of Bread Making	Cauvain, Stanley P, and Young, Linda S.,	Aspen publication. Maryland	Second Edition, 1999
2	Technology and Engineering	Matz, Samuel A., akery	Chapman & Hall, London.	Third Edition
3	Bakery Products Science And Technology	Zhou.W,HuiY,H;(2014),	Wiley Blachwell Publishers,	2nd Edition, (2014),

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Chocolate, Cocoa, and confectionery" (Science and Technology)	Bernard. W. Minifie., PhD	CBS publishers and Distributors	3rd edition
2	The art of confectioner, sugar work and pastillage	Ewald Notter	Wiley Publishers	
3	Bakery products science and technology	Y H Hui	Blackwell Publishers	

Video Links (NPTEL, SWAYAM…)	
Module No.	Link ID
1	https://nios.ac.in/departmentsunits/vocational-education/stand-alone-courses/bakery- and-confectionery.aspx

SEMESTER S4

FOOD BIOTECHNOLOGY

Course Code	PEFTT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the Fundamentals of Molecular Biology and tools and techniques of Genetic Engineering
2. Analyze the fermentation processes and their applications in the production of food
3. Evaluate the potential risks and benefits of biotechnological advancements in the food industry.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Basics of Molecular Biology**: – Chemical nature of the genetic material - DNA, RNA and proteins, DNA replication, transcription and translation in prokaryotes.Application of enzymes in bakery, meat, fruit,vegetable and dairy Industries	9
2	**Expression of foreign genes:-** Promoter enzymes; Recombinant DNA technology: Restrictionenzymes, cloning vectors, Basics of strain improvement techniques, cloning procedure, cloning of specific gene and their identification.	9
3	F**ermented Foods:-** Dairy products, Oriental fermentations, Alcoholic Beverages. Processof production of commercially important organic acids- Citric Acid, Lactic Acid, Aminoacids- glutamic acid, lysine. Bioproducts for food industries- SCP, Mushroom. Natural Bio preservatives -Nisin.	9
4	**Concepts working, significance of Nutrigenomics** Nutraceuticals and Functional Foods. Genetically Modified Foods- Plant and Animal origin. Ethical issues concerning GM Foods,testing for Genetically modified organisms, labeling and traceability, Biosafety- Publicperception of GM foods, GMO Act- 2004	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	60

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the basic principles of molecular biology and applications of enzymes in food	**K2**
CO2	Comprehend the fermentation processes and their applications in the production of food additives, enzymes, and other metabolite	**K2**
CO3	Apply different recombinant DNA techniques used in the development of genetically modified foods.	**K3**
CO4	Analyze the potential risks and benefits of biotechnological advancements in the food industry.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2	2	1				
CO2	2		2			2	2					
CO3	2	2	2			2	2	2				
CO4	3	2	2				2	2				

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Molecular Biology of the Gene	James D. Watson	Benjamin Cummings, San Francisco, USA.	7th Ed. 2013
2	Biotechnology	B.D. Singh	Expanding Horizons. Kalyani Publishers, New Delhi.	2014
3	Food Biotechnology Principles and Practices	Joshi,V.K and Singh,R.S	IK International Publishing House Pvt.Ltd	2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Basic Molecular and Cell Biology	David Latchman	BMJ Publishing group,	3rd Edition,1997 `
2	Gene cloning and DNA Analysis-An Introduction	T.A. Brown	Blackwell Sciences Ltd. UK	4th Edition,2001
3	Food Science and Food Biotechnology	Lopez G.I.G and Canovas,G.V.B	CRC Press Florida,USA	2003
4	Biotechnology: Food Fermentations, Vol 1 and 2.	Joshi .V.K and Pandey	Education Publishers	2002
5	Food Biotechnology.	Rita Singh,	Global Vision Publication House Delhi,	2004

<table>
<tr><th colspan="2">Video Links (NPTEL, SWAYAM…)</th></tr>
<tr><th>Module No.</th><th>Link ID</th></tr>
<tr><td>1</td><td rowspan="4">https://archive.nptel.ac.in/courses/102/103/102103045/</td></tr>
<tr><td>2</td></tr>
<tr><td>3</td></tr>
<tr><td>4</td></tr>
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SEMESTER S4

REFRIGERATION AND COLD CHAIN

Course Code	PEFTT414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To the study the application of various air conditioning and refrigeration systems in food industry
2. To understand psychrometric terms and cold chain processes and their application in food processing

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Basics of Refrigeration and Air-conditioning:** Refrigeration-units of refrigeration - COP- Reversed Carnot cycle T-S diagram – P-H chart; Types of refrigerants and their properties; commonly used refrigerants; Psychrometric properties-psychrometric chart; Saturation line-relative humidity line-constant specific volume lines – constant thermodynamic wet bulb temperature lines-constant enthalpy lines.	9
2	**Refrigeration Components:** Compressor -Reciprocating, Rotary and Centrifugal Compressors. Condenser- Aircooled& watercooled; Evaporative condensers- Cooling tower, Spray Pond. Evaporator-Classification-Flooded evaporator-Dry expansion evaporator. Refrigeration Controls: Low sand high side float valves, capillary tube, thermostatic expansion valve, automatic expansion valve.	9
3	**Refrigeration & AC systems:-** Simple vapour compression and vapour absorption. Industrially used refrigeration systems .T hree fluid refrigeration, Lithium Bromide Refrigeration System .Cryogenics-Cryogenic Freezers. Air Conditioning-Human and Industrial air conditioning systems	9

4	**Cold chain:-** Need for cold chain- shelf- life- Q10 Value - just -in-time deliveries. Temperature limits Chilling and freezing injury, cold – shortening; PPP and TTT concepts. Temperature monitoring: Temperature-time indicators and correlation-Transportation regulations-Role of packaging in cold chain-Thaw Indicators-Cooling Load-Cold storages.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Familiarize concepts of basic refrigeration cycles, Air conditioning and psychrometry	**K2**
CO2	Identify the components of Refrigeration system and compare different types of refrigeration systems	**K3**
CO3	Identify the components of Air Conditioning and compare different types of Air conditioning systems	**K3**
CO4	Analyse the components of cold chain	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									
CO2	3	2	3	1								
CO3	3	2	3	1								
CO4	2	1	2									

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Refrigeration and Air Conditioning	Arora, and Chandra	Prentice Hall India LearningPrivateLimited	(2010), ISBN-13: 978-8120339156.
2	Refrigeration and Air conditioning	D.K. Chavan and G.K. Pathak	STANDARD BOOK HOUSE	1st Edition (2016), ISBN-13: 978-8189401528.
3	Refrigeration and Air Conditioning	C.P Arora	McGraw Hill Education	3rd edition,(2017), ISBN-13: 978-9351340164

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Textbook of Refrigeration and Air-conditioning	R .S Khurmi and J.K. Gupta	S Chand Publications	2006, ISBN-13: 978-8121927819
2	Refrigeration and Air-conditioning	Ahmadul Ameen	PHI Learning Pvt. Ltd.	2006, ISBN: 8120326717, 9788120326712
3	Postharvest Technology and Food Process Engineering	Amalendu Chakraverty, R. Paul Singh	CRC Press	2016, ISBN: 1466553219, 9781466553217
4	Unit Operations of Agricultural Processing	Sahay K.M. & Singh K.K	Vikas Publishing House Pvt Limited	2nd Edition, 2004, ISBN: 8125911421, 9788125911425
5	Food Process Engineering and Technology	Zeki Berk	Academic Press	3rd Edition (2018), ISBN: 9780128120187

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/107/112107208/
2	https://archive.nptel.ac.in/courses/112/107/112107208/
3	https://archive.nptel.ac.in/courses/112/107/112107208/
4	https://archive.nptel.ac.in/courses/112/107/112107208/

SEMESTER S4

EXTENSION AND TRANSFER OF TECHNOLOGY

Course Code	**PEFTT416**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To develop knowledge of extension and technology transfer with respect to food processing.
2. To understand the role of intellectual property legislation in the food business and also learn how to manage the production, processing, and marketing of food and agricultural products.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Current technologies and practices in food processing sector, Value chain in Food processing sector, Product process service, Study of market/ research, Marketing channel, Food business stakeholders, Market size barriers, Players in the Indian food industry	9
2	**Strategy:** Strategy- Nash equilibrium. Social, environmental, legal obligation for technology, Intellectual Property Rights, WIPO, Intellectual Properties: Patent, Trademarks, Copyright, Geographical Indications. Filing Process and Procedures in India	9
3	**Positioning:** Positioning the technology for the end user, Hypothesis- Types and characteristics, Procedure for testing hypothesis, SWOT Analysis, Determine strength weakness, opportunities and threats.	9
4	**Prototype:** Design of prototype, testing – Quality standards, Finding the target, Market alignment, Technology alignment, Risk analysis, Revenue expense valuation Evaluation of technology: Inventory management, EOQ, EBQ, ABC and VED analysis, CPM and PERT network analysis, Role of Government and non-Government organizations in technology transfer	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	60

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the current technologies and practices in food processing sector	**K2**
CO2	Describe different strategies in food business	**K2**
CO3	Explain how to position a technology for the end user	**K2**
CO4	Understand different technology and market alignment techniques	**K2**
CO5	Elaborate on different technology evaluation methods	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2					3		3			3	3
CO2	2					3		3			3	3
CO3	2					3		3			3	3
CO4	2					3		3			3	3
CO5	2					3		3			3	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Art and science of technology transfer	Speser	John & Wiley	2008
2	IPR: Protection and Management	NithyanandaK.V	Cengage India	
3	The Change Book: A blueprint for technology transfer	ATTC Network	Author House Publisher	2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Changing the World by Technology Transfer: Licensing and Commercializing of Intellectual Properties.	M Rashid Khan	Xlibris	2011
2	Entrepreneurship Development in Food Processing	KP Sudheer, Indira V		Ist Edition, 2017

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc23_lw01/preview

SEMESTER S4

PHYTOCHEMICALS IN FOOD

Course Code	**PEFTT417**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Students will learn to identify major classes of phytochemicals present in commonly consumed foods, such as flavonoids, carotenoids, and phenolic acids.
2. Students will investigate how various food processing techniques affect the concentration and bioavailability of phytochemicals in foods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Introduction to a variety of food plants & health promotion, Methods used to study connections between phytochemicals and health, ROS and Antioxidant hypothesis, Total antioxidant activity assays,Methods to study phytochemicals and health promotion. Introduction, morphological, microscopical, physical, chemical, biological analysis of herbal drugs.	**9**
2	**Introduction to different sugars, organic acids and soluble fibres:** Sugars, Soluble fibres, Ascorbic acid, Other organic acids, Amino acids.Flavonoids, anthocyanins and polyphenolics - Flavonoid biosynthesis, Anthocyanins, Condensed tannins, Tea polyphenolics, Adulteration of crude drugs and theirdetection by evaluation methods.	**9**
3	**Types, functions and mode of action of different fats and oils:** Carotenoids, Seed oils, Terpenoids, Glucosinalates Alkaloids and Seed storage proteins- Alkaloids – Capsaicinoids, Alkaloids – Caffeine, Essential amino acids, Seed storage proteins, Plant-Animal interactions	**9**

4	**Different alkaloids and their mode of action:** Morphine and a brief account of its derivatives and analogues, Ergot alkaloids and semisynthetic derivatives, Caffeine and Theophylline, Reserpine, Quinine and Quinidine, Atropine, Hyoscyamine and Scopolamine, Structure and use of Homatropine. Vincristine and Vinblastine, Taxol, Camptothecin, Podophyllotoxin and Semi-synthetic derivatives of these compounds, Quality control of herbal drugs as per WHO Guidelines.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Learn the methods used to study phytochemicals and health.	**K1**
CO2	Gain knowledge on different sugars, organic acids and soluble fibres.	**K2**
CO3	Gain knowledge about different alkaloids and their mode of action.	**K3**
CO4	Gain knowledge on quality control of herbal drugs as per WHO guidelines	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2						2	2				
CO2	2											
CO3	2	2	2	2				2			2	2
CO4	3	2					2	2	2	2	2	

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Phytochemical methods	Harborne JB	Chapman and Hall. Lea PJ, Leegood RC	1993
2	Biochemistry and Molecular Biology of Plants, 2nd Edition	Russell L. Jones (Editor), Wilhelm Gruissem (Editor), Bob B. Buchanan (Editor)	Wiley	2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Plant Biochemistry and Molecular Biology.	Thomas Colin Campbell, Thomas Campbell	Oxford University PressCampbell	3rd Ed. 2006
2	Organic chemistry- Spectroscopic identification of organic compounds	Morrison and Boyd		11th edition

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_bt08/preview

SEMESTER S4

SPICES AND PLANTATION CROPS TECHNOLOGY

Course Code	PEFTT418	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Familiarize the students about the processing techniques of various major and minor spices, coffee and tea, cocoa and cocoa products
2. To gain the knowledge on quality analysis, grading and packaging of spices

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Major spices** :- Post Harvest Technology, composition, processed products of - Pepper, Cardamom, onion, ginger and turmeric – Oleoresins and essential oils – Method of manufacture – Chemistry of the volatiles – Enzymatic synthesis of flavour identical - Quality control, Flavour of major spices - Spice oil and oleoresins	9
2	**Minor spices**:- Post Harvest Technology, composition, processed products of - Cumin, Coriander, Cinnamon, fenugreek, pepper, Garlic, Clove and Vanilla - Oleoresins and essential oils – Method of manufacture – Chemistry of the volatiles – flavours, Quality control.	9
3	**Tea :-** Occurrence – chemistry of constituents – harvesting – types of tea – green, oolong tea ctc – Chemistry and technology of CTC tea – Manufacturing process and equipment involved – Green tea manufacture – Instant tea manufacture – Grading of tea, Processing and quality control. **Coffee**:- Occurrence –harvesting – fermentation of coffee beans – changes taking place during fermentation – drying – roasting – Process flow sheet for the manufacture of coffee powder – Instant coffee, methods, process, and equipment involved– Chicory chemistry - Quality grading of coffee.	9
4	**Chemistry of the cocoa bean** :– changes taking place during fermentation of cocoa bean – Processing of cocoa bean – cocoa powder – cocoa liquor manufacture Chocolates – Types – Chemistry and technology of chocolate	9

	manufacture **Processing of plantation crops** :– production and importance – processing of coconut, oil-palm, arecanut, cashew– harvesting and stages of harvest – drying, cleaning and grading – production of value added products – packaging and storage of produces	

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the processing steps involved for different plantation crops	**K1**
CO2	Acquire knowledge on the processing of major and minor spices	**K1**
CO3	Define the different unit operations and its equipment's involved in coffee, tea and cocoa processing	**K2**
CO4	Processing steps involved for cocoa and cocoa products and processing of coconut, oil palm, arecanut and cashew	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2				2						
CO2	2	2				1						2
CO3	2	2				1						2
CO4	2	2				1						
CO5												

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Spices and Condiments – Major Spices of India	J S Pruthi	National Book Trust, New Delhi	2001
2	Spices and Condiments – Minor Spices of India	J S Pruthi	National Book Trust, New Delhi	2001
3	Handbook of Postharvest Technology: Cereals, Fruits, Vegetables, Tea, and Spices, CRC Press, 2003.	Amalendu Chakraverty, Arun S. Mujumdar, G. S. Vijaya Raghavan & Hosahalli S. Ramaswamy	CRC Press	2003
4	Post Harvest Engineering of Horticultural Crops through objectives	Pandey P H	SarojxPrakasam, Allahabad.	2002

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Tea and Tea Products: Chemistry and Health-Promoting Properties, Nutraceutical Science and Technology	Chi-Tang Ho, Jen-Kun Lin, Fereidoon Shahid	CRC Press	2008
2	Handbook Seasoning, Spices &Flavoring	Susheela Raghavan	CRC Press 2nd Ed	2006
3	Handbook of herbs & spices	K V Peter	Elsevier	2012

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_ag13/preview https://archive.nptel.ac.in/courses/126/105/126105023/

SEMESTER S4

UNIT OPERATIONS IN FOOD PROCESSING

Course Code	PEFTT415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To acquaint and equip the students with different unit operations of food industries and their design features

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Size reduction**:- Benefits, classification, determination and designation of the fineness of ground material, sieve/screen analysis, principle and mechanisms of comminution of food, Rittinger's, Kick's and Bond's equations, work index, energy utilization; Size reduction equipment: Principal types, crushers (jaw crushers, gyratory, smooth roll), hammer mills and impactors, attrition mills, tumbling mills, ultra fine grinders, fluid jet pulverizer, cutting machines (slicing, dicing, shredding). **Mixing of fluids:** – blending – types of blenders – flow patterns in fluid mixing – energy input in fluid mixing - kneading – in flow mixing -mixing of particulate solids – mixing and segregation – quality of mixing – mixedness – emulsification and homogenization -equipments	9
2	**Filtration:** – filter media – types and requirements - constant rate filtration – constant pressure filtration – filter cake resistance - filtration equipments – filter aids – expression – introduction – mechanisms – applications and equipment. **Membrane separation**:- General considerations, materials for membrane construction, ultra-filtration, processing variables, membrane fouling, applications of ultra-filtration in food processing, reverse osmosis, mode of operation, and applications; Membrane separation methods, demineralization by electro-dialysis, gel filtration, ion exchange, per-evaporation and micro filtration	9

3	**Food freezing**:- Introduction, freezing point curve for food and water, freezing points of common food materials, Principles of food freezing, freezing time calculation by using Plank's equation; Freezing systems; Direct contact systems, air blast immersion; Changes in foods; Frozen food properties; freezing time, factors influencing freezing time ,freezing/thawing time; Freeze concentration: Principles, process, methods; Frozen food storage: Quality changes in foods during frozen storage; Freeze drying: Heat mass transfer during freeze drying, equipment and practice. **Dehydration:**– Thermodynamics of moist air – Convective drying – drying curve- constant rate – falling rate – total drying time – Conductive drying – Dryers in the food processing industry – Cabinet, tunnel, belt, belt – trough, rotary – spray dryers – Fluidized bed dryer – Pneumatic dryer – drum dryer, sun, solar drying – Issues in food drying technology.	**9**
4	**Extraction:**– solid – liquid extraction (leaching) – material balance – equilibrium – multistage extraction – stage efficiency – solid – liquid extraction systems – Super Critical Fluid extraction – principles – solvents – Super critical extraction systems – application – Liquid – liquid extraction – principle – application **Evaporation**:- Principles of evaporation, mass and energy balance, factors affecting rate of evaporation, Heat and mass transfer in evaporator, factors influencing the overall heat transfer coefficient, influence of feed liquor properties on evaporation; Evaporation equipment: Natural circulation evaporators, horizontal/vertical short tube, long tube, forced circulation; single effect, multiple effect evaporators, feeding methods of multiple effect evaporation systems, feed preheating,; Fouling of evaporators and heat exchanges; Recompression heat and mass recovery and vacuum creating devices.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Characterize different unit operations in food processing	**K2**
CO2	Identify and evaluate size reduction mechanism and mixing of fluids	**K3**
CO3	Describe the working of filtration/membrane separation equipments used in food industry	**K3**
CO4	Explain the mechanisms of freezing and dehydration operation	**K3**
CO5	Analyse different types of evaporators and extraction systems	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	2			1							
CO3	3	1	1		1							
CO4	3	2	1		1							
CO5	3	2	1		1							

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Food Engineering	R. Paul Singh and Dennis R. Heldman	Elsevier, Amsterdam, Netherlands	2014, 5th Ed.
2	Unit Operations in Food Processing	Earle, R.L.	Pergamon Press. Oxford. U.K.	2003
3	Food Process Engineering and Technology.	Zeki Berk.	Academic Press. 360 Park Avenue South, New York, USA.	2009.
4	Unit Operations of Chemical Engineering,.	McCabe, w.l. Julian Smith, Peter Harriott	McGraw-Hill, Inc., NY, USA.	7th Ed 2004.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Unit Operations in Food Engineering.	Albert Ibarz and Gustavo V. Barbosa-Canovas.	CRC Press, Boca Raton, FL, USA	2003.
2	Fundamentals Of Food Process Engineering,	Toledo R T,	Springer, Fourth Edition (2018), ISBN: 9783319900971	
3	Food Process Engineering Operations,	Saravacos G D,	Taylor and Francis, 2011,ISBN: 9781420083538	
4	Food Processing Technology Principles And Practice		Fellows P J, Elsevier, Fourth Edition (2020), ISBN 9789351073918	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc20_ag01/preview

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	**UCHUT346**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	**6**
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return –- Payback – Accounting Rate of Return.	**6**

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	**15**	**12.5**	**12.5**	**50**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	**50**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	**K2**
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	**K3**
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	**K2**
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7^{TH} Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: UCHUT346

Course Name: Economics for Engineers

Max. Marks: 50		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 8 marks

Module 1

9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$.. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNPfc from GDPmp.	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDPmp and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)
*****				

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Fundamentals of ethics** - Personal vs. professional ethics, Civic Virtue, Respect for others, **Profession and Professionalism,** Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, **Technology and digital revolution**-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, **High technologies: connecting people and places**-accessibility and social impacts, **Managing conflict**, Collective bargaining, **Confidentiality**, Role of confidentiality in moral integrity, **Codes of Ethics**. **Basic concepts in Gender Studies** - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, **Gender disparity and discrimination in education**, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations**, Ethical values and practices in**	6

	connection with gender - equity, diversity & gender justice, **Gender policy and women/transgender empowerment initiatives.**	
2	**Introduction to Environmental Ethics:** Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). **Sustainable Engineering Principles:** Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. **Ecosystems and Biodiversity:** Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. **Landscape and Urban Ecology:** Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	**6**
3	**Hydrology and Water Management:** Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. **Zero Waste Concepts and Practices:** Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. **Circular Economy and Degrowth:** Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. **Mobility and Sustainable Transportation:** Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	**6**
4	**Renewable Energy and Sustainable Technologies:** Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. **Climate Change and Engineering Solutions:** Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. **Environmental Policies and Regulations:** Overview of key environmental policies and	**6**

	regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. **Case Studies and Future Directions:** Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	

Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	**Item**	**Particulars**	**Group/Individual (G/I)**	**Marks**
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
		Total Marks		**50**

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis**: Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts**: Ability to apply course concepts to real-world problems and local contexts.
- **Creativity**: Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills**: Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	**K3**
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	**K4**
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	**K5**
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	**K4**
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessmen	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:
Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.

- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

FOOD ANALYSIS AND QUALITY EVALUATION LAB

Course Code	**PCFTL407**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Lab

Course Objectives:

1. To develop the knowledge of students in the area of Food analysis
2. To evaluate different food products using various chemical analysis techniques.

Expt. No.	**Experiments**
1	Determination of total ash content in food products
2	Determination of acid-insoluble ash and acid-soluble ash in food products
3	Determination of water-soluble and water-insoluble ash in food products
4	Separation of amino acids by TLC.
5	Estimation of the protein content of food by Lowry's method
6	Determination of fat content of food by Soxhlet extraction method
7	Determination of starch in food by Anthrone method
8	Determination of crudefiber content of food
9	Sensory evaluation of food sample – Descriptive and Discriminative method
10	Quality analysis of Beverages - tea/coffee/cocoa,chicory
11	Quality analysis of sugar and confectionery products
12	Quality analysis of fruit and vegetable products
13	Quality analysis of spices
14	Quality analysis of fat and oils
15	Quality analysis of fish and meat
16	Quality analysis of milk and milk products

17	Quality analysis of cereal and cereal products
18	Determination of Antioxidant activity of food by Spectroscopic method
19	Determination of pigments present in food – chlorophyll/ carotenoid /anthocyanin

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	**25**	**20**	**50**

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	**15**	**10**	**10**	**5**	**50**

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	To understand the principles behind food analysis	**K2**
CO2	To gain Knowledge of basic equipment for food sample analysis	**K2**
CO3	Calculate and interpret nutrient composition of foods	**K2**
CO4	Evaluate data generated by experimental methods for quality evaluation of food sample	**K3**
CO5	Interpret the quality parameters of food sample in relation to food standards and regulation	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	2				2	2	2	2			3
CO3	3	2	2									3
CO4	3	2				2	2	2	2			3
CO5	3	2	2						2			3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The chemical analysis of foods and food products	Morris B. Jacobs	CBSPublishers and distributors, New Delhi.	2018
2	Hand book of analysis and quality control for fruit and vegetable products	Ranganna S	Tata McGraw Hill Publishing	2017
3	Methods in Food Analysis	Rui M. S. Cruz, Igor Khmelinskii, Margarida Vieira	CRC Publishers	2014

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Food Analysis Laboratory Manual	Neilsen S S	Springer	2017
2	AOAC Methods of Food Analysis		AOAC International	
3	ISI Handbook of Food Analysis		New Delhi Indian StandardsInstitution	1980

Video Links (NPTEL, SWAYAM…)

Module No.	Link ID
1	onlinecourses.swayam2.ac.in
2	https://onlinecourses.swayam2.ac.in/cec20_ag06
3	https://nptel.ac.in/courses/126105027
4	https://fssai.gov.in/cms/manuals-of-methods-of-analysis-for-various-food-products.php

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

   - Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
   - Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

   - Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
   - Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
   - Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

   - Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
   - Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**
   - Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

   - Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
   - Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

FOOD PROCESS ENGINEERING LAB

Course Code	**PCFTL408**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Lab

Course Objectives:

1. To understand the principle behind the various food processing unit operations.

Expt. No.	Experiments
1	Experimentation on the size reduction of food material using milling
2	Determination of Particle size distribution of food material using sieve analysis
3	Development of Food Product using single screw extrusion
4	Determination of the rate of heat transfer and heat transfer coefficient of heat exchanger
5	Determination of rate of filtration for constant pressure filtration
6	Perform the simple distillation for studying separation of mixture
7	Determination of Mixing index of various Food samples
8	Experimentation of Pasteurisation of milk/ fruit juices and microbial inactivation
9	Determination of the rate of drying for the given food grains in Rotary dryer
10	Experimentation of Osmotic dehydration of food materials
11	Determination of the percentage of actual recovery for single stage leaching
12	Determination of separation efficiency of an inclined belt separator
13	Experimentation on microwave heating of food materials
14	Experimentation on drying with a Fluidised bed dryer
15	Verification of the applicability of Freundlich equation for adsorption for the given system and determination of the values of adsorption constants ‘k’ and ‘n’ at room temperature
16	Development of a food product using Freeze drying
17	Determination of the separation efficiency of a spiral separator
18	Performance of a Spray dryer in food product development.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe operational principles of milling equipment, dryers, extruders, heat exchangers, distillation, and separators in food engineering processes.	**K2**
CO2	Explain underlying concepts of particle size distribution analysis, heat transfer mechanisms, filtration principles, distillation processes, and adsorption dynamics in food engineering.	**K2**
CO3	Interpret experimental data from sieve analysis, heat transfer coefficients, filtration rates, adsorption constants, and separation efficiencies to assess process performance and equipment effectiveness in food engineering.	**K2**
CO4	Apply theoretical knowledge to operate and conduct experiments with milling equipment, dryers, extruders, distillation apparatus, and separators to achieve specific experimental goals in food engineering.	**K3**
CO5	Conduct experiments in food engineering to determine fluidization velocity, settling velocity, and recovery curves, integrating theoretical principles with practical applications relevant to food processing and manufacturing.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3		3				2	3	3		3
CO2	3	3		3				2	3	3		3
CO3	3	3		3				2	3	3		3
CO4	3	3		3				2	3	3		3
CO5	3	3		3				2	3	3		3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Engineering	Richardson & Coulson	Elsevier	5th edition
2	Unit operations in food engineering	Ibarz and Canovas	CRC Press	2002
3	Experiments in Unit Operations and Processing of Foods	Cecila Lenzi	Elsevier	2008

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cereal grains laboratory references & procedure manual	Sergio O Serna Saldivar	Taylor and Francis Group	2012
2	Food Processing Technology: Principles and Practice	PJ Fellows	CRC Press	2nd Edition, 2000
3	Unit Operations in Chemical Engineering.	Warren McCabe, Julian Smith, Peter Harriott	Mc Graw Hill Book Co.	5th edition 1993

Video Links (NPTEL, SWAYAM…)

Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_ag01/preview
2	https://onlinecourses.nptel.ac.in/noc20_ag01/preview
3	https://onlinecourses.nptel.ac.in/noc20_ag01/preview
4	https://onlinecourses.nptel.ac.in/noc20_ag01/preview

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

FOOD TECHNOLOGY

SEMESTER S5

PROCESSING OF FRUITS AND VEGETABLES

Course Code	PCFTT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	4:0:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1.The students will gain knowledge about methods used for preserving fruits and vegetables.

2.Students will get a thorough Knowledge and understandings of the specific processing technologies used for different foods and the various products derived from these materials.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Scope of fruit and vegetable processing industry in India and world present status-constraints-prospective-fruit and vegetables. Classification-overview of principles of preservation-Drying–dehydration-freezing-concentration-canning-chemical preservation. Hurdle technology-intermediate moisture foods, Irradiation–application in fruit and vegetable industry	11
2	**Characteristics of Fruits and Vegetables:** Physical, Textural characteristics, structure and composition. Maturity standards; Importance, methods of Maturity determinations maturity indices for selected fruits and vegetables. Harvestingof important fruits and vegetables. Fruit ripening- chemical changes-regulations, methods. Storagepractices: Control atmospheric- Bead atmosphere-hypotactic storage- cool store-Zero emerge cool chamber-stores striation. Commodity pre-treatments–chemicals, wax coating,pre-packaging,phytonutrients in fruits and vegetables grading, cleaning. Physiological post-harvest diseases chilling injury and disease.	11

	Handling and packaging of fruits and vegetables.	
3	**Processing technology of fruits, vegetables & Beverages:** Technology of Jam, jelly, flow chart-equipment Tests for end point determination. Marmalade- method- flowchart- equipment Tests for endpoint determination. Pectin- commercial pectin manufacturing-Theories of gel formation- fruit preserves. Types of beverages- Juice-RTS-Nectar-Squashsyrups-concentrates-cordials-specifications. Minimal processing of vegetables-processing and packaging off reshcut fruits and vegetables. Processing technology of vegetable wafers, vegetable soup powder- sauces and ketchups .Differences – types tomato and soy sauce-problems in sauce making. Preparation of chutney-pickles-types of pickling- dry salting- brining-vine garpickling Sauer kraut technology-spoil age in sauer kraut.	**11**
4	**Fermented products from fruits and vegetables:** Technology of Vinegar fermentation—types of vinegar- methods-slow and quick process- Technology of fermented fruit wines-champagne cider-fortified wines- sherry- vermouth- Nera- Toddy. Hygiene and sanitation in Fruit and vegetable processing industry Quality evaluation –Rules and regulations related to fruit and vegetable products. FSSAI specifications for fruit and vegetable products.	**11**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To study the processing and preservation technologies of fruits and vegetables	K2
CO2	To give knowledge on the value addition on fruit and vegetable	K2
CO3	To give knowledge on the Processing technology of fruit beverages	K3
CO4	To give knowledge on the Fermented products from fruits and vegetables	K3

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	1				-	-	-	-	-	-
CO2	1	2	3				-	-	-	-	-	-
CO3	3	1	2	1			-	-	-	-	-	-
CO4	3	2	1	2			-	-	-	-	-	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Food Processing Technology: Principles and Practice.	Fellows, P J.	CRC/ Woodhead, ISBN0849308879, ISBN1855735334	2nd Edition, 1997.
2	Handbook of Vegetable Preservation and Processing	Y. H. Hui, S. Ghazala, D.M. Graham, K.D. Murrell & W.K. Nip	Marcel Dekker ISBN:0824743016	2003
3	Food Processing and Preservation,	Sivasankar, B.	Prentice Hall of India, ISBN 10:8120320867ISBN13:9788120320864	2002
4	Hand Book of Fruit Science and Technology: Production,Composition,Storage and Processing.	Salunke, D. K and S. S Kadam	Prentice Hall of India, 2002. ISBN 10:8120320867 ISBN13:9788120320864	1995

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fruit and vegetable processing- Organisations and Institutions;	SumanBhatti, Uma Varma,	ISBN10: 8123904045 ISBN13:9788123904047	Commonwealth Publishers, 2007
2	Post-Harvest Technology of Fruits and Vegetables: Handling,Processing, Fermentation and Waste Management vol. I & II,	Varma L. R. and Joshi V.K,	Indus Publishing, ISBN8173871086,9788173871085	2000
3	Fruitand vegetableprocessing- Improvingquality;	Jongen,Wim	Wood head publishing, ISBN:9781855735484,ISBN:9781855736641	1st Edition 2002
4	Fruitand Vegetable Preservation: Principles &Practices	R.P.Srivastava&SanjeevKumar	International book distributing Co.Lucknow, ISBN 10:8123924372,ISBN13:9788123924373	3rd Edition 2019

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/u8CVwxLtD14?list=PLhWyf911na-XNr-totysuQubXQLXlt6BN
2	https://youtu.be/YhJ_JGp2wxM?list=PLhWyf911na-XNr-totysuQubXQLXlt6BN
3	https://youtu.be/DMZeJr9Ow2Q
4	https://youtu.be/m2E5ZAoOqFM

SEMESTER S5

PROCESSING OF MILK & MILK PRODUCTS

Course Code	**PCFTT502**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To understand the production and processing of milk and milk products.
2. To familiarize with the technological advancements in milk industry.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Milk-Types-Composition-factors affecting composition of milk. Physical-Chemical and Thermal Properties- Reception and storage-cooling of milk - Cleaning-basic principles-can washing - can washers-cleaning-in centralised and decentralized CIP system - cleaning of various equipment.	11
2	Pasteurization - principles, objectives and methods - LTLT/holding pasteurization-types, advantages and disadvantages - HTST pasteurization-components and function of HTST pasteurizer, advantages and disadvantages. Sterilization - In-bottle sterilization, UHT processing.	11
3	Homogenization theory, effects of homogenization of milk, Types of homogenizers-stages of homogenization-importance. Centrifugation-clarification-clarifiers and separators-separation efficiency- Bactofugation. New technologies in dairy industry-membrane separation- Ultrafiltration, Reverse osmosis process, electrodialysis.	11

4	Technology of traditional Indian milk products- Khoa, Channa. Butter manufacture, cheese manufacture-methods. Basic principles of spray drying and roller drum drying- Skimmed milk powder, fermented milk products.	**11**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Gain a thorough knowledge on the properties, reception and storage of milk in dairy industries.	**K1**
CO2	Get familiarized with the principle, importance and various methods of sterilization and pasteurization followed in dairy industry.	**K2**
CO3	Study the effect of homogenization of milk and its efficiency	**K2**
CO4	Comprehend the importance of centrifugation and membrane separation in dairy industry	**K2**
CO5	Impart the ideas on the manufacture of different types of milk and milk products	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	-	-	-	-	-	-	-	-	-
CO2	3	3	3	3	-	-	-	-	-	-	-	-
CO3	3	3	3	2	-	-	-	-	-	-	-	-
CO4	3	3	3	3	2	-	-	-	-	-	-	-
CO5	3	3	3	3	2	-	-	-	-	-	-	-

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Dairy Plant Engineering and Management	Tufail Ahmed	CBS Publishers and Distributors,New Delhi	2001
2	Dairy Science and Technology	Pieter Walstra, Jan T M Wouters, Tom J Geurts,	Taylor Francis Group, CRC press	2006
3	Outlines Of Dairy Technology	De Sukumar	Oxford University Press	2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Milk and Dairy Product Technology	Spreer, Edgar	Marcel Dekkar	2005
2	Dairy Technology: Milk and Milk Processing	Shivashraya Singh	New India Publishing Agency	2014
3	Novel Dairy Processing Technologies: Techniques, Management, and Energy Conservation (Innovations in Agricultural & Biological Engineering)	Megh R. Goyal, Anit Kumar, Anil K. Gupta	Apple Academic Press	2018
4	Engineering Aspects of Milk and Dairy Products	Selia, Jane dos Reis Coimbra and Jose A. Teixeir	CRC Press	2009

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	www.onlinecourses.nptel.ac.in
2	www.archive.nptel.ac.in

SEMESTER S5

FOOD SAFETY AND QUALITY REGULATION

Course Code	PCFTT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hr. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. This course will deal with the food safety, quality and regulations practiced in food industry.
2. The course will focus on providing graduate students with a detailed knowledge of risk analysis in food analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	INTRODUCTION: Principles of food safety, need for food safety culture, consequences of unsafe food, challenges in food safety, recent statistics of foodborne illness, Design and construction of food premises, cleaning and disinfection facilities, Control of rats and rodents, insects and microbes, Personal hygiene,SOP,SSOP. Hazard and risk analysis: Hazard, types, risk, risk analysis, risk assessment, food recall -significance food recall plan, allergen control plan, food traceability, significance, methods available.	9
2	Definition of food quality- ways for describing food quality, factors affecting food quality, Quality aspect in food supply chain and food processing industry Different quality control tools, Food authentication – significance, methods available for food authentication.	9
3	HACCP – History, introduction and significance, Principles of HACCP and steps in HACCP,HACCP worksheet – preparation with case study Significance, CCP determination, ISO, FSMS, TQM. Food Standards and Specification –Global Food Safety Initiative, CODEX, CODEXINDIA, National Codex Committee of India, ToR,	9

	Shadow Committee FSSA 2006- its origin, Salient features of Act, Regulations,FSSAI,its activities for ensuring food safety in India, BIS,BRC,FAO, AGMARK,Other agencies involved in regulating food business.	
4	Risk assessment studies: Risk management, risk characterization and communication. Voluntary Quality Standards and Certification GMP, GHP, HACCP, GAP, Good Animal Husbandry Practices, Good Aquaculture Practices. An Overview and structureof9001:2000/2008,ISO9001:2000,ISO22000:2005,ISO9000,ISO22000,ISO14000ISO17025,FSSC22000,BRC,BRCIOP,IFS,SQF1000,SQF 2000.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recognize the concept of food safety and standards and rules established to enforce the safety and purity of food products	**K2**
CO2	Summarize different types of food hazards and contaminations and know about food safety aspects of novel methods of food processing	**K3**
CO3	To gain knowledge of the food safety aspects of novel methods of food processing	**K2**
CO4	Distinguish various food safety regulations, risk assessments and voluntary quality standards and certification	**K3**
CO5	Relate different types of food acts and food standardization	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	2	2									
CO3	2	3		3		2	2					
CO4	2	3		3								
CO5	2	2	2			3	3					

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Quality Assurance for the Food industry	Andres Vasconcellos J.	CRC press.	2005
2	Food quality assurance-Principles & practices	Inteaz Alli.	CRC press.	2004
3	HACCP -A practical approach	Sara Mortimore and Carol Wallace	Chapman and Hall	Thirdedition,2013
4	International Standards for Food Safety	Rees,NaomiandDavidWatson	AspenPublica tion	2000

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Total Quality Assurance for the Food Industries	Gould WA and GouldR W	CTI Publications Inc. Baltimore	1998
2	Food Safety Handbooks	Schmidt,RonaldH.and Rodrick, G.E	WileyInter science	2005
3	Handbook of indices of food quality and authenticity	Singa lRS	Woodhead Publishing	1997
4	Food Safety Regulations, Concern sand Trade:The Developing Country Perspective	Mehta,RajeshandJ.Geo rge	Macmillan	2005
5	The Prevention of Food Adulteration Act,1954D	Gould WA and GouldR W	Commercial Law Publishers India) Pvt. Ltd.	2016

ideo Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
2	https://shorturl.at/CZwmy
3	https://www.youtube.com/watch?v=B_Q8ZW_qVIQ
4	https://shorturl.at/aX93z

SEMESTER S5

FOOD ADDITIVES AND FLAVORINGS

Course Code	**PBFTT504**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	4	**ESE Marks**	40
Credits	3:0:0:1	**Exam Hours**	2 Hrs. 30 min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To impart knowledge about various categories of food additives their functions, permissible limits to be used in various food products.
2. To gain knowledge on flavour isolation methods and application of flavouring agents in food processing

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Food Additives – definition, types, examples, functions, rules and regulations responsible for use of food additives, GRAS, maximum permissible limits, AD1, Risk Assessment, Toxicity Evaluation of food additive , Application of food additive in food processing and preservation	9
2	Different types of food additives –functions ,mode of action, examples and applications in food industry – Preservatives, Antioxidants, Acidulants, Colouring agents, Emulsifiers, Sweeteners and thickening agent	9
3	Food flavour – Food flavour industry – scope and significance, flavouring agents, example, natural, natural identical and artificial flavour, methods of flavour isolation, different flavour products, spray drying of flavour, different types of flavour products, application of flavouring agents	9
4	Flavour creation ,Sensory analysis of flavour, Subjective and objective methods of flavour analysis, Scoville unit Seasoning –development and optimization process, application of seasonings flavour enhancer, flavour quality analysis, flavour organizations certifications	9

Suggestion on Project Topics

1. Market study on the food additive – variety and types of food additive in different processed foods
2. Case study on the food additive – Review of literature on health risk caused by food additive
3. Application of food additive in selected processed food item
4. Isolation of flavour from selected spices/herbs
5. Development of seasoning agent and evaluation of seasoning using subjective and objective methods

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand diverse food additives used in the food industries and its technological function	**K2**
CO2	Knowledge on the rules and regulations prevailing for the use of food additives	**K2**
CO3	Application of different food additives in food processing and preservation	**K3**
CO4	Application of flavouring agents and flavour products in food industry	**K3**
CO5	Identification of different methods of flavour isolation and development of seasonings	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									2
CO2	3					2						2
CO3	3		2			2						
CO4	3											
CO5	3								2			

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Additives	Larry Branen,P. Michael Davidson,Seppo Salmine, John H. Thorngate III	Marcel Dekker, Inc.	Second 2002
2	Chemistry of Food Additives and Preservatives	Titus A. M. Msagati	John Wiley & Sons, Ltd.	2013
3	Food Flavour Technology	Robert S. T. Linforth Andrew J. Taylor	Wiley-Blackwell	2010
4	Natural Food Flavors and Colorants	Mathew Attokaran	Wiley-Blackwell	2017

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Saltmarsh's Essential Guide to Food Additives	Mike Saltmarsh	Royal Society of Chemistry	2020
2	Food Additives Data Book	Lily Hong-Shum, Jim Smith	Wiley-Blackwell	2011

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_ag15/preview
2	https://onlinecourses.swayam2.ac.in/cec21_ag12/preview

PBL Course Elements

L: Lecture (3 Hrs.)	**R: Project (1 Hr.), 2 Faculty Members**		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

1. **Project Planning and Proposal (5 Marks)**
    - Clarity and feasibility of the project plan
    - Research and background understanding
    - Defined objectives and methodology

2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**
    - Individual contribution to the presentation
    - Effectiveness in answering questions and handling feedback

3. **Involvement in the Project Work and Team Work (3 Marks)**
    - Active participation and individual contribution
    - Teamwork and collaboration

4. **Execution and Implementation (10 Marks)**
    - Adherence to the project timeline and milestones
    - Application of theoretical knowledge and problem-solving
    - Final Result

5. **Final Presentation (5 Marks)**
    - Quality and clarity of the overall presentation
    - Individual contribution to the presentation
    - Effectiveness in answering questions

6. **Project Quality, Innovation, and Creativity (3 Marks)**
    - Overall quality and technical excellence of the project
    - Innovation and originality in the project

Creativity in solutions and approaches

SEMESTER S5

MODELLING & SIMULATION IN FOOD PROCESSING

Course Code	**PEFTT521**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To introduce students to the principles and applications of statistical methods in the food processing industry fundamental principles of food process modelling and simulation.
2. To provide students with a comprehensive understanding of the different types of food process models the role of food process modelling and simulation in the food industry and its potential for improving food quality and safety, reducing costs and waste, and promoting sustainability.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Statistical Analysis in Food Processing: Definition of statistical analysis and its importance in the food processing industry, Understanding the various types of statistical methods used in food processing analysis, Overview of hypothesis testing and its use in food analysis, Steps involved in conducting hypothesis tests and interpreting the results.	9
2	Fundamentals of Linear Regression, Introduction to regression analysis, Simple linear regression and its assumptions, Estimation of regression coefficients, Hypothesis testing in linear regression, Model evaluation metrics: R-squared, adjusted R-squared, Mean Squared Error (MSE), Introduction to multiple linear regression	9

3	Introduction to Food Process Modelling: Overview of food process modelling and its applications, Types of models and their use in food processing, Empirical Model Development: Factorial, fractional factorial and rotatable central composite experimental design, developing empirical equations using experimental data.	**9**
4	Mass transfer models: such as liquid-liquid extraction, distillation, multicomponent separation, evaporator. The main steps in model building of separation equipment. Major types of available heat exchange equipment. Shell-andtube heat exchangers, Required Heat Duty. The main steps in model building of heat-exchange equipment.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the statistical techniques and their applications in food processing, including ANOVA, and hypothesis testing	**K2**
CO2	Understand the fundamental principles of food process modelling.	**K2**
CO3	Know the different types of food process models, including empirical models, analytical models, and numerical models	**K3**
CO4	Develop and validate food process models using experimental data and statistical techniques, such as regression analysis and optimization methods.	**K4**
CO5	Know the different types of simulation software and their applications in the food industry, including computational fluid dynamics (CFD) software, such as ANSYS Fluent, MATLAB	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									
CO2	3	2										
CO3	3	3	2	2								
CO4	3	3	2	2	2	1						
CO5	3	2	2	2	2							

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Statistical Methods	Donna L. Mohr, Freund and W.J. Wilson	Academic Press	4th Edition, 2021
2	Food Processing Operations Analysis	H. Das	Asian Books Private Limited	2005
3	Modeling Food Processing Operations	Serafim Bakalis, Kai Knoerzer and Peter J. Frye	Woodhead Publishing	2015
4	Process Modelling and Simulation	César de Prada, Constantinos Pantelides, José Luis Pitarch	Mdpi AG	2019

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Processing Operations Modeling: Design and Analysis	Soojin Jun, Joseph M. Irudayaraj	Routledge Taylor &Francis Group	Second Edition, 2001
2	Food process modelling and control: chemical engineering applications	Ozilgen, M	Gordon and Breach Science Publishers	1998
3	Mathematical Modelling, Vol. 1:A Chemical Engineering Perspective (Process System Engineering)"	Aris R	Academic Press	1999

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/103/107/103107096/
2	https://archive.nptel.ac.in/courses/103/107/103107096/
3	https://archive.nptel.ac.in/courses/103/107/103107096/
4	https://archive.nptel.ac.in/courses/103/107/103107096/

SEMESTER S5

NANOTECHNOLOGY IN FOOD

Course Code	PEFTT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 min.
Prerequisites (if any)	Nil	Course Type	PE

Course Objectives:

1.Develop students' knowledge in nanoparticles, covering their synthesis, characterization, and unique properties.

2.Explore the potential applications of nanotechnology in the food industry by leveraging the distinctive features of nanomaterials.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to nanotechnology** Introduction to nanotechnology- Definition, History. Nanoparticles - Physical, Chemical andBiological properties. Quantum dots, nanocomposites, nanofibers, nanowires, nanotubes, nanoforms of carbon. **Environmental toxicology of engineered nanoparticles**. Monitoring nanoparticles - Risk governance, General regulations on safety aspects, Regulation aspects of nano scale food ingredients.	9
2	**Synthesis and characterization of nanomaterials:** Bottom up Synthesis and Top down Approach – Physical, chemical and Biological methods ofnanoparticles synthesis. Characterization of nanomaterials – UV Visible Spectroscopy, X-ray diffraction technique, Scanning Electron Microscopy, Transmission Electron Microscopy including high-resolution imaging, Surface Analysis techniques -AFM, SPM, STM, SNOM, ESCA, SIMS.	9

3	**Nano-engineered food ingredients and additives:** Nano materials for food applications - functionalized nanomaterials, nano additives. Foodprocessing and food safety and bio-security - Nanotechnology for improving food quality, detection of contaminants, Electrochemical sensors for food analysis	**9**
4	**Nanotechnology in food packaging and food preservation:** Nanomaterials for bacterial, fungal and viral inhibition – properties, mode of action. Nanopackaging for enhanced shelf life -nano composites, nano coatings, nanomembranes.Role in active packaging -Smart/ Intelligent packaging. **Potential hazards and regulations:** Potential hazards - health risks, toxicity of nanoparticles, effects of inhaled particles -respiratory systems, skin exposure to nanoparticles, hazards and risks of exposure tonanoparticles.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the history of nanotechnology, nanoparticles and its properties.	**K1,K2**
CO2	Develop a basic knowledge in synthesis and characterization of nanoparticles.	**K2**
CO3	Recognize nano-engineered food ingredients and additives.	**K3**
CO4	Understand the application of various nanoparticles in food packaging and nano-sensing	**K3**
CO5	Understand the toxic effects and risk associated with engineered nanoparticles	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										1
CO2	3	2	2									1
CO3	2	2	3									1
CO4	2	2										1
CO5	3	3										1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to nanotechnology	Charles P. Poole and Frank J. Owens.	Wiley Publishing	2008
2	Nanotechnology: An Introduction to Synthesis, Properties and Applications of Nanomaterials	Thomas Varghese & K.M. Balakrishna.	Atlantic Publishing	2012
3	Nanotechnology in Agriculture and Food Science	Monique A. V. Axelos and Marcel Van de Voorde.	Wiley VCH	2017
4	Nanotechnologies in Food	Qasim Chaudhary, Laurence Castle, Richard Watkins.	RSC Publishing	2017
5	Nanotoxicity: From In Vivo and In Vitro Models to Health Risks.	Saura C. Sahu	Wiley-Blackwell	2009

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Nanotechnology in the Agri-Food Sector: Implications for the Future	Prof. Dr. Lynn J. Frewer, Prof. Dr. Willem Norde, Arnout Fischer and Dr. Frans Kampers.	Wiley-VCH Verlag GmbH & Co. KGaA	2011
2	Nanotechnology in the Food,Beverage and Nutraceutical Industries	Qingrong Huang.	Woodhead Publishing,	2012
3	Advances in Nanosensors for Biological and Environmental Analysis	Akash Deep and Sandeep Kumar	Elsevier	2019

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/ebO38bbq0_4
2	https://archive.nptel.ac.in/courses/118/107/118107015/#, https://youtu.be/w6DD-Yw4t60, https://youtu.be/rR7GwTqxFOE
3	https://archive.nptel.ac.in/courses/118/107/118107015/, https://archive.nptel.ac.in/courses/118/107/118107015/
4	https://archive.nptel.ac.in/courses/118/107/118107015/, https://archive.nptel.ac.in/courses/118/107/118107015/

SEMESTER S5

FOOD RHEOLOGY

Course Code	**PEFTT523**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To understand the rheological properties of food and their significance in food processing.
2. To apply the knowledge of viscoelastic properties in handling equipments and processing of foods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Basic concepts of Food Rheology**: Introduction and scope, food rheological properties, Various types of stress, strain, modulus and viscosity, Poisson ratio evaluation,Physical state of a food material, various flows, Classical idealmaterials, Time effects (viscoelasticity), Rheological equations, Rheological models	9
2	**Rheological Properties of food:** Types of Deformation;Newtonian and Non-Newtonian fluids, Creep, Stress relaxation, Viscometry, Dynamic tests, Various behaviour: Force deformation behaviour, Stress strain behaviour (Basic &Bulk), Elastic plastic behaviour, Viscoelastic behaviour, Flow behaviour of Pseudoplastic, Thixotropic, Dilatant, Rheopectic and Plastic; Phasetransition in foods.	9
3	**Measurement of Rheological properties:** Different viscosity measurement and viscometers: Rotational viscometers, Vibrational (Oscillation) viscometer, Bostwick consistometer, Pressure-Driven	9

	Flow Viscometers, Capillary tubes and Falling sphere type, Torsion Gelometer for Solid Foods, Texture of Foods: Dough testing instruments.	
4	**Application of Rheology in fluid food handling and processing:** Textural properties of fruits & vegetables; Dough, Pasta and Baked products; dairy products; Meat; Fat and fat products; and their instrumental Measurements. Rheology of chocolate, Texturalcharacteristics of foodemulsions, Functions of emulsifiers inrelation to food texture,Sensory measurement of food texture and texture profile.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the rheological properties of food and their significance in food processing	**KL2**
CO2	Apply the knowledge of viscoelastic properties in handling and processing of foods	**KL3**
CO3	Differentiate different types of viscometers, understand their working and applications in food industry	**KL2 ,KL4**
CO4	Analyse the rheological properties and texture in food handling and processing	**KL4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		2	1									
CO2	3	2	2	1		2						
CO3	2				3	2						
CO4	1	2	2	1	3	2						

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Physics Physical Properties- Measurement and Applications	Ludger O Figura	Springer.	2007
2	Rheological Methods in Food Process Engineering	James F. Steffe	Freeman Press.	1992
3	Rheology: principles, measurements, and applications	Macosko W. Christopher	Wiley	1994
4	Food Process Engineering	Heldman, D. R.	AVI Publications.	2007

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Dough Rheology and Baked Products:	Faridi, H. and Faubion, J. M.	CBS Publications, New Delhi.	1997
2	Engineering Properties of Foods:	Rao, M. A., Rizvi, S. S. H. and Datta A. K.	CRC Press.	2005.

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/126/105/126105011/
2	https://archive.nptel.ac.in/courses/126/105/126105011/
3	https://archive.nptel.ac.in/courses/126/105/126105011/
4	https://archive.nptel.ac.in/courses/126/105/126105011/

SEMESTER S5

SENSORY EVALUATION OF FOOD PRODUCTS

Course Code	PEFTT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To familiarize the different methods used in the sensory evaluation of the food product
2. To understand the procedure and selection aspects of evaluation panel and the method to be used for the evaluation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction** Definitions of sensory evaluation, importance of sensory evaluation. Factors affecting food acceptance- sensory, psychological and physiological aspects. Sensory assessment of food quality Appearance- Visual perception, colour of foods, Flavour, Taste, Aroma & odour- Perception of odour, sniffing, texture.	9
2	**Selection of panel-** Types of panel, methodology for sensory evaluation. **Considerations for testing sensory evaluation-** Testing area & setup, lighting, testing schedule, preparation of samples, cooling & order of presentation, choosing of panels. Sensory tests for food- Types of test- Analytical - Discriminative (difference test, sensitivity test) Descriptive test – (Category scaling, radio scaling, Flavour Profile analysis, texture profile analysis, quantitative descriptive analysis) Affective test – Paired performance, Ranking, Rating- hedonic & food action scale	9

3	**Texture & optical properties:** Subjective measurements – physiological aspects, psychological aspects, mechanical aspects.Imitative measurements- equipments used, texture profile analysis, dynamic test for evaluation of food texture, colour, measuring equipment, L*a*b system.	**9**
4	**Introduction to Statistical Analysis in Food Processing**: Definition of statistical analysis and its importance in the food processing industry, Understanding the various types of statistical methods used in food processing analysis. Data analysis- concept of mean, median & standard deviation, certification of laboratories for sensory analysis, application of sensory evaluation	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand different aspects of sensory science & evaluation & their applications	**K2**
CO2	To understand the principles of physiological and psychological aspects of sensory evaluation.	**K2**
CO3	To familiarize the different test methods involved in discrimination& descriptive tests	**K3**
CO4	To understand how a sensory evaluation provides important information in the product development process	**K2**
CO5	To examine the types of statistical analyses used to summarize sensory data and to draw conclusions and make recommendations about product characteristics.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2			3					3				
CO3	2											
CO4	2											
CO5	3											

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Sensory Properties of Foods	Brich G,Brennan J & Parker K J	Mcmillan Publishing Company.	1977
2	Foodscience	Srilakshmi B	New age International Ltd	2000
3	Unit operations of Agricultural Processing, Second edition,	K M Sahay and K K Singh,	Vikas Publishing India Pvt Ltd,	2004
4	Modeling Food Processing Operations	Serafim Bakalis, Kai Knoerzer and Peter J. Frye	Woodhead Publishing	2015
5	Process Modelling and Simulation	César de Prada, Constantinos Pantelides, José Luis Pitarch	Mdpi AG	2019

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food science	Charley H	Mcmillan Publishing Company.	-
2	Sensory evaluation of food, Statistical methods & procedure	Mahony M	-	-
3	, "Engineering Properties of Biological Materials". .	Singhal,O.P. and Samuel,D.V.K	Saroj Prakasan, Allahabad	2003
4	"Physical properties of foods"	Peleg, M.andBagelayE.B.,	AVI publishing Co. USA	1983
5	Food Processing Operations Modeling: Design and Analysis	Soojin Jun, Joseph M. Irudayaraj	Routledge Taylor &Francis Group	Second Edition, 2001

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://shorturl.at/AUdPr
2	https://shorturl.at/7V4l3
3	https://shorturl.at/aaQcx
4	https://archive.nptel.ac.in/courses/103/107/103107096/

SEMESTER S5

ENGINEERING PROPERTIES OF FOOD MATERIALS

Course Code	**PEFTT526**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To acquire knowledge about properties of food materials that are important in processing

2. To study the measurement methods of engineering properties and analysis of food materials

SYLLABUS

Module No.	**Syllabus Description**	**Contact Hours**
1	Introduction to engineering properties of biological materials, classification of food properties, food quality and safety **Physical properties**: Shape and size – criteria for describing shape and size Volume and methods of measurement of volume, Density, Types, Method of measurement of apparent density, material density, particle density, bulk density, true density Porosity, types, methods of measurement ofporosity Surface area measurement for fruits and vegetables, egg surface area, pore surface area.	**9**
2	**Frictional properties**: Laws of friction, effect of load and properties of contacting bodies. Effect of sliding velocity and contact surface temperature, effect of water film and surface roughness. Rolling resistance, angle of repose, angle of internal friction, pressure distribution in storage structures and compression chambers	**9**
3	**Thermal properties**:Thermal conductivity, measurement of thermal conductivity, Specific heat, measurementof specific heat, enthalpy, thermal diffusivity ,boiling point, freezing point ,dielectric properties	**9**

	Aerodynamic properties: drag coefficient,terminal velocity,measurement of terminal velocity; application the properties in the harvesting, post harvest handling, processing, storage and how it affects consumer's perception of food quality and safety.	
4	**Texture &optical properties:** Subjective measurements – physiological aspects, psychological aspects, mechanical aspects.Imitative measurements- equipments used, texture profile analysis, dynamic test for evaluation of food texture, mechanical test applicable to food materials – firmness and hardness, mechanics of hardness, dynamic hardness, effect of age, water content and temperature on texture of foods; colour, measuring equipment, L*a*b system.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the significance and evaluation of physical properties of food materials	**K2**
CO2	Acquire knowledge of frictional properties of food materials	**K2**
CO3	Recognize thermal, aerodynamic properties food and their measurement techniques.	**K2**
CO4	Understand methods for the measurement of texture and colour of food material	**K2**
CO5	Apply the knowledge of various properties to improve the qualitative and quantitative attributes of food.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										1
CO2	3	2	2									1
CO3	3	2	3									1
CO4	3	2										1
CO5	3	3										1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Physical properties of plants animal materials	Mohensenin N N	Gordon and Breach publishers, New York	1980
2	Engineering properties of foods	Rao M A , Rizvi S SH,Azim K Datta and Jasim Ahmed,	4th Ed., CRC Press	2014
3	Unit operations of Agricultural Processing, Second edition,	K M Sahay and K K Singh,	Vikas Publishing India Pvt Ltd,	2004
4	Physical Properties of Foods Novel Measurement Techniques and	Ignacio Arana	Applications, 1st edition, CRC Press,	2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Engineering Properties of Biological Materials".	Singhal,O.P. and Samuel,D.V.K	Saroj Prakasan, Allahabad	2003
2	"Physical properties of foods"	Peleg, M.andBagelayE.B.,	AVI publishing Co. USA	1983

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	http://digimat.in/nptel/courses/video/126105011/L36.html
2	https://ggsestc.digimat.in/nptel/courses/video/126105011/L09.html
3	http://digimat.in/nptel/courses/video/126105011/L36.html
4	https://ggsestc.digimat.in/nptel/courses/video/126105011/L09.html

SEMESTER S5

SEPARATION PROCESSES IN FOOD TECHNOLOGY

Course Code	PEFTT527	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Differentiate conventional and novel separation techniques
2. Apply novel separation techniques in various fields of Food Technology

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to separation techniques: Separation factor and its dependence on process variable. Equilibrium and rate governed process. Classification of separation process: conventional methods like sedimentation, centrifugal separation, distillation, absorption, adsorption etc. Limitations of conventional methods. Theory of cascades and its application in single and multistage operation for binary and multi component separations, thermodynamic analysis of separation systems.	9
2	Membrane Separation: Membrane configurations – Plate and Frame, Tubular, Spiral wound and Hollow fibre systems. Types of membranes, Materials for membrane, design considerations, performance of membrane, electro dialys is ,reverse osmosis, Nanofiltration, ultrafiltration, Microfiltration and Donnan dialysis, External field induced membrane separation processes, Applications in food processing.	9
3	Chromatographic Separations: Theory of chromatographic separation, Selectivity or separation factor, Efficiency of chromatographic system. Principle, method and applications of different chromatographic techniques in Food Processing: Liquid chromatography, Liquid-solid chromatography, Affinity chromatography, Immunochromatography, Ion exchange	9

	chromatography Advantages and disadvantages of chromatographic separations.	
4	Adsorption and Ionic Separations: Types and choice of adsorbants, Adsorptive Membranes, adsorption techniques, batch adsorption, stage-type and continuous adsorption; fixed-bed adsorption, adsorption isotherms. Ion exchange resins, properties, Binary ion exchange, Ion movement theory, Controlling factors, electrophoresis, Di-electrophoresis, Zonal electrophoresis, Applications of Ion Exchange separation in the Food Industry. Other major techniques: Freeze crystallization, Prevaporation and permeation techniques for solids, liquids and gases. Osmotic distillation, Zone melting: Equilibrium diagrams, Controlling factors ,Apparatus and applications, Foam Separation: Surface adsorption, Nature of foams, Apparatus, Applications in food processing, and Controlling factors.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Differentiate conventional and novel separation techniques	**K2**
CO2	Analyze and design novel membranes for food industry application	**K2**
CO3	Paraphrase different chromatographic separation techniques	**K2**
CO4	Analyze adsorption and ion exchange based separation processes	**K3**
CO5	Describe pervaporation, freeze crystallisation and other separation processes	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2			3									
CO3	3											
CO4		3										
CO5	3											

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Bioseparations–Principles and Techniques	Sivasankar	Prentice Hall of India Pvt.Ltd, New Delhi.	2005
2	Food Process Engineering and Technology	Zeki Berk	Academic Press	2006
3	Handbook of process chromatography	Gunter Jagschies Gail Sofer and Lars Hagel	Academic Press	2007
4	Essentials in Modern HPLC Separations	Serban C. Moldoveanu and Victor David	ElsevieR	2012
5	Downstream processing for Biotechnology	Belter, P.A. and Cussler, E.L. Hu, W.S	Wiley, New York.	1988

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Separation Process Principles	Seader J.D., Ernet J. Henlay, and Keith, D.	Wiley	2010
2	Membrane Separation processes	Kaushik Nath	Prentice Hall of India Pvt.Ltd	2008
3	Basic Principles of Membrane Technology	Marcel Mulder	2 Ed., Springer Publications	2007
4	Membrane Processes: Pervaporation, Vapor Permeation and Membrane Distillation for Industrial Scale Separations	S. Sridhar and Siddhartha Moulik	Wiley-Scrivener, 1st edition	2018
5	Downstream processing for Biotechnology	Belter, P.A. and Cussler, E.L. Hu, W.S	Wiley, New York.	1988
6	Bioseparation Engineering: Principles, Practice and Economics	Ladisch, M.R	Wiley, Interscience.	2001
7	Novel Catalytic and Separation Processes Based on Ionic Liquids	Dickson Ozokwelu, Suojiang Zhang, Obiefuna Okafor, Weiguo Cheng and NicholasLitombe,	Elsevier	2017
8	Essentials in Modern HPLC Separations	Serban C. Moldoveanu and Victor David	Elsevier	2012

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/102106022
2	https://nptel.ac.in/courses/102106022
3	https://nptel.ac.in/courses/102103044
4	https://nptel.ac.in/courses/102103044

SEMESTER S5

FOOD FERMENTATION TECHNOLOGY

Course Code	**PEFTT528**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	40
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. Impart knowledge and skills related to process technologies in fermented food products.
2. Learn about the fermentation metabolism involved in the production of various fermented food products.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to food fermentation-**Definition and significance of food fermentation, historical background and cultural aspects of fermented foods. Importance of fermentation in food preservation. Microbiology of fermented foods- types of microorganisms in food fermentation (yeast, bacteria, molds), role of microorganisms, starter cultures and their selection.	9
2	**Fermentation metabolism, equipment and technology-**fermentation pathways and metabolic processes- glycolysis, TCA cycle, anaerobic respiration. Production of metabolites- ethanol, lactic acid, acetic acid. Factors affecting microbial growth and fermentation. Types of fermentation systems- batch, fed-batch, continuous. Design and operation of bioreactors.	9
3	**Fermented food products and its manufacturing-** fermented dairy products- yogurt, cheese, kefir- microorganisms and process involved. Fermented fruits and vegetable products- sauerkraut, kimchi, pickles- lactic acid bacteria and fermentation processes. Fermented cereal and legume product- bread, rice-based and soy-based products- biochemical changes during fermentation. Fermented meat and fish products- sausages, salami, fish sauce, fish paste. Storage and shelf-life of the products.	9

4	**Health benefits of fermented foods-** Probiotics and their health benefits, fermented foods in nutrition and health. **Quality control and safety in fermented foods-** Quality control parameters and testing methods, regulatory aspects of fermented foods, Safety aspects-pathogenic microorganisms, toxins and allergens. Future prospects of food fermentation and advances in fermentation technology.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of microbial fermentation.	**K2**
CO2	Understand the different fermentation metabolism process and equipments	**K3**
CO3	Gain knowledge on manufacturing of different fermented food products.	**K4**
CO4	Critically assess quality control,safety measures and nutritional values of fermented foods.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2	1					
CO2	3		2		1							
CO3	2		2				2					
CO4	1					2		2				

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Microbiology and Technology of Fermented Foods"	Robert W. Hutkins.	Blackwell	2^{nd} Edition, 2016
2	"Food, Fermentation and Micro-organisms"	Charles W. Bamforth	Wiley- Blackwell	2^{nd} Edition, 2019
3	"Fermented Foods and Beverages of the World"	Jyoti Prakash Tamang	CRC Press	1^{st} Edition, 2010
4	" Fermented Foods Part ii Technological interventions"	Ramesh C. Ray	CRC Press	1^{st} Edition, 2017

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Fermented Foods Part 1"	Ramesh C. Ray	CRS Press	1st Edition, 2016
2	"Handbook for Fermented functional foods"	Farnworth, Edward R	CRS Press	2nd Edition, 2008
3	"Fundamentals of food biotechnology"	Byong H. Lee	Wiley- Blackwell	2nd Edition, 2015

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/103/107/103107088/
2	http://nitttrc.edu.in/nptel/courses/video/102105058/L25.html
3	https://archive.nptel.ac.in/noc/courses/noc20/SEM2/noc20-bt21/
4	http://acl.digimat.in/nptel/courses/video/102105058/L41.html

SEMESTER 5

NOVEL FOOD PROCESSING TECHNOLOGY

Course Code	**PEFTT525**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Students will be learning the latest technologies in food processing and its applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction and HPP Introduction to Food Process Engineering** Scope and importance. High Pressure processing – principle, equipment and process of HPP treatment, Application of HPP in food processing.	**9**
2	**Membrane Technologies and PEF** Membrane technologies in food processing – Overview of Membrane technology. Microfiltration, Ultra filtration (UF), Nano filtration(NF) and Reverse Osmosis (RO) and theirindustrial applications. Pulsed electric field processing - Principle, process and application in food processing sector	**9**
3	**Ultrasonic processing** Properties and application of ultrasonic processing techniques. Microwave and radio frequency processing. Ohmic heating, IR heating, dielectric heating and Food irradiation, Effects on foods Applications of irradiation.	**9**
4	**Hurdle technology and Nanotechnology** Hurdle technology - Concept of hurdle technology and its application. Nanotechnology, its principles and applications in foods	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	*Internal Ex*	*Evaluate*	*Analyse*	*Total*
5	**15**	**10**	**10**	**40**

Criteria for Evaluation(Evaluate and Analyse): 20 marks

1. **Problem Definition (2 points)**
   - Check for clarity and completeness of the problem definition.
   - Check for clear objectives and divide into sub objectives
2. **Problem Analysis (4 points)**
   - Examine the problem and identify the variables and responses
   - Divide the problem to manageable parts
3. **Implementation of software (2 points)**
   - Use suitable software for getting the solution
   - Check for the appropriateness of tools used
4. **Evaluate (4 points)**
   - Evaluate the proposed solution.
   - Assess the correctness of solution.
5. **Validation of the results (4 points)**
   - Validate the results using available data.
   - Do the error analysis of the results obtained.
6. **Conclusion and presentation (4 points)**
   - Summarize the results obtained
   - Do critical thinking of the results obtained and evaluate.
   - Submit a technical report

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Carry out the high-pressure processing in food industry	**K2**
CO2	Explain in detail about all membrane technology so that it can be applied in processing industry.	**K2**
CO3	Process the food items through novel heating methods.	**K3**
CO4	Apply nanotechnology to food processing to produce novel foods.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	1	-	-	-	-	-	-	-
CO2	3	3	2	-	1	-	-	-	-	-	-	-
CO3	3	2	2	-	2	-	-	-	-	-	-	-
CO4	3	2	3	-	1	-	-	-	-	-	-	-

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food processing Handbook,	James G Brennan, Alistair S Grandison (Ed.),	Wiley – VCH	2011
2	Introduction to Food Process Engineering	P G Smith	Springer	2011
3	Food process engineering and technology	Zeki Berk	Elsevier	2013

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Transport Processes and Separation Process Principles	Geankoplis, C.J.	Prentice Hall	4th Edition, 2003
2	Coulson & Richardson's Chemical Engineering Vol.2 (Particle Technology & Separation Processes")	Richardson, J.E. et al	Butterworth – Heinemann / Elsevier,	5th Edition, 2003
3	Unit Operations in Chemical Engineering	McCabe W.L., Smith J.C.	McGraw – Hill Int.,	7th Edition,2001
4	Transport Processes and Separation Process Principles	Geankoplis, C.J.	Prentice Hall	4th Edition, 2003

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/a4aKLHCLyD8?list=PLbRMhDVUMngdSyw7OUHJJNYPfo_p4Ysln
2	https://youtu.be/ZGfNTI0fGsU?list=PLbRMhDVUMngdSyw7OUHJJNYPfo_p4Ysln
3	https://youtu.be/zwBVB5MICFA?list=PLbRMhDVUMngdSyw7OUHJJNYPfo_p4Ysln
4	https://youtu.be/jebsSUpUlqw?list=PLbRMhDVUMngdSyw7OUHJJNYPfo_p4Ysln

SEMESTER 5

FOOD PRESERVATION LAB

Course Code	**PCFTL507**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0-0:3-0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Lab

Course Objectives:

1. To understand basic preservation methods in food manufacturing
2. To evaluate quality parameters of processed food products

Details of Experiment

Expt. No	Experiment
1	Specifications suggested by fssai-specification of fruit beverages
2	Preservation of pine apple squash
3	Preservation by drying
4	Preparation of pine apple syrup
5	Preparation of RTS and analysis of yield, TSS and acidity
6	Preparation of mango bar and shelf life study
7	Preparation of nutmeg pickle
8	Preparation of sauce
9	Preparation of grape wine
10	Sulphating of vegetables
11	Preparation of synthetic syrup
12	Manufacture of pan bread
13	Preservation meat by curing and analysis of color and water holding capacity

14	Candling of egg and estimation of egg quality
15	Blanching of vegetable and checking the adequacy

Course Assessment Method

(CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Demonstrate the understanding about various methods for food preservation of fruits and vegetables	**K1**
CO2	Evaluate the raw material-product ratio . Analyse various FSSAI specification of food products	**K5**
CO3	Analyse various FSSAI specification of food products	**K4**
CO4	Identification of traditional products	**K1**
CO5	Understand water holding capacity of animal products	**K2**

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2		2		1	3			3
CO2	3	2		2		2		1	3			3
CO3	3	2		2		2		1	3			3
CO4	3	2		2		2		1	3			3

1: Slight (Low),2: Moderate (Medium),3: Substantial (High), : No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Trends in food chemistry, nutrition and technology in Indian sub-continent	Suni Mary Varghese , Salvatore Parisi , Rajeev K. Singla , A. S. Anitha Begu	Springer	2022
2	New methods of food preservation	G. W. Gould	Springer-science-business media	1995
3	Hand book of food preservation	Mohammed shafiur rahman	CRC press	2020

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food processing and preservation (food science and technology)	Neelam khetarpaul	Daya books	2005
2	Essential of food science	Vickie A. Vaclavik &Eliza beth W. Christian	Springer	2008

Video Links (NPTEL, SWAYAM…)	
Sl. No.	**Link ID**
1	https://www.youtube.com/watch?v=uq2wjuIIBxk
2	https://www.youtube.com/watch?v=7gquYRxLMFI https://www.youtube.com/watch?v=9wlJbsNBUgc https://www.youtube.com/watch?v=Ts8EuIZZf1A

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

FOOD PROCESSING LAB

Course Code	PCFTL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0-0-3-0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Lab

Course Objectives:

1. To study processing methods of different types of food
2. To learn the effect of different processing methods on the nutritional quality preservation of the foods

Details of Experiment

Expt. No	Experiment
1	Study on Freeze Drying
2	Study on Spray Drying
3	Blanching of Vegetables with and without acid pre-treatment
4	Dehydration study of vegetables
5	Stages of sugar cooking/ cookery
6	Preparation of fruit candy/Toffee
7	Preparation of Chocolate
8	Osmotic Dehydration of Amla/Gooseberry
9	Preparation of Jam, Jelly
10	Preparation of Paneer
11	Preparation of tomato puree/ ketchup
12	Preparation of sauerkraut
13	Preparation of squash
14	Preparation of Dahi/Yoghurt

15	Sensory Evaluation of food products by 9 point hedonic Scale

Course Assessment Method (CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Apply the principles and methods involved in the processing of different foods	K1
CO2	Analyse various preservation techniques of foods	K5
CO3	Develop skills to analyze food quality	K4
CO4	Formulate various nutritive products and work in team	K1
CO5	Evaluate different food products	K2

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2		2			3			3
CO2	3	2		2		2		1	3			3
CO3	3	2		2		2		1	3			3
CO4	3	2		2		2		1	3			3
CO5	2	2		2		2		1	3			3

1: Slight (Low),2: Moderate (Medium),3: Substantial (High), : No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Processing Technology and Practices	P.J Fellows, Woodhead Publishing	Woodhead Publishing	4th Edition, 2016
2	Food Science	Norman N Potter, Joseph H. Hotchkiss , Aspen Publishers Inc.,U.S.	Science Technology	5th Edition 1999
3	Food Facts and Principles:	Shakuntala Manay.	Publisher: new age	4th Edition,2021
4	Food Chemistry	H.D. Belitz, W. Grosch, P. Schieberle,	springer	3rd Edition, 2004

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food processing and preservation	Neelam Kedhar Paul, Daya Publications	Elsevier	2005
2	Food Science	B. Srilakshmi	New age publishers	8th edition, 2023
3	Textbook of food science and technology,	Avantina Sharma	CBS Publishers and distribution	3rd edition, 2019
4	Food Safety and Standards	Universal publishers	Universal publishers	2021

	Act, 2006			

Video Links (NPTEL, SWAYAM…)	
Sl. No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/cec19_ag05/preview
2	https://onlinecourses.swayam2.ac.in/cec19_ag01/preview

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

FOOD TECHNOLOGY

SEMESTER S6

DESIGN OF FOOD PROCESSING EQUIPMENT & PLANT LAYOUT

Course Code	**PCFTT601**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To acquire basic understanding of design parameters and knowledge of design procedures for commonly used food processing equipment.
2. To acquaint and equip the students with different unit operations of food industries and their design features

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction and Design of vessels:** Nature of equipment, general design procedure, equipment classification. Properties of materials -Types of stresses and strains - Poisson's ratio- Hooke's Law – stress strain diagram-factor of safety, Materials of construction, Corrosion Pressure vessel- design of cylindrical and spherical shell subject to internal and external pressure, Silo design- Rankine Theory-Airy equation-Janssen equation	**9**
2	**Design of heat exchangers, evaporators and dryers:** Design of heat exchangers –Double pipe heat exchangers, plate heat exchanger, shell and tube heat exchangers, design of evaporators-mass and energy balance-single effect and multiple effect evaporators, Design of dryers-mass and energy balance, Psychrometrics-drying time calculation.	**9**
3	**Design of supplemental processes:** Design of food extruders – power requirements, Design of cold storage – factors to be considered –estimation	**9**

	of cooling load – construction and operation. Design of Agitation Vessels- power requirements, Design of centrifugal separator- rate of separation, Design considerations of cleaners, Design considerations of Belt conveyor, bucket elevator, screw conveyor, chain conveyor, pneumatic conveyor.	
4	**Plant layout and Design:** Basic concepts of plant layout and design with special reference to food process industries, Plant location- factors- location theory and models, plant layout-characteristics- types, SLP and its application in development of plant layout, Development of food plant layout and design	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the basics concepts related to design of equipment.	**K2**
CO2	Determine the design parameters of heat exchangers, evaporators and dryers	**K3**
CO3	Estimate the design parameters of extruders, cold storage, agitation vessel and centrifugal separator	**K3**
CO4	Explain the basic concepts in development plant layout	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2								3	
CO2	3	3	3				2			1	3	2
CO3	3	3	3				1			1	3	1
CO4	3	3	1				2			1	3	2
CO5						3		3				

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Process Equipment Design	Joshi, M.V and V.V.Mahajani	New India Publishing Agency, New Delhi.	2004, (3 rd edition).
2	Introduction to Food Engineering	R. Paul Singh and Dennis R. Heldman	Elsevier, Amsterdam, Netherlands	2014, 5th Ed.
3	Plant Layout and Design	James M Moore	Mcmillan & Co.,	2009.
4	Unit Operations of Chemical Engineering,.	McCabe, w.l. Julian Smith, Peter Harriott	McGraw-Hill, Inc., NY, USA.	7th Ed 2004.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of Food Processing Equipment	George D. Saravacos, Athanasios E. Kostaropoulos,	Springer	2012, 1461352126, 9781461352129
2	Food Process Design,	Zacharias B. Maroulis, George D. Saravacos,	CRC Press	2003, ISBN: 0203912012, 9780203912010
3	Postharvest Technology and Food Process Engineering	Amalendu Chakraverty, R. Paul Singh	CRC Press	2016, ISBN: 1466553219, 9781466553217
4	Fish Handlingand Processing.	Borgess,G.H.O.,Cutting,C.L.,Lovern,J.A.andWaterman,U	Chemical Publishing Co	1967

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc21_ch18/preview
2	
3	
4	http://ecoursesonline.iasri.res.in/course/view.php?id=529

SEMESTER S6

MEAT, FISH AND POULTRY PROCESSING

Course Code	**PCFTT602**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. Provide an understanding of various chemical and instrumental methods for analyzing food.
2. Equip graduate students with detailed knowledge of modern techniques used in research, development, and inspection of food products in industry, analytical laboratories, and government agencies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction** Current level of production, consumption and export of meat products. Meat composition from different sources; muscle structure and compositions; Effect of feed, breed and management on production and quality, post-mortem muscle chemistry; meat colour and flavours; meat microbiology and safety	9
2	**Types of Meat** Beef, mutton, pork as human foods. Modern abattoirs, typical layout and features, Ante- mortem handling and design of handling facilities; stunning methods; steps in slaughtering and dressing; and inspection; inedible by-products; effects of processing on meat tenderization; abattoir equipment and utilities. **Processing technologies** Chilling and freezing of carcass and meat; canning, pickling, curing and smoking; prepared meat products like salami, kebabs, sausages;	

	intermediate moisture and dried meat products; radiation processing, canning of meat	
3	**Handling and Transportation** Care in Handling and transportation of fish at sea; Handling and transportation on land; Ship board refrigeration equipment's; Cold chain. design of refrigerated and insulated trucks; **Cured and smoked fish** Salt curing of fish; Storage of salted fish; Quality of finished products; Production of cold smoke fish; Hot smoked fish and light salted and light smoked fish; smoked fish using curing liquid	**9**
4	**Poultry processing** Introduction, Reception of birds, antemortem examination of poultry, steps involved in poultry processing, post mortem examination, chilling, grading, cutting, packaging. Meat yield	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Categorize various equipment involved in production, processing and acceptance of meat and poultry.	**K2**
CO2	Investigate the processed products and by-product utilization of meat and poultry.	**K2**
CO3	Evaluate about the care in handling and transportation of fish at sea and on land	**K3**
CO4	Distinguish the quality of cured and smoked fish in preservation	**K3**
CO5	Understand the various processes involved in poultry processing.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			1							
CO2	2					2						
CO3	3					3		3				
CO4	3					3		3				
CO5	3					3		3				

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Meat Science	Lawrie,R.A	PergamonPress, Oxford,UK	SecondEdition, 1975
2	Egg Science and Technology	Stadelmen,W.J.andCotterill,O.J.,	AVI, Westport	SecondEdition, 1977
3	The complete technology book on Meat, Poultry and Fish Processing.	NPCS Board of consultants &Engineers	NIIR consultancy	Second edition, 2013

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Meat Processing	JosephKerry,JohnKerryandDavidLedwood	Woodhead Publishing Limited, England, CRC press	2002
2	Poultry Meat Processing and Quality	Mead,G	Woodhead Publishing Limited, England, CRC press	2004
3	Techno-Chemical Control of fish Processing Industry	. Gerasimov,G.V. andAntonova,MT.	Amerind Publishing Co. Pvt. Ltd	1979
4	Fish Handling and Processing.	Borgess,G.H.O.,Cutting, C.L.,Lovern,J.A.andWaterman,U	Chemical Publishing Co	1967

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://shorturl.at/veaNK
2	https://shorturl.at/u7lyt
3	https://shorturl.at/MkzFy
4	https://shorturl.at/IyCIf

SEMESTER S6

NON-THERMAL PROCESSING

Course Code	**PEFTT631**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To analyze the principles and methods of non-thermal processing technologies and their application in food processing industries.
2. A thorough understanding of the specific novel processing technologies for different food items

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Emerging technologies in food processing – necessity and advantages of non-thermal technologies. High Pressure Processing – Principle, equipment and process of HPP treatment, mechanism of microbial inactivation, application of HPP in food processing. Concept of hurdle technology.	**9**
2	Pulsed Electric field processing - Principle, process, equipment, principle of microbial inactivation, application in food processing. Ultrasonic processing – principle of operation, effect on microorganisms, applications. Ozone processing and cold plasma.	**9**
3	Radiation processing of foods – food irradiation, microwave processing, Infrared heating, radiofrequency heating – principle, working and microbial inactivation. Pulsed light technology and Oscillating magnetic field in preservation – principle and working.	**9**
4	Food extrusion technology – theory, equipment and its functioning, Types of extruders, Effect of extrusion on foods. Membrane technologies in food processing – Overview of Membrane technology. Microfiltration, Ultrafiltration (UF), Nanofiltration (NF) and Reverse Osmosis (RO) and their industrial applications.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain different emerging technologies in food industry	**K1**
CO2	Discuss the fundamentals of HPP processing and its applications	**K2**
CO3	Describe the concepts behind PEF, ultrasound, ozone and cold plasma technologies	**K3**
CO4	Explore food preservation techniques using magnetic fields and radiation of different forms	**K3**
CO5	Familiarise with the concepts of extrusion and membrane technology	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1				2						2
CO2	3					2						2
CO3	3					2						2
CO4	3					2						2
CO5	3					2						2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food processing Handbook	James G Brennan, Alistair S Grandison	Wiley	2011
2	Introduction to Food Process Engineering	P G Smith	Springer	2011
3	Food process engineering and technology	Zeki Berk	Elsevier	2013

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Transport Processes and Separation Process Principles	C J Geankoplis	Prentice Hall	4th Edition, 2003
2	Unit Operations in Chemical Engineering	McCabe W.L., Smith J.C	McGraw – Hill Int	2001
3	Food Processing Technology	Fellows, P	Ellis Horwood International Publishers, Cambridge	1988

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/126105015
2	https://nptel.ac.in/courses/126105011
3	https://onlinecourses.nptel.ac.in/noc24_ch60/preview
4	https://nptel.ac.in/courses/126103017

SEMESTER S6

BEVERAGE PROCESSING

Course Code	**PEFTT632**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To develop the knowledge of students on various concepts, procedures and manufacturing of different beverages.
2. To understand the quality control and quality assurance systems in beverage processing industry

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Basic Ingredients in Beverages:** Beverage definition-ingredients-water, CO2, Bulk and Intense Sweeteners, Water Miscible and Water dispersible flavouring agents, Colours-natural and artificial, micro and nano emulsions of flavours and colours in beverages, Preservatives, Emulsifiers and Stabilizers, FSSAI standards for beverages, quality control analysis for beverages	**9**
2	**Beer and Wine Manufacture:** Ingredients-Malt, Hops-adjuncts-water, yeast. Beer manufacturing process-malting, preparation of sweet wort, brewing, fermentation, pasteurization and packaging. Beer defects and spoilage. Wine fermentation-types-red and white. Wine defects and spoilage	**9**
3	**Manufacture of Distilled Fermented Beverages:** Malt and Grain, Whisky-types, raw materials and processing. Design-wash still operation-spirit still operation-continuous distillation-design and operation of grain whisky stills-maturation and blending- co products. Whiskies of the world and their regulations	**9**

4	**Carbonated and Non-carbonated Beverages:** Procedures - Carbonation equipment - ingredients, preparation of syrups - filling systems - packaging - containers and closures. Coffee bean preparation-processing-brewing-decaffeination-Instant Coffee-Tea-types-Black, Green and oolong. Dairy based beverages. Fruit juices, Nectar, Squash, RTS beverages and Isotonic Beverages. Flash Pasteurization, Canning and Aseptic packaging of beverages	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To study the basic ingredients of beverages and classification	**K3**
CO2	To study the process techniques and equipment used for the beer and wine manufacture	**K3**
CO3	To study the process techniques and equipment used for the manufacture of distilled fermented beverages	**K3**
CO4	To study the process techniques and equipment used for carbonated beverages	**K3**
CO5	To study the quality parameters of beverages	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3										
CO2	3	3	2	2								
CO3	3	3	2	2								
CO4	3	3	2	2								
CO5	2	2	3									

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Chemistry and Technology of Soft Drinks and Fruit Juices.	Ashurst,P.R	Blackwell Publishing, 2005.	2nd Edition, 2005
2	Carbonated soft drinks-Formulation and Manufacture	Steen,D.P and Ashurst,P.R,	Blackwell Publishing, 2000.	Blackwell Publishing, 2000.
3	Food-Facts and Principles	Shahkunthala Manay,V and Shadakdharaswamy,M,	", New Age International Pvt. Ltd, 3	3Revised Edition,2000.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of post-harvest technology.	Amalendu Chakraverty et al,	Ed: Marcel Dekker Inc, (Special Indian Edition), 2000.	Special Indian Edition2000.
2	Microbiology and Technology of fermented Foods.	Robert W Hukins,	IFT Press, Blackwell Publishing Ltd	2006.
3	Food, Fermentation and Microorganisms	Charles,W, Bamforth	Blackwell Science Publishing Ltd	2005
4	” Whisky-Technology, production and marketing”,	Inge Russel, Graham Stewart and Charlie Bamforth,	Elsevier	2003

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1 **2** **3** **4**	https://nptel.ac.in/courses/126105023 https://onlinecourses.swayam2.ac.in/cec19_ag01/preview https://elearn.nptel.ac.in/shop/nptel/dairy-and-food-process-and-products-technology/?v=c86ee0d9d7ed

SEMESTER S6

FOOD STORAGE ENGINEERING

Course Code	PEFTT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To provide students with a fundamental understanding of traditional and modern storage structures
2. To acquire knowledge on the importance of safe and scientific storage structures and different factors affecting storage.
3. To design storage structures and cold stores based on the available product information.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Food and grain storage -** Scope, importance, basic requirements, safe and scientific storage, pre and post-storage operations; post-harvest physiology of semi-perishables and perishables, climacteric and non-climacteric fruits, respiration, ripening, changes during ripening. Direct damages, indirect damages, causes of spoilage in storage (moisture, temperature, humidity, respiration loss, heat of respiration, sprouting), destructive agents (rodents, birds, insects, etc.), sources of infestation and control.	9
2	**Traditional and small-scale storage methods -**Traditional storage structures, bag and bulk storage - mud bin, drums, gunny bags etc - method of stacking. Small-scale storage structures, brick, concrete types. Local storage structures, Morai, Bukhari, Kothar, Kuthla, improved storage structures Pusa Bin, Brick and Cement Bin, Bunker storage structure, Cover and plinth storage structure, Factors affecting storage, Temperature and Moisture migration in storage structures.	9

3	**Modern and large-scale storage -** Characteristics of Bulk materials, Bulk storage, Silos; types- deep, shallow. Physical factors influencing flow characteristics, mechanics of bulk solids, flow through hoppers, openings, and ducts - Bulk density, True density, Apparent density, Angle of repose, Angle of rupture, Plane of rupture, Angle of repose, Internal and external friction, Pressure distribution in silos, Rankine's earth pressure coefficient, Airys, Janssens equations, numerical. Warehouses - Selection of site for storage types, specific volume of bagged grains and importance, Percentage utilization of warehouse-numerical. Economic aspects of storage.	**9**
4	**Storage of perishables -**Cold storage – construction and operation, design of cold storage; controlled and modified atmospheric storage, membrane storage, hypobaric storage, evaporative cooling storage,control of temperature and relative humidity inside storage. Storage conditions for raw and processed fruits, vegetables, meat, dairy etc. Factors affecting storage life. Material handling equipment - belt, chain, screw, roller, pneumatic conveyors and bucket elevators	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the requirements of safe and scientific storage of grains and the operations involved	**K2**
CO2	Study the types and the methods for traditional and small-scale safe storage of food grains	**K2**
CO3	Explain the modern storage structures and grain flow characteristics	**K2**
CO4	Analyse and design silos using different pressure distribution theories	**K3**
CO5	Explore the storage techniques for different food commodities and their handling	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3					2						2
CO2	3					1						1
CO3	3					1						1
CO4	3	2	2			2						1
CO5	3	1	1			2						1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Drying and Storage of Cereal grains	Bala,B.K	Oxford and IBH Publishing Co.	1998
2	Preservation and Storage of Grains, Seeds and their By-products	Multon, J.L	CBS	1989
3	Unit Operations of agricultural Processing	K.M. Sahay & K.K Singh	Vikas Publishing House Pvt Ltd	2015
4	Grain Handling and Storage	G. Boumans	Elsevier	1985
5	Food Processing Handbook	James G. Brennan & Alistair S. Grandison eds	Wiley-VCH	2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Post-Harvest Food Grain Storage	Himangshu Barman	Agrobios	2008
2	Handling and Storage of Food Grains in Tropical and Sub-tropical Areas	Hall, C.W.	FAO Publ. Oxford and IBH	1970
3	Agricultural Process Engineering	Henderson, S. and Perry, S.M	AVI Publ	1976
4	Post Harvest Handling – A System Approach	Shefelt, R.L. and Prussi, S.E	Academic Press	1992
5	Grain Storage Engineering and Technology	Vijayaraghavan, S	Batra Book Service	1993

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://elearn.daffodilvarsity.edu.bd/course/view.php?id=14376
2	https://nptel.ac.in/courses/127106237
3	https://nptel.ac.in/courses/113105104
4	

SEMESTER S6

FOOD TOXICOLOGY

Course Code	**PEFTT634**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To develop an understanding of the nature and properties of toxic substances in foods and their implications for health

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to food toxicology:** Definition, scope, Principles of toxicology, General mechanisms by which toxic chemicals causes disease in humans - Absorption, distribution, metabolism and excretion of toxic chemicals, Classification of food toxicants, factors affecting toxicity of compounds.	**9**
2	**Natural and Derived Food toxicants:** Toxicants and allergens derived from plants, animals, marine algae, mushrooms etc. Microbial toxins –Mycotoxins Impacting Food Production and Manufacturing-, Guidance and regulations on mycotoxins in food, Toxicants generated during food processing and packaging such as nitrosamines, acrylamide, benzene, dioxins, furans etc., persistent organic pollutants, food carcinogen and mutagens and their mode of action, Transgenic approaches, Bioterrorism.	**9**
3	**Food allergens:** Types, Chemistry of food allergens, Food allergy and hypersensitivity, mechanisms of allergic-type reactions and allergenic components in foods. Food disorders associated with metabolism, lactose intolerance, and asthma, handle of food allergies.	**9**

4	**Determination of toxins and risk assessment:** Quantitative and qualitative analysis of toxicants in foods; Biological determination of toxicants, Assessment of food safety – Risk assessment and risk benefit indices of human exposure, acute toxicity, mutagencity and carcinogenicity, reproductive and developmental oxicity, neurotoxicity and behavioural effect, immunotoxicity. Laws and regulation of safety assessment of foods	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Acquire an idea about scope, basic principles of food toxicology, classifications and the factor affecting toxicity	**K3**
CO2	Familiarize about natural and derived food toxicants and their mode of action	**K3**
CO3	Develop knowledge about Food allergens, toxicity of additives and difference between safety, hazard and toxicity	**K3**
CO4	Explain the methods for the Determination of toxins and their risk assessment	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3										
CO2	3	3	2	2								
CO3	3	3	2	2								
CO4	3	3	2	2								

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Toxicology	Helferich, William and Carl K.Winter	CRC Press	2001
2	Introduction to Food Toxicology	Shibamoto, Taka yuki and Leonard F.Bjeldanzes	Academic Press	2nd Edition. 2009

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Toxicology	Debasis Bagchi, Anand Swaroop	CRC Press	2017
2	Food Forensics and Toxicology	Titus A. M. Msagati	John Wiley & Sons Ltd	2017
3	Introduction to Food Toxicology	Takayuki Shibamoto, Leonard F. Bjeldanes	Elsevier	2009

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/126105023
2	https://onlinecourses.swayam2.ac.in/cec19_ag01/preview
3	https://elearn.nptel.ac.in/shop/nptel/dairy-and-food-process-and-products-technology/?v=c86ee0d9d7ed

SEMESTER S6

FOOD SUPPLY CHAIN MANAGEMENT

Course Code	PEFTT636	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To understand the fundamentals on logistic and supply chain management in food industries.
2. To study the various aspects of food supply chain management globally, and the pervasiveness of Information Technology in SCM.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Logistics and supply chain management**: Scope, Significance and Drivers; Basic Model – Primary and Secondary Activities; Role and Challenges of Logistics and supply chain management in food industry.	9
2	**Demand and supply management**: Forecasting techniques, Strategic planning for material sourcing, Outsourcing strategies, Warehouse strategies, Inventory models and control techniques.	9
3	**Introduction to Refrigerated Cold Chain**: Importance and challenges in postharvest management- Major technologies in maintaining cold chain. Factors influencing refrigerated cold chain. Storage requirements of fruits and vegetables, milk and milk products, meat and meat products and marine products during transit.	9
4	**Applications of Packaging in logistics**: Types of packaging and packaging materials, Export & import packaging and labelling details, Containerization, Pervasiveness of IT in Supply Chain Management – ERP, Bar-coding, RFID, GPS, E-Procurement**.** Export and import procedure and Documentation, Risk management in global logistics, Customer relationship management in LSCM Module.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand logistics and supply chain management	**K1**
CO2	Know about the procurement and warehousing strategies in SCM	**K2**
CO3	Gain knowledge on use of application of refrigeration in food supply chain	**K1**
CO4	Understand the importance of information technology in packaging and supply	**K1**
CO5	Know about the relevance and concept of documentation in supply chain management	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				2		1					
CO2	3					2	2					
CO3	3		3				3	1				
CO4	3	2								1		1
CO5	3							2	1		2	

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Logistics and supply chain management	D K Agarwal	Macmillan Publishers India Ltd., Eighth Impressions	2010
2	Total Supply Chain Management	Ron Basu. J. Nevan Wright	Elsevier, Ist Ed	2008
3	Supply Chain Management Theories and Practices	R. P. Mohanty, S.G. Deshmukh	DreamtechPress	2005

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The Complete book on Cold Storage, Cold Chain and Warehouse (with controlled Atmosphere storage and rural Godowns)	Ajay Kumar Gupta	NIIR project consultancy services	2022

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	www. archive.nptel.ac.in
2	www.nptel.ac.in

SEMESTER S6

ADVANCED EXTRUSION TECHNOLOGY

Course Code	**PEFTT637**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To impart knowledge on the working and principle of extruders in food technology.
2. To formulate, process and understand quality parameters of varied combinations in food extruder.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Food Extrusion**: Definition, introduction to extruders, principles and types, Uses of extruders in the food industry, Pre-conditioning of raw materials used in extrusion process, Extruder Selection, Design, and Operation for Different Food Applications.	9
2	**Single screw extruder:** Principle of working, Net Flow, Operations, manufacturing of pasta and vermicelli. Twin screw extruder: Counter rotating and co-rotating twin screw extruder. Rheological Properties of Materials During the Extrusion Process, Advantages of Twin Screw Extruder.	9
3	**Extrusion Technology:** Classification of Breakfast cereals - Raw materials, process and quality testing for Ready to eat breakfast cereals. Texturized vegetable protein: Definition, Manufacturing process and quality parameters of TVP	9
4	**Effect of extrusion on food products:** Chemical and nutritional changes in food during extrusion, factors affecting extrusion, Net Flow, Packaging materials for extruded product. Recent Advances in extrusion technology: Carbon dioxide or Nitrogen assisted extrusion technology	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic fundamentals, design of equipment and processing of different extruded products.	**KI**
CO2	Explain the types of extruders and its impact on extrusion process, rheological behaviour and product quality.	**K2**
CO3	Discuss about the future trends and aspects in food extrusion	**K3**
CO4	Perceive on the chemical and nutritional changes occurring in extrusion process and its technological advancements	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2		2	2	1					
CO2	3		2		2	2	1					
CO3	3				2	2	1					
CO4	3				2	2	1					

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Extruded foods	SMatza	Springer	2000
2	Textbook of Technology of Extrusion Cooking	N DFrame	4^{th} Ed.Springer Science and BusinessMedia	2012
3	Extrusion of Foods	Judson M. Harper	1^{st} Ed.CRCPress	2019

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Extruders in Food Application	Riaz M.N.	CRC Press	2000
2	Advances in Food Extrusion Technology	Maskan and Altan	CRC Press	2012

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	www.archive.nptel.ac.in
2	www.onlinecourses.nptel.ac.in

SEMESTER S6

PROCESS INSTRUMENTATION AND CONTROL

Course Code	**PEFTT638**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To provide basic understanding of designing and implementing practical control strategies in process industries.
2. To acquire knowledge on instrumentation performance characteristics
3. To understand measuring principle of different temperature and pressure measuring instruments

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to process dynamics and control:** – definition of terms- Laplace transform – transform of simple functions- derivatives and integrals- properties of Laplace transforms - final value theorem -initial value theorem- transition of transforms and functions- examples - inversion by partial fraction – solution of differential equation- qualitative nature of solutions- linear loop open system – first order systems -mercury thermometer- liquid level and mixing processes – response of these to different types of forcing functions- systems in series – interacting and non-interacting types	**9**
2	**Linear open loop systems**: – second order systems – mercury thermometer in well and manometer – impulse and step response of under damped, critically damped and over-damped system, their derivation – closed loop system – servo and regulator problems – block diagram development – controllers- types, basic principles and transfer functions -PID, PI and PD	**9**
3	**Introduction to stability of linear system**: - Routh Hurwitz criterion for stability – root locus technique- plotting the root locus diagram –	**9**

	transportation lag and its effect on root locus diagram- Introduction to frequency response – Bode diagram for first order system – bode stability criterion , gain margin and phase margin	
4	**Instrumentation:** – definition, performance characteristics of an instrument – static and dynamic characteristics- temperature measurement – three types of thermometers -liquid filled, vapour pressure and gas type – bimetallic thermometer – resistance thermometer – principles of thermoelectricity like Seebeck effect, Peltier effect, Thomson effect - radiation methods like pyrometry, optical pyrometry, thermistor. Pressure measurement – manometer of Utube, well type and inclined type – bourdon tube, flat and corrugated diaphragm, capsules.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain open loop systems	**K2**
CO2	Analyse and apply the knowledge of linear closed loop system	**K4**
CO3	Acquire working knowledge of control system by frequency response	**K3**
CO4	Explain the performance characteristics of instruments	**K1**
CO5	Summarize different temperature and pressure measuring instruments	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2										
CO2	2			2								
CO3	2		2									
CO4	2											
CO5	2											

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Process Systems Analysis and Control	Ronald R.Coughnowr	Mc Graw Hill	1991
2	Industrial instrumentation	Eckman DP	Wiley Eastern	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Chemical Process Control, An introduction to theory and practice	Stephanopoulose G	Prentice Hall	
2	Mechanical and industrial Measurements	Jain RK	Khanna	
3	Process control	Harriot P	Tata Mc Graw Hill	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/103101142
2	https://nptel.ac.in/courses/103106148
3	https://nptel.ac.in/courses/126105020
4	https://nptel.ac.in/courses/103105064

SEMESTER S6

ADVANCES IN FOOD PACKAGING

Course Code	**PEFTT635**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. Able to understand with the different packaging materials and packaging systems used in food industry.
2. Understand the properties of different packaging material and its manufacturing processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Food packaging:** need of food packaging, Role of packaging in extending shelf life of foods packaging materials and various package forms, Food packaging systems, Product characteristics and package requirements.	9
2	**Different forms of packaging:** Rigid, semi-rigid, flexible forms of packaging, wooden boxes and crates, Paper and paperboards, Glass, Metal container, Types of cans- Tinplate containers, Tin free steel (TFS), Aluminium containers, Composite containers, Lacquers. Properties, advantages and limitations of the following packaging materials: Glass, aluminium, its foil, metal tin containers, Paper and paperboard.	9
3	**Advances in Packaging Technology:** Introduction, Active packaging, Vacuum Packaging, Controlled atmospheric packaging, Modified atmosphere packaging, Aseptic packaging, Biodegradable plastics, Edible gums, Coatings. Gas packaging, seal and shrink packaging. Form & fill sealing, Aseptic packaging systems, Retort pouches, RFID indicators.	9
4	**Packaging systems for- Fruit & vegetable products:** Dehydrated foods, Frozen foods, fish and fish products, meat and meat products, Dairy	9

	products, carbonated beverage, tea, coffee, alcoholic beverages, confectionery- fat and oil, biscuits, cakes, bread, food grains, storage and handling packaging materials. Physical properties and barrier properties of packaging materials.	

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	*Internal Ex*	*Evaluate*	*Analyse*	*Total*
5	**15**	**10**	**10**	**40**

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. Visit to relevant industries
2. Students can collect data with the latest trends in packaging consulting the web sites and magazines
3. Identification of different types of packaging and packaging materials
4. Determination of tensile strength of given material.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the different uses of packaging materials its manufacturing process and properties	**K1**
CO2	Understand the principle and working set up of different packaging systems	**K1**
CO3	Evaluate the physical and barrier properties of different packaging materials	**K2**
CO4	Implement different packaging materials for food products	**K3**
CO5	Identify the criteria to design packaging material based on type of foods	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1									
CO2	3	2	1	2								
CO3	3	2	2	1	1							
CO4	1	1	2	3	1	1						
CO5	3	1	3	2		1						

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Packaging: Principles and Practice	Robertson,G.L	Taylor & Francis, 2nd Ed.	2006
2	Principles of Food Packaging	S. Sacharow and R.C. Griffin	AVI Pub. Co.,	1988
3	Handbook of Food Packaging Technology	Frank Albert Paine, Heather Y. Paine	Blackie Academic &Professional	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Innovations in Food Packaging	Han, Jung H.	Elsevier	2005
2	Novel Food Packaging Techniques	Ahvenainen, Raija	Wood Head Publishing	2003

Video Links (NPTEL, SWAYAM…)	
Module No.	Link ID
1	www.nptel.edu
2	www.onlinecourses.nptel.ac.in
3	www.nptel.edu
4	www.nptelhrd

SEMESTER S6

FOOD PACKAGING TECHNOLOGY

Course Code	**PBFTT604**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	**ESE Marks**	40
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course objectives:

1. Understand the importance and role of packaging in the food industry.
2. Explore various types of food packaging materials and their properties.
3. Learn about packaging requirements for different food categories

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Packaging Scope & Importance**: Packaging of foods, requirement, importance and scope, frame work of packaging strategy, environmental consideration. Factors affecting shelf life of food material during storage, Interactions of spoilage agents with environmental factors as water, oxygen, light, pH, etc. and general principles of control of the spoilage agents; Difference between food infection, food intoxication and allergy	9
2	**Packaging Systems And Packaging Materials**: 15Packaging systems, types: flexible and rigid; retail and bulk; levels of packaging; special solutions and packaging machines, technical packaging systems and data management packaging systems, Different types of packaging materials, their key properties and applications, Metal cans, manufacture of two piece and three-piece cans, Plastic packaging, different types of polymers used in food packaging and their barrier properties. manufacture of plastic packaging materials, profile extrusion, blown film/ sheet extrusion, blow molding, extrusion blow molding, injection blow molding, stretch blow molding, injection molding	9

3	**Glass containers:** types of glass used in food packaging, manufacture of glass and glass containers, closures for glass containers. Paper and paper board packaging, paper and paper board manufacture process, modification of barrier properties and characteristics of paper/ boards. Relative advantages and disadvantages of different packaging materials; effect of these materials on packed commodities, Nutritional labelling on packages, CAP and MAP, shrink and cling packaging, vacuum and gas packaging; Active packaging, Smart packaging, Packaging requirement for raw and processed foods, and their selection of packaging materials, Factors affecting the choice of packaging materials, Disposal and recycle of packaging waste, Printing and labelling, Lamination	**9**
4	**Packaging from starches and other plant and animal sources**: Biodegradable packaging systems, Non-Migratory Biopolymers, Testing methods for flexible materials, rigid materials and semi rigid materials; Tests for paper (thickness, bursting strength, breaking length, stiffness, tear resistance, folding endurance, ply bond test, surface oil absorption test, etc.), plastic film and laminates (thickness, tensile strength, gloss, haze, burning test to identify polymer, etc.), aluminum foil (thickness, pin holes, etc.), glass containers (visual defects, colour, dimensions, impact strength, etc.), metal containers (pressure test, product compatibility, etc.).	**9**

Suggestion on Project Topics

1. **Select a Focus Area:**
    - Choose sustainable packaging materials and technologies as the focus area for your project. Explore materials like biodegradable polymers, active packaging solutions, or advanced recycling techniques.
2. **Literature Review:**
    - Conduct a comprehensive literature review on sustainable packaging materials and technologies.
    - Gather information on theoretical frameworks, principles of sustainable packaging, regulatory standards (e.g., FDA guidelines), and case studies of successful implementations.

3. **Define Objectives:**
    - Clearly define the objectives of your project, such as evaluating the effectiveness of biodegradable packaging materials in extending shelf life or comparing the environmental impact of different packaging materials.
4. **Plan Experimental Design:**
    - Design experiments or simulations to achieve your objectives.
    - Specify factors like packaging material selection (e.g., biopolymers, compostable materials), processing conditions (e.g., manufacturing techniques like extrusion or injection molding), and performance metrics (e.g., barrier properties, biodegradability).
5. **Data Collection:**
    - Conduct experiments or gather data using appropriate methods (e.g., barrier tests, degradation studies, mechanical properties testing).
    - Document data from trials with different packaging materials and processing techniques.
6. **Data Analysis:**
    - Organize collected data systematically.
    - Perform statistical analysis or use appropriate analytical methods (e.g., comparison of mechanical properties, degradation rates) to interpret your data.
    - Present data analysis results using tables, graphs, or charts to illustrate trends and relationships.
7. **Compare and Contrast Techniques:**
    - Compare different sustainable packaging techniques within your focus area.
    - Analyze factors such as effectiveness in preserving food quality, environmental impact, and economic feasibility.
8. **Evaluate Results:**
    - Evaluate the results of your experiments or simulations.
    - Discuss how well your findings align with theoretical frameworks and existing practices in sustainable food packaging.
    - Propose recommendations for improvements or modifications based on your findings.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. • Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the basics and importance of food packaging	**K2**
CO2	Interpret the food qualities preserved by using different packaging technologies	**K2**
CO3	Apply the knowledge to design and manufacture food packages	**K6**
CO4	Apply the knowledge for testing different packaging materials	**K5**
CO5	Develop sustainable food packaging solutions	**K6**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3							3	2		3	3
CO2	3							3	2		3	3
CO3	3							3	2		3	3
CO4	3							3	2		3	3
CO5	3						3	3	2		3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Packaging Technology	Walter Soroka	Institute of Packaging Professionals	4th Edition, 2019
2	Food Packaging: Principles and Practice	Gordon L. Robertson	CRC Press	3rd Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Packaging Technology: Fundamentals, Materials and Processes	Anne Emblem, Henry Emblem	Woodhead Publishing	1st Edition, 2012
2	Innovations in Food Packaging	Jung H. Han	Academic Press	2nd Edition, 2014
3	Handbook of Food Packaging Technology	Martin Yam	Wiley-Blackwell	2nd Edition, 2011

Video Links (NPTEL, SWAYAM…)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_ge32/preview
2	
3	
4	

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. **Involvement in the Project Work and Team Work (3 Marks)**

- Active participation and individual contribution
- Teamwork and collaboration

4. **Execution and Implementation (10 Marks)**

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. **Final Presentation (5 Marks)**

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. **Project Quality, Innovation, and Creativity (3 Marks)**

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S6

UNIT OPERATIONS IN FOOD PROCESSING

Course Code	**OEFTT611**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To acquaint and equip the students with different unit operations of food industries and their design features

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Size reduction**: Benefits, classification, determination and designation of the fineness of ground material, sieve/screen analysis, principle and mechanisms of comminution of food, Rittinger's, Kick's and Bond's equations, work index, energy utilization; Size reduction equipment: Principal types, crushers (jaw crushers, gyratory, smooth roll), hammer mills and impactors, attrition mills, tumbling mills, ultra-fine grinders, fluid jet pulverizer, cutting machines (slicing, dicing, shredding).	9
2	**Membrane separation**: General considerations, materials for membrane construction, ultra-filtration, processing variables, membrane fouling, applications of ultra-filtration in food processing, reverse osmosis, mode of operation, and applications; Membrane separation methods, demineralization by electro-dialysis, gel filtration, ion exchange, per-evaporation and micro filtration	9
3	**Food freezing**: Introduction, freezing point curve for food and water, freezing points of common food materials, Principles of food freezing, freezing time calculation by using Plank's equation; Freezing systems;	9

	Direct contact systems, air blast immersion; Changes in foods; Frozen food properties; freezing time, factors influencing freezing time, freezing/thawing time; Freeze concentration: Principles, process, methods; Frozen food storage: Quality changes in foods during frozen storage; Freeze drying: Heat mass transfer during freeze drying, equipment and practice.	
4	**Extraction** – solid – liquid extraction (leaching) – material balance – equilibrium – multistage extraction – stage efficiency – solid – liquid extraction systems – Super Critical Fluid extraction – principles – solvents – Super critical extraction systems – application – Liquid – liquid extraction – principle – application	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Characterize different unit operations in food processing	**K1**
CO2	Identify and evaluate size reduction mechanism and mixing of fluids	**K3**
CO3	Explain the working of filtration/membrane separation equipments used in food industry	**K3**
CO4	Explain the mechanisms of freezing and dehydration operation	**K3**
CO5	Analyse different types of evaporators and extraction systems	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	2			1							
CO3	3	1	1		1							
CO4	3	2	1		1							
CO5	3	2	1		1							

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Food Engineering	R. Paul Singh and Dennis R. Heldman	Elsevier, Amsterdam, Netherlands	2014, 5th Ed.
2	Unit Operations in Food Processing	Earle, R.L.	Pergamon Press. Oxford. U.K.	2003
3	Food Process Engineering and Technology.	Zeki Berk.	Academic Press. 360 Park Avenue South, New York, USA.	2009.
4	Unit Operations of Chemical Engineering,.	McCabe, w.l. Julian Smith, Peter Harriott	McGraw-Hill, Inc., NY, USA.	7th Ed 2004.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Unit Operations in Food Engineering.	Albert Ibarz and Gustavo V. Barbosa-Canovas.	CRC Press, Boca Raton, FL, USA	2003.
2	Fundamentals Of Food Process Engineering,	Toledo R T,	Springer, ISBN: 9783319900971	Fourth Edition (2018)
3	Food Process Engineering Operations,	Saravacos G D,	Taylor and Francis, ISBN: 9781420083538	2011
4	Food Processing Technology Principles And Practice	Fellows P J,	Elsevier, ISBN 9789351073918	Fourth Edition (2020)

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc20_ag01/preview
2	https://archive.nptel.ac.in/courses/103/107/103107127/#
3	https://onlinecourses.nptel.ac.in/noc20_ag03
4	https://archive.nptel.ac.in/courses/103/103/103103155/

SEMESTER S6

REFRIGERATION & AIR CONDITIONING IN FOOD INDUSTRY

Course Code	OEFTT612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To the study the application of various air conditioning and refrigeration systems in food industry
2. To understand psychrometric terms and freezing processes and their application in food related industry and operations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Refrigeration:** Thermodynamics & Principles, reversed Carnot cycle, heat pump, unit of refrigerating capacity, coefficient of performance; Production of low temperatures, refrigerants- classification, nomenclature and properties (physical, chemical, safety, thermodynamic and economical); types of refrigeration systems and their components.	**9**
2	**Ice manufacture**: principles and systems of ice production, Treatment of water for making ice, brines, freezing tanks, ice cans, air agitation, quality of ice, Cold storage design for different categories of food resource (size and shape, construction and material, insulation, vapour barriers, floors, frost-heave, interior finish and fitting.)	**9**
3	**Air-conditioning:** Psychrometry principles, factors affecting comfort air-conditioning, classification, sensible heat factor, industrial air-conditioning Winter/summer/year-round air-conditioning, unitary air-conditioning systems, central air-conditioning, Cooling load calculations.	**9**
4	**Applications of Refrigeration and Air-conditioning in different food products:** fruits, vegetables, meat, fish and poultry, dairy products, Handling	**9**

	and distribution of food through Refrigerated and Air-conditioned transport, Food freezing, Frozen food storage.	

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Familiarize concepts of refrigeration and types, components and properties of refrigerants	**K2**
CO2	Apply the refrigeration for production of ice and ice- related cold process in food industry	**K3**
CO3	Identify the various components of Air Conditioning and compare different types of Air conditioning systems	**K3**
CO4	Analyze the various application of Refrigeration and Air-conditioning in different foods and food related operations	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									
CO2	3	2	3	1								
CO3	3	2	3	1								
CO4	2	1	2									

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Refrigeration and Air Conditioning	Arora, and Chandra	Prentice Hall India Learning Private Limited	(2010), ISBN-13: 978-8120339156.
2	Refrigeration and Air conditioning	D.K. Chavan and G.K. Pathak	Standard Book House	1st Edition (2016), ISBN-13: 978-8189401528.
3	Refrigeration and Air Conditioning	C.P Arora	McGraw Hill Education	3rd edition,(2017), ISBN-13: 978-9351340164

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Textbook of Refrigeration and Air-conditioning	R .S Khurmi and J.K. Gupta	S Chand Publications	2006, ISBN-13: 978-8121927819
2	Refrigeration and Air-conditioning	Ahmadul Ameen	PHI Learning Pvt. Ltd.	2006, ISBN: 8120326717, 9788120326712
3	Postharvest Technology and Food Process Engineering	Amalendu Chakraverty, R. Paul Singh	CRC Press	2016, ISBN: 1466553219, 9781466553217

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/107/112107208/
2	https://archive.nptel.ac.in/courses/112/107/112107208/
3	https://archive.nptel.ac.in/courses/112/107/112107208/
4	https://archive.nptel.ac.in/courses/112/107/112107208/

SEMESTER S6

INSTRUMENTAL METHODS IN FOOD ANALYSIS

Course Code	**OEFTT613**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)		**Course Type**	Theory

Course Objectives:

1. Provide an understanding of various chemical and instrumental methods for analyzing food.
2. Equip graduate students with detailed knowledge of modern techniques used in research, development, and inspection of food products in industry, analytical laboratories, and government agencies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Spectroscopy:** Food Analysis, Nature and Concept of Food analysis, Spectroscopy, Electromagnetic spectrum, Beer Lamberts law, Absorbance, Construction of calibration curve, Principle, instrumentation of single and double beam UV-Visible spectrophotometer, IR, NMR and Raman Spectroscopy **Microscopy:** Scanning electron microscopy (SEM) – Principle Instrumentation, applications, Transmission electron microscopy (TEM)- Principle, Instrumentation, applications	**9**
2	**Chromatography:** Basis of chromatography (mobile and stationary phases, Chromatography: Theory & Principle, chromatographic parameter (partition coefficient, capacity factor, retention & dead time, Resolution& their calculation), Types of chromatographic techniques, HPLC, GC – Instrumentation and application	**9**
3	**Hyphenated technique:** Mass Spectroscopy and Chromatography coupling technique - Basics of mass spectrometry, components of mass spectrometer,	**9**

	applications of mass spectroscopy, Gas chromatography, GC MS, LCMS – instrumentation and applications	
4	**Biological Techniques:** DNA and protein-based methods, Fundamental principle and instrumentation of the system, measurement techniques and interpretation of Polymerase Chain Reaction, Real Time Polymerase Chain Reaction, Enzyme Linked Immunosorbent Assay, Radio Immuno Assay, Use of PCR for detection of genetically modified organisms, Electrophoresis techniques. **Innovative Analytical Techniques:** Biosensor, nanotechnology and biosensor – applications in food industry, Instrumental measurement of texture of food, viscometer, viscoanalysis, texture analysis	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of spectroscopy principle, instrumentation and applications	**K2**
CO2	Identify chromatographic techniques and the application of these techniques to the separation and analysis of multi-component samples	**K3**
CO3	Summarize different hyphenated techniques of instrumental methods in food analysis	**K2**
CO4	Knowledge on application of different biological methods in food analysis	**K3**
CO5	Describe the application of innovative instrumental techniques for food analysis	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3										
CO2	3	2	2									
CO3	2	3		3								
CO4	2	2	3									
CO5	2	2		2								

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Analysis	Nielsen, Suzanne (Ed.)	Springer US	2010
2	Food Analysis	HPLC. Leo M.L. Nollet (Editor)	Second Edition, CRC Press	2000

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of Food Analysis Instruments	Semih Otles,	CRC Press	2016
2	Spectroscopic Methods in Food Analysis	Adriana S. Franca and Leo M.L. Nollet	CRC Press	2018
3	Modern Methods Of Food Analysis	. Kent K. Stewart and John R. Whitaker	AVI PUBLISHING COMPANY, INC,	
4	Chromatography: Concepts and Contrasts.	James M. Miller	second edition, Wiley Interscience	2004
5	Applications of Vibrational Spectroscopy in Food Science	Eunice Li-Chan (Editor), John Chalmers and Peter Griffiths (Co-Editors).	2 Volume Set. John Wiley & Sons Ltd. Chichester, England.	2010
6	Infrared and Raman Spectroscopy of Biological Materials (Practical Spectroscopy)	HansUlrich Gremlich and Bing Yan (Eds). Marcel Dekker	NY, USA	2001
7	Handbook of Food Analysis Instruments	Semih Otles,	CRC Press	2016
8	Spectroscopic Methods in Food Analysis	Adriana S. Franca and Leo M.L. Nollet	CRC Press	2018

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/m33OeLsp8o0 , https://youtu.be/hvPZ3d8YVjs
2	https://youtu.be/PMq02umihQk
3	http://nptel.iitm.ac.in/
4	https://nptel.ac.in/courses/102103044

SEMESTER S6

POST HARVEST HANDLING OF FOOD MATERIALS

Course Code	**OEFTT614**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. Provide an understanding of various handling techniques and storage methods.
2. Equip graduate students with detailed knowledge of traditional and modern storage techniques, changes occur during storage.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Basic post- harvest physiology, definition, respiration and gas exchange, Effect of respiration on physiology, factors affecting respiration, Respiration control methods, Changes during post- harvest, physical and chemical changes, transpiration, water stress	**9**
2	**Factors affecting post-harvest physiology and mechanical damages**: Preharvest factors on postharvest life, harvesting and handling injuries, storage conditions; temperature, RH, composition and its modification, Ethylene, Role of ethylene in ripening, Climacteric and non-climacteric fruits, ethylene biosynthesis, ethylene generators, ethylene control methods. Mechanical damage, chilling injury, freezing injury, quality and safety factors	**11**
3	**Handling and storage:** Maturity indices, Assessment of crop maturity, Changes during ripening, physical and chemical changes, change in colour, texture, flavour during storage,	**11**

	Harvesting methods, Storage conditions and types, Storage characteristics of different fruits and vegetables, MAP and CAP**:** Biotic and abiotic factors; temperature, insect infestation, microbes; fungi, virus, bacteria etc	
4	**Quality Improvement techniques:** Harvesting and handling techniques, coatings and treatments, insect control and microbial control, Post-harvest treatments**,** GAP, GHP, GMP, HACCP, measurement of product quality methods; destructive and non-destructive tests; physical chemical, biological, visual methods	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the physiological and biochemical changes in horticultural produce after harvest and apply suitable control methods	**K2**
CO2	Use suitable ripening techniques for the development of safe and quality ripened products Acquire insight on various factors involved in spoilage and the techniques for its control	**K3**
CO3	Identify various storage methods for the shelf-life extension of horticultural produce	**K2**
CO4	Acquire knowledge about different factors which affect the post-harvest life of fruits and vegetables.	**K3**
CO5	Apply the knowledge of quality control for the quality improvement of fruits and vegetables	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3											
CO2	3		3					3				
CO3	3											
CO4	3											
CO5	3											

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Post-Harvest physiology and pathology of vegetables		Marcel Dekker Inc. NY	2 nd edition,
2	Post-Harvest Technology of Horticulture crops	Sudheer K P and Indira V	New India Publishing Agency	-

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fruit and Vegetables Harvesting, Handling and Storage	A.K Thomson	Blackwell Publishing	**-**
2	Post Harvest Technology	Dhruba Raj Bhattarai, Prof.Durga Mani Gautam	Heritage Publishers & Distributors Pvt. Ltd	2020

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/-NyDCWuAGfk
2	https://youtu.be/XNVOmQ-50xM
3	https://shorturl.at/AOPFg
4	https://shorturl.at/ur2mz

SEMESTER S6

DAIRY TECHNOLOGY

Course Code	**OEFTT615**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of milk types, the various factors affecting these compositions, as well as the knowledge of milk's properties and sterilization techniques
2. To provide students with an in-depth understanding of centrifugation and membrane separation techniques
3. To equip students with the knowledge and skills necessary for the manufacture of various dairy products

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Properties, Reception and Storage of Milk:** Milk-Types-Composition-factors affecting composition of milk. Physical-Chemical and Thermal Properties- Reception and storage-cooling of milk - Different types of coolers and cooling systems. Cleaning-basic principles-can washing - can washers-cleaning-in centralised and decentralized CIP system - cleaning of various equipment, corrosion control.	**9**
2	**Pasteurization and Sterilization of milk:** Pasteurization - principles, objectives and methods - LTLT/holding pasteurization-types, advantages and disadvantages - HTST pasteurization-components and function of HTST pasteurizer, advantages and disadvantages. Sterilization - In-bottle sterilization, UHT processing – vacreation - Indirect heating systems using plate heat exchangers, Direct heating – steam injection and infusion	**9**

3	**Homogenization:** Homogenization theory, effect of homogenization of milk, Homogenizer components - valves. Pumps - functions and efficiency of process-operation and maintenance. Types of homogenizers-stages of homogenization-importance **Centrifugation and membrane separation:** Centrifugation-clarification-clarifiers and separators-separation efficiency-factors affecting fat percentage in cream-fat loss in skim milk. Construction of separator components- bactofuge treatment. Ultra filtration - Reverse osmosis process - Electro dialysis.	9
4	**Manufacture of dairy products:** Butter manufacture – methods - cheese manufacture-methods. Yogurt, paneer. Skimmed milk powder Drum dryer-spray dryer-construction, powder recovery systems agglomeration. Ice-cream manufacture-overrun-types of freezers.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	To understand the different types of milk and their compositions, the factors affecting these compositions and effectively apply principles of cleaning.	**K2**
CO2	To comprehend different pasteurization and sterilization techniques and selecting appropriate methods for ensuring the safety of dairy products	**K3**
CO3	To implement various processes to improve the quality and safety of dairy products.	**K4**
CO4	To evaluate and apply the methods for manufacturing different dairy products	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	2	2	2									
CO3	2	2										
CO4	3	2	2		2	2						

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Dairy Plant Engineering and Management	Tufail Ahmed	CBS Publishers and Distributors, New Delhi	2001, ISBN-13: 978-8122501186
2	Dairy Science and Technology	Pieter Walstra, Jan T M Wouters, Tom J Geurts	Taylor Francis Group, CRC press	2006. ISBN-13: 978-0824727635
3	Outlines Of Dairy Technology	De Sukumar	Oxford University Press	2015, ISBN-13: 978-0195611946
4	Food Engineering and Dairy Technology	Kessler, H.G.	Publishing House Verlog A. Kessler, Germany	1st ed. 1981

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Milk and Dairy Product Technology	Spreer, Edgar	Marcel Dekker	2005. ISBN-13: 978-0824700942
2	Engineering Aspects of Milk and Dairy Products	Selia, Jane dos Reis Coimbra and Jose A. Teixeir	Jane Selia dos Reis Coimbra & Jose A. Teixeir, CRC Press	2009, ISBN13: 978-1420090222
3	Handbook of Milk Processing Dairy Products and Packaging Technology	Eiri Board	Engineers India Research Institute	2008. ISBN-13: 978-8186732960
4	Dairy Technology: Vol.01 Milk and Milk Processing	Shivashraya Singh	New India Publishing Agency	2014. ISBN-13: 978-9383305094
5	Novel Dairy Processing Technologies: Techniques, Management, and Energy Conservation (Innovations in Agricultural & Biological Engineering)",	Megh R. Goyal, Anit Kumar, Anil K. Gupta	Apple Academic Press	2018. ISBN-13: 978-1771886123

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/126/105/126105013/
2	https://youtu.be/PAJ5VPG46Nw

SEMESTER S6

THERMAL PROCESSING OF FOOD

Course Code	**OEFTT616**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	40
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. Explain the fundamental concepts of thermal processing.
2. Identify and describe different thermal processing methods.
3. Analyse the effects of thermal processing on food quality and microbial safety.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to thermal processing:** Overview of food preservation methods, food properties pertaining to thermal operations: thermal, electrical, electromagnetic properties of food, Importance of thermal processing in food and dairy industry, basic principles of heat transfer, fundamentals of heat transfer in food, thermal properties of food, Shirkange and expansion.	9
2	**Thermal processing methods and equipment's:** Microbial death kinetics, D- value, z-value and F-value concepts, interpretation of thermal death times in curves. Thermal processing methods: Pasteurisation- Principles and objectives of pasteurization, types of pasteurization processes (e.g., HTST, LTLT), Equipment used for pasteurization, Effects on food quality and nutrition. Sterilization-Principles and objectives of sterilization, thermal destruction of microorganisms, types of sterilization processes (e.g., UHT, aseptic processing), equipment used for sterilization, effects on food quality and nutrition. Blanching- Principles and objectives of blanching, Blanching equipment and	9

	methods, Effects of blanching on food quality and enzyme inactivation	
3	**Drying** - Mechanisms and methods of drying, Types of dryers: tray, drum, spray, and freeze dryers, Moisture content and drying kinetics. Freeze Drying- Principles and stages of freeze drying, Equipment and applications Quality retention and rehydration characteristics. Freezing - Principles of freezing and thawing, Freezing rate and its impact on food quality, Types of freezers: blast, plate, and cryogenic freezers Advanced Thermal Processing Techniques and its principle, application and impact on food quality and safety- Microwave and Radio Frequency Heating, Ohmic Heating, High-Pressure Processing (HPP), Novel Thermal Processing Techniques- Infrared heating, Pulsed electric field (PEF).	**9**
4	**Quality and Safety of Thermally Processed Foods -** Impact of thermal processing on food quality attributes (colour, texture, flavour), Nutritional changes due to thermal processing, Safety considerations and regulations. Packaging for Thermal Processing - Packaging materials and their properties, Role of packaging in thermal processing, Compatibility of packaging materials with thermal processes.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand the principles and importance of thermal processing in the food industry.	**K2**
CO2	Gain knowledge about various thermal processing techniques such as pasteurization, sterilization, and blanching.	**K3**
CO3	To learn about different methods of thermal processing and their impact on food quality and safety.	**K4**
CO4	Study the effects of thermal processing on food quality, safety, and nutritional value and packing.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3					2	1					
CO2	3		2		1							
CO3	2		2				2					
CO4	3					2		2				

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Introduction to Food Engineering"	R. Paul Singh	Academic Press	5nd Edition, 2013
2	"Thermal Food Processing: New Technologies and Quality Issues"	Da-wen Sun	CRC Press	1st Edition, 2005
3	"Food Processing Technology: Principles and Practice"	P J Fellows	Woodhead Publishing	4th Edition, 2016
4	"Fundamentals of Food Process Engineering"	Romeo T. Toledo	Springer-Verlag New York Inc.	2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Food process engineering and technology"	Zeik Brek	Academic Press	5th edition 2013
2	"Phytochemical aspects of food engineering and processing"	Sakamon Devahadtin	CRC Press	2011
3	"Food process engineering and technology"	Zeik Brek	Academic Press	5th edition 2013

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	Thermal Processing of Foods - Course (nptel.ac.in)
2	NOC | Thermal Processing of Foods (nptel.ac.in)
3	https://www.digimat.in/nptel/courses/video/126103017/L01.html
4	Thermal Operations in Food Process Engineering: Theory and Applications - Course (nptel.ac.in)

SEMESTER S6

ENGINEERING PROPERTIES OF FOOD MATERIALS

Course Code:	**OEFTT617**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To acquire knowledge about properties of food material that are important in processing
2. To study the measurement methods of engineering properties and analysis of food materials

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction** to engineering properties of biological materials, classification of food properties, food quality and safety **Physical properties**: Shape and size – criteria for describing shape and size Volume and methods of measurement of volume, Density, Types, Method of measurement of apparent density, material density, particle density, bulk density, true density, Surface area measurement for fruits and vegetables, egg surface area, pore surface area.	**9**
2	**Frictional properties**: Laws of friction, effect of load and properties of contacting bodies. Effect of sliding velocity and contact surface temperature, effect of water film and surface roughness. Rolling resistance, angle of repose, angle of internal friction, pressure distribution in storage structures and compression chambers	**9**
3	**Thermal properties**: Thermal conductivity, measurement of thermal conductivity, Specific heat, measurement of specific heat, enthalpy, thermal diffusivity, boiling point, freezing point, dielectric properties **Aerodynamic properties:** drag coefficient, terminal velocity, measurement of terminal velocity; application the properties in the harvesting, post-harvest	**9**

	handling, processing, storage and how it affects consumer's perception of food quality and safety.	
4	**Texture & optical properties:** Subjective measurements – physiological aspects, psychological aspects, mechanical aspects. Imitative measurements- equipment's used, texture profile analysis, dynamic test for evaluation of food texture, mechanical test applicable to food materials – firmness and hardness, mechanics of hardness, dynamic hardness, effect of age, water content and temperature on texture of foods; colour, measuring equipment, L*a*b system.	**8**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the significance and evaluation of physical properties of food materials	**K2**
CO2	Acquire knowledge of frictional properties of food materials	**K2**
CO3	Recognize thermal, aerodynamic properties food and their measurement techniques.	**K2**
CO4	Explain methods for the measurement of texture and colour of food material	**K2**
CO5	Apply the knowledge of various properties to improve the qualitative and quantitative attributes of food.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										1
CO2	3	2	2									1
CO3	3	2	3									1
CO4	3	2										1
CO5	3	3										1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	, Physical properties of plants animal materials	Mohensenin N N	, Gordon and Breach publishers, New York,	1980
2	Engineering properties of foods	Rao M A , Rizvi S S H,Azim K Datta and Jasim Ahmed,	4th Ed., CRC Press	, 2014
3	Unit operations of Agricultural Processing, Second edition,	K M Sahay and K K Singh,	Vikas Publishing India Pvt Ltd,	2004
4	Physical Properties of Foods Novel Measurement Techniques	Ignacio Arana	Applications, 1st edition, CRC Press,	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	, “Engineering Properties of Biological Materials”. .	Singhal,O.P. and Samuel,D.V.K	Saroj Prakasan, Allahabad	2003
2	“Physical properties of foods”	Peleg, M.and BagelayE.B.,	AVI publishing Co. USA	1983

Video Links (NPTEL, SWAYAM…)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_ag01/preview
2	http://ecoursesonline.iasri.res.in/mod/page/view.php?id=95508

SEMESTER S6

MODELLING AND SIMULATION OF FOOD PROCESSES LAB

Course Code	**PCFTL607**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Lab

Course Objectives:

1. To provide an overview of the possibilities of process simulation as a tool for computer systems analysis, which minimizes risks and costs in experimentation.
2. To familiarise students with the techniques of modelling of engineering processes and of the developed model optimization

Details of Experiment

Expt. No	Experiment
1	Simulation of extraction of multicomponent using DWSIM software.
2	Simulation of distillation column using DWSIM software
3	Simulation of heat exchanger using DWSIM software
4	Simulation of CSTR using DWSIM software
5	Simulation of plug flow reactor using DWSIM software
6	Simulation of refrigeration gas plant using DWSIM software
7	Simulation of Component Separator using DWSIM software
8	Simulation of Solids Separator using DWSIM software
9	Simulation of Continuous Cake Filter using DWSIM software
10	Regression and curve-fitting of data from reaction and enzyme kinetics using MATLAB software
11	Simulation of Mixer using DWSIM software
12	Simulation of rotary dryer using MATLAB-SIMULINK software
13	Numerical analysis of Newton Raphson method using MATLAB software
14	Bisection Methods for Solving a Nonlinear Equation in MATLAB software
15	Gaussian Elimination Method for Solving Simultaneous Linear Equations in MATLAB

Course Assessment Method

(CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	To describe food engineering processes in mathematical form and create simulation models of various types	K1
CO2	To apply simulation techniques to solve complex system issues and to select feasible, if not optimum, solutions and configurations amongst competing designs	K3
CO3	Use modern computing tools necessary for analysis of the experimental data	K3
CO4	Practice ethical approaches in experimental investigation, collection and reporting of data and adhering to the safety ethics set by the laboratory.	K2
CO5	Communicate through oral and writing skills through viva and preparing reports of experimental work.	K3

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3			2	2			2
CO2	3	3	3	3	3			2	2			2
CO3	3	3	3	3	3			2	2			2
CO4	3	3	3	3	3			2	2			2
CO5	3	3	3	3	3			2	2	2		2

1: Slight (Low),2: Moderate (Medium),3: Substantial (High), : No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Process Modelling, Simulation and Control for Chemical Engineers	W.L. Luyben	McGraw Hill Book Co., New York	2nd Edn., 1990.
2	Process Modelling and Model Analysis	K. M. Hangos and I. T. Cameron	Academic Press	2001
3	Process Modelling and Simulation	César de Prada, Constantinos Pantelides, José Luis Pitarch	Mdpi AG	2019

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals and Modelling of Separation Processes	Holland C. D	Prentice Hall	1975
2	Process Modelling and Control in Chemical Engineering	Najim K	CRC Press	1990
3	Mathematical Modelling, Vol. 1:A Chemical Engineering Perspective (Process System Engineering)"	Aris R	Academic Press	1999

Video Links (NPTEL, SWAYAM…)	
Sl. No.	**Link ID**
1	https://archive.nptel.ac.in/courses/103/107/103107096/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**
    - Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
    - Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
    - Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
    - Creativity and logic in algorithm or experimental design.
2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**
    - Setup and Execution: Proper setup and accurate execution of the experiment or programming task.
3. **Result with Valid Inference/Quality of Output (10 Marks)**
    - Accuracy of Results: Precision and correctness of the obtained results.
    - Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.
4. **Viva Voce (10 Marks)**
    - Ability to explain the experiment, procedure results and answer related questions
    - Proficiency in answering questions related to theoretical and practical aspects of the subject.
5. **Record (5 Marks)**
    - Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

FOOD TECHNOLOGY

SEMESTER S7

FOOD INFORMATICS

Course Code	PEFTT741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To define and explain the scope, importance, and applications of food informatics in various sectors of the food industry.
2. To apply fundamental data analytics techniques, including descriptive, predictive, and prescriptive methods, using industry-specific tools and software.
3. To analyze the role and implementation of information systems in the food industry, focusing on ERP, supply chain management, and their impact on food safety and quality.
4. To understand and apply food safety management systems, standards, and regulations, emphasizing microorganism detection and the use of informatics in compliance and certification processes.
5. To explore and evaluate the application of emerging technologies such as block chain, IoT, sensor technologies, and AI in enhancing food traceability, quality, safety, and personalized nutrition.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Food Informatics- definition, scope, importance and application of various food industry, Data Sources in Food Informatics- Sources of data in food production, processing, and distribution, challenges in data collection and integration, errors in data representation, Information Systems in Food Industry, Enterprise Resource Planning, Supply chain management, Role of information systems in food safety and quality assurance.	9

2	Fundamentals of Data Analytics, different data analysis techniques(descriptive, predictive, prescriptive), Tools and software for data analysis in the food industry, Big data in food industries, role and its application in food production and supply chain management, Challenges and ethical considerations in data analytics for food systems. Data representation errors and detection and troubleshooting procedures.	**9**
3	Food Safety and Traceability, Food Safety Management Systems, food safety standards and regulations, Microorganism detection and analysis: Emerging Foodborne Pathogens, Investigation of food borne illnesses outbreaks, Pathogen Detection Methods, Role of informatics in compliance and certification processes, Traceability in Food Supply Chains, Using informatics to identify and mitigate risks in food production and distribution.	**9**
4	Emerging technologies (block chain, IoT) enabling traceability in food supply chains, cloud computing, Sensor Technologies, Applications of IoT and sensors in monitoring food quality and safety, Artificial Intelligence in Food Informatics: food processing, packaging, and distribution, AI-driven decision support systems, Emerging trends such as precision agriculture and personalized nutrition, Merits and demerits of incorporation of AI in Food industries.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Demonstrate comprehensive understanding and applications of food informatics across various sectors of the food industry	**K2**
CO2	Analyze different sources of data in food production and supply chains, identify challenges in data collection and integration	**K3**
CO3	Applying descriptive, predictive, and prescriptive analytics techniques to analyze food industry data using relevant software tools, with an emphasis on addressing data representation errors and troubleshooting procedures.	**K3**
CO4	Evaluate the role of informatics in implementing and managing Food Safety Management Systems (FSMS)	**K5**
CO5	Explore the emerging technologies and their implications in food industries	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2				2							3
CO2	2	3			3							3
CO3	3	3	2	2	3							3
CO4	3	2			3	2						3
CO5	2	2	2	3	3	3						3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Supply Chain Management and Logistics: From Farm to Fork	Samir Dani	Kogan Page	1st edition, 3 June 2015
2	Fundamental Food Microbiology	B. Ray.	CRC Press 2005.	3rd edition, 2005
3	Operations and Supply Chain Management Essentials You Always Wanted To Know (Self Learning management Series)	Ashley McDonough	Vibrant Publishers	2020
4	Perspectives of Artificial Intelligence in Agriculture	Dr. Harvinder Kaur Sidhu	Gyan Geeta Prakashan	2023
5	Big Data and Business Analytics	Jay Liebowitz	Auerbach Publications;	1st edition, 23 April 2013

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Management Information System	C. Laudon Kenneth and P. Laudon Jane	Peasrson Education	2018

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/106/104/106104189/
2	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
3	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
4	https://onlinecourses.nptel.ac.in/noc22_cs44/preview

SEMESTER S7

FOOD LAWS AND REGULATIONS

Course Code	**PEFTT742**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To provide a general overview of food laws and regulations
2. To assist students in finding sources of information pertaining to food laws and regulations

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Concept of food safety, food security and nutrition security, Indian food regulatory regime (existing & old), FSSAI, PFA Act and Rules, food licensing and registration system, AGMARK, Legal metrology act 2009, Export (Quality Control and Inspection) Amendment Act, 1984, Voluntary Standards and Certification, BIS, food licensing and registration system, food import clearance system, essential commodity act, 1995.	**9**
2	Food hazard contamination-biological (bacteria, viruses and parasites), chemical (toxic constituents / hazardous materials) pesticides residues / environmental pollution / chemicals) and physical factors, Risk assessment studies: risk management, risk characterisation and communication, concept and implementation of HAACP in food premises, use of permitted additives like colours, preservatives, emulsifiers, stabilisers, antioxidants, etc. and their regulations.	**9**
3	Regulations of nutrition labelling, nutrient level claims, and health claims, Regulations of standard of identity and quality, Introduction to the Food, Drug, and Cosmetic Act, Introduction to OIE & IPPC, Other International	**9**

	Food Standards (e.g. European Commission, USFDA etc), Introduction to WTO Agreements: SPS and TBT Agreement, Export & Import Laws and Regulations, WHO/FAO Expert Bodies (JECFA/JEMRA/JMPR)	
4	Voluntary quality standards and certifications-GMP, GHP, GAP, Good Animal Husbandry Practices, ISO 9000, ISO 22000, 1SO 14000, ISO 17025, PAS 22000, FSSC 22000, BRC, Halal & Kosher standards, concept of six sigma, TQC, NABL, NABH, NBQP	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concept of food laws and regulations and major regulatory bodies in India	**K2**
CO2	Apply the concept of food contamination, risk assessment and HACCP application in food	**K3**
CO3	Understand regulations related to labelling and various international food safety standards and agreements	**K2**
CO4	Analyse sanitary regulations and ISO specifications	**K4**
CO5	Explain the basic concept of food laws and regulations and major regulatory bodies in India	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	2	3				1
CO2	3					3	3	3				1
CO3	3					2	2	3				1
CO4	3					3	2	1				1
CO5	3					1	2	1				1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of indices of food quality and authenticity	Singal R S	Woodhead publications, Cambridge, UK	1st edition, 1997
2	Food safety and standard act, 2006, Rules ang regulations, Volume 1, 2 & 3	Rajan Nijhawan	ILBCO	25th edition, 2024
3	Principles and practices of safe processing of foods	Shapton DA	Butterworth Publication	1st edition, 2013
4	Techniques of food analysis	Winton AL	Allied Science Publications, New Delhi	1999

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The chemical analysis of food and food products	Jacob MB	CBS Publications, New Delhi	1999
2	Food Safety Handbooks	Food Safety Handbooks	Wiley Interscience, UK	2005
3	Food Safety Regulations, Concerns and Trade: The Developing Country Perspective	Mehta, Rajesh and J. George	Macmillan	2005
4	Guide to US Food Laws and Regulations	Patricia A. Curtis	Wiley Online Library Online Books	2nd edition, 2022

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
2	https://onlinecourses.swayam2.ac.in/cec22_ag01/preview
3	https://onlinecourses.swayam2.ac.in/nou22_ag11/preview
4	https://onlinecourses.nptel.ac.in/noc24_lw03/preview

SEMESTER S7

NANOTECHNOLOGY IN FOOD

Course Code	PEFTT743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the applications of nanotechnology in food industry
2. Design and tailoring of advanced nanotechnological methods to facilitate emerging methods for sustainable food security

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Module 1: Introduction to Nanotechnology in Food Introduction to Nanotechnology – Definitions – Historical development – Scope and applications in the food industry. Nanomaterials – Types – Properties – Synthesis methods. Applications of Nanotechnology in Food – Food packaging – Food safety – Food processing – Nutrient delivery systems.	9
2	Module 2: Nanomaterials in Food Nanomaterials – Characteristics – Functional properties – Mechanisms of action. Types of Nanomaterials used in Food – Organic nanomaterials – Inorganic nanomaterials – Composite nanomaterials. Safety and Toxicity of Nanomaterials – Regulatory aspects – Risk assessment – Toxicological studies.	9
3	Module 3: Nanotechnology in Food Processing and Preservation Nanotechnology in Food Processing – Enhancement of food texture – Flavor improvement – Stability and shelf-life extension. Nanotechnology in Food Preservation – Antimicrobial properties – Controlled release of preservatives	9

	– Nano-encapsulation techniques. Nanotechnology in Food Packaging – Intelligent packaging – Active packaging – Barrier properties enhancement	
4	Module 4: Future Trends and Innovations in Food Nanotechnology Emerging Trends in Food Nanotechnology – Nano-biosensors – Nano-coatings – Smart delivery systems. Innovations in Food Nanotechnology – Recent advancements – Case studies – Future prospects. Ethical and Environmental Considerations – Public perception – Ethical issues – Environmental impact.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of optimization, including various types and formulations of optimization problems.	**K2**
CO2	Apply classical optimization techniques to both single and multi-variable problems, utilizing methods such as the simplex method and Lagrange multipliers.	**K3**
CO3	Implement numerical methods and algorithms for solving complex optimization problems, including gradient descent and interior point methods.	**K4**
CO4	Utilize advanced optimization techniques like dynamic programming, genetic algorithms, simulated annealing, and particle swarm optimization for real-world applications.	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3									1
CO2	3	2										1
CO3	3	2	1									1
CO4	3	2	1									1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Nanotechnology in Food Industry: Advances in Processing, Preservation, and Packaging	V. K. Joshi	Wiley	1st Edition, 2020
2	Nanotechnology Applications in Food: Flavor, Stability, Nutrition, and Safety	Alexandru Mihai Grumezescu	Academic Press	1st Edition, 2017
3	Food Nanotechnology: Principles and Applications	Seid Mahdi Jafari	Springer	1st Edition, 2018
4	Nanofood Engineering: Processing, Applications, and Potential Health and Safety Implications	Umesh Hebbar	CRC Press	1st Edition, 2020

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of Food Nanotechnology: Applications and Approaches	Seid Mahdi Jafari	Academic Press	1st Edition, 2020
2	Nanotechnology in Edible Food Packaging	Jorge Barros-Velazquez	Springer	Edition, 2019
3	Nanoengineering in the Beverage Industry	Alexandru Mihai Grumezescu	Academic Press	1st Edition, 2020
4	Nanomaterials for Food Applications	Nitin Mantri, Anjali Karpe, Swati Ravindra	Wiley	1st Edition, 2021

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_bt25/preview
2	https://archive.nptel.ac.in/courses/118/107/118107015/
3	https://archive.nptel.ac.in/courses/102/104/102104069/
4	https://archive.nptel.ac.in/courses/126/105/126105015/

SEMESTER S7

FOOD INDUSTRY WASTE MANAGEMENT

Course Code	**PEFTT744**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Understand the extent of wastes produced in different food industries
2. Identify the various by-products produced in food industries and their utilization
3. Analyze the strategies adopted for disposal of solid and liquid wastes
4. Distinguish various wastewater treatment methods in food industries

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Sources, classification and characterization of food industry waste from different industries - fruit and vegetable processing industry, beverage processing industry, sugar industry, dairy industry, meat and fish processing industry and sugar industry.	**9**
2	Characterization and utilization of by-products from cereals, pulses, oilseeds, fruits, vegetables, plantation, dairy, eggs, meat, fish and poultry processing industries. Elements of importance in efficient management of wastes from aforesaid food industries. Standards for emission or discharge of environmental pollutants from food processing industries. Characterization of food industries effluents, in terms of parameters of importance.	**9**

3	Physical, chemical and biological treatment strategies for solid and liquid wastes. Biological composting, Biogas and Vermicomposting. Unit concept of treatment of food industry effluents: Screening, sedimentation, floatation as per and primary treatments, biological oxidations: objectives, organisms, reactions, oxygen requirements, aeration devices.	**9**
4	Effect on characteristic parameters of effluents in treatments using lagoons, trickling filters, activated sludge process, oxidation ditches, UASB, rotating biological contractors and theirs variations and advanced modifications. Advanced wastewater treatment systems: physical, physicochemical and chemical treatments. Coagulation and flocculation, disinfection, handling and disposal of sludge and treated effluents conforming to EPA provisions.	**9**

Course Assessment Method

(CIE: 40 Marks, ESE: 60 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24Marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 Marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate comprehensive understanding of wastes produced by different food industries.	**K1, K2**
CO2	Analyze the various by-products produced in food industries and their utilization.	**K3,K4**
CO3	Applying various strategies for disposal of solid and liquid wastes.	**K2,K3**
CO4	Evaluate the different strategies adopted for effluent treatment	**K4, K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2				2							3
CO2	2	2			2	2						3
CO3	2	2			2							3
CO4	3	2			3	2						3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Water and Wastewater Technology	Hammer. M. J	Prentice Hall India Learning Private Limited	7th edition, 2012
2	Managing Food Industry Waste	Robert R Zall	Blackwell Publishing	2004
3	Drying and Valorization of Food Processing Waste	Chong C H	CRC Publishers	2024
4	The Complete Book on Managing Food Processing Industry Waste	H. Panda	Asia Pacific Business Press	2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Current Developments in Biotechnology and Bioengineering	Sunita Varjani, Ashok Pandey, Edgard Gnansounou, Samir Kumar Khanal and Sindhu Raveendran	Elsevier	1st edition, 2020
2	Wastewater Treatment and Reuse in the Food Industry	Marcella Barbera and Giovanni Gurnari	Springer	1st edition, 2018

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	http://acl.digimat.in/nptel/courses/video/126105023/L52.html
2	http://acl.digimat.in/nptel/courses/video/127101014/L11.html
3	https://archive.nptel.ac.in/courses/126/105/126105014/
4	https://archive.nptel.ac.in/courses/105/105/105105178/

SEMESTER S7

EMERGING TECHNIQUES IN FOOD QUALITY & SAFETY

Course Code	**PEFTT746**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To explore the area of emerging technologies involved in food quality as well as food safety
2. To understand the development of new approaches to food quality evaluation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Criteria for quality control: Principles of food safety Historical developments - need for quality control and safety- strategy and criteria for food safety - microbiological criteria for safety and quality- Ways of describing food quality: Composition, appearance, kinesthetic and flavour attributes Nutritional quality of foods and its assessment (content and quality of nutrients). Definition- Aspects of quality - Quality control tools. Methods of food quality evaluation - Reference and standard methods for chemical and microbiological analysis, Comparison of newer and rapid methods	9
2	The role of food quality and safety evaluation. Sensory quality and its evaluation, instrumental measurement of sensory attributes such as colour, viscosity, texture, specific gravity. Rheological and textural characteristics, Use of rheological properties in food science and engineering. Texture profile analysis. Sensory evaluation using artificial intelligence, Application of Computer Vision Systems for Objective Assessment of Food Qualities- Evaluation of Color- Evaluation of Texture, Evaluation of shape and size. Nanotechnology in Food Quality and Safety Evaluation System	9
3	NIR Spectroscopy for Chemical Composition and Internal Quality in Foods-	9

	Principle of the Technique- Device and Apparatus Applications. Electronic Nose for Detection of Food Flavor and Volatile Components. Biosensors for Evaluating Food Quality and Safety	
4	Quality Measurements Using Nuclear Magnetic Resonance and Magnetic Resonance Imaging- Principles of Magnetic Resonance Equipment and applications. Ultrasound Systems for Food Quality Evaluation. Hyperspectral and Multispectral Imaging Technique for Food Quality and Safety Inspection	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the historical developments and strategies in food safety and quality control.	**K2**
CO2	Evaluate the nutritional quality of foods through the assessment of nutrient content and quality.	**K2**
CO3	Apply sensory evaluation techniques and instrumental measurements to assess food quality attributes like texture, color, and flavour.	**K3**
CO4	Analyse the role of advanced technologies such as NIR spectroscopy, electronic noses, and biosensors in food quality and safety evaluation.	**K2**
CO5	Explain the integration of artificial intelligence and computer vision systems to objectively evaluate food qualities such as color, texture, and size.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		2										3
CO2		2										3
CO3	3	2										3
CO4	3	2			3							3
CO5	3	2			3							3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Emerging Technologies for Food Quality and Food Safety Evaluation	Yong Jin Cho	Taylor and Francis group	1st Edition, 2011
2	Quality control for the food industry	A Krammer & BA Twigg	Westport	3rd Edition, 1970

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Principles of Sensory Evaluation of Food	Amerine, Pangborn & Rosslos	Academic Press	1st Edition, 1965
2	Guide to Quality Management Systems for Food Industries	R Early	Backie Academic	1995
3	Sensory Evaluation of Food - Theory and Practice	Jellinek G.	Ellis Horwood	1985
4	Quality control in Food Industry	A Krammer & BA Twigg	AVI Publ.	3rd Edition, 1973

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
2	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
3	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
4	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview

SEMESTER S7

FOOD PLANT SANITATION & HYGEINE

Course Code	PEFTT747	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Food Plant Layout	**Course Type**	Theory

Course Objectives:

1. To Gain knowledge of the fundamental principles underlying food safety and sanitation practices in food processing plants.
2. To Understand the importance of maintaining high standards of sanitation to prevent foodborne illnesses and ensure consumer safety.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Food Plant Sanitation Fundamentals of Food Safety and Sanitation. Definition and importance of sanitation in food processing. Principles of food hygiene and sanitation practices. Overview of foodborne illnesses and their impact on public health. Sanitary Design and Layout of Food Plants. Principles of sanitary design in food processing facilities. Layout considerations for minimizing cross-contamination. Importance of proper drainage, ventilation, and lighting. Cleaning and Sanitizing Procedures Types of cleaning agents and sanitizers used in food plants. Cleaning schedules and protocols for equipment, surfaces, and utensils. Validation and verification of cleaning effectiveness.	9
2	Food Plant Hygiene Practices: Personal Hygiene and Employee Practices. Importance of personal hygiene for food handlers. Requirements for protective clothing, handwashing, and personal behavior. Employee training and education on hygiene practices. Pest Control and Waste Management	9

	Integrated pest management strategies in food facilities. Waste handling and disposal practices. Role of sanitation in preventing pest infestations. Water Quality and Control Importance of water quality in food processing. Water source selection, treatment, and monitoring. Hygienic design of water supply systems.	
3	Sanitation in Food Processing Equipment and Facilities: Equipment Cleaning and Maintenance Procedures for disassembly, cleaning, and reassembly of equipment. Maintenance schedules and preventive maintenance programs. Hygienic design principles for equipment and utensils. Facility Sanitation and Environmental Monitoring Cleaning protocols for floors, walls, and ceilings. Air quality management and ventilation systems. Environmental monitoring programs for microbial and allergen control.	**9**
4	Regulatory Compliance and Quality Assurance. Food Safety Regulations and Standards Overview of global and local food safety regulations (e.g., FDA, USDA, EU standards). Compliance requirements for food processing facilities. Role of quality assurance in maintaining sanitation standards. Auditing and Inspection Procedures Internal and external audits for sanitation and hygiene. Preparation for regulatory inspections and audits. Corrective actions and continuous improvement in sanitation practices. Emerging Trends and Technologies Innovations in sanitation equipment and technologies. Adoption of automation and digital tools for sanitation monitoring. Future challenges and opportunities in food plant sanitation.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe principles of food hygiene and sanitation practices to ensure food safety.	**K2**
CO2	Evaluate the effectiveness of cleaning and sanitizing procedures in food processing environments.	**K3**
CO3	Explain the regulatory requirements and standards for food plant sanitation and hygiene.	**K2**
CO4	Elaborate emerging trends and technologies in food plant sanitation for future advancements.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1					2						3
CO2	1					2						3
CO3	1					2						3
CO4	1					2						3
CO5	1					2						3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Plant Sanitation: Design, Maintenance, and Good Manufacturing Practices	Michael M. Cramer, Ph.D.; Frederick J. Orthlieb	CRC Press	2^{nd} Edition,2019
2	Principles of Food Sanitation	Norman G. Marriott, Robert B. Gravani	Springer	6th Edition, 2018
3	Food Safety and Sanitation	David McSwane, Richard Linton, Francis Fung, Brenda Harris	Springer	Ist Edition, 2006

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of Hygiene Control in the Food Industry	H. L. M. Lelieveld, John Holah, David Napper	Woodhead Publishing	2nd Edition, 2016
2	Food Plant Engineering Systems	Theunis Christiaan Robberts	Academic Press	1st Edition, 2021
3	Food Safety and Quality Systems in Developing Countries: Volume II: Case Studies of Effective Implementation	Andre Gordon	Academic Press	Academic Press

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
2	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
3	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
4	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview

SEMESTER S7

FAT AND OIL PROCESSING TECHNOLOGY

Course Code	**PEFTT748**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To comprehend the nutritional roles of fats and oils in the human diet, including their impact on health
2. To gain proficiency in the extraction, refining, and processing techniques of fats and oils

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Fats and Oils: Natural sources of vegetable oils: coconut, palm, peanut, rice bran, sesame, mustard, sunflower seeds. Production status and oil content. Physical properties: viscosity, melting point. Chemical properties: hydrolysis, hydrogenation, oxidation, polymerization. Importance of fatty acids and their classifications. Role of oils in nutrition. Impact of fatty acids on health	**9**
2	Oil Extraction Techniques: Mechanical Extraction Methods Ghani, power Ghani, rotary, hydraulic press, screw press, expellers, filter press. Principles of operation and maintenance. Solvent Extraction Process steps and continuous solvent extraction for rice bran, soybean, and sunflower oils. Extraction methods for groundnut and cottonseed oils. Special Oils Production processes for palm oil, virgin coconut oil, and other specialty oils.	**9**
3	Refining and Processing of Oils: Refining Processes Objectives and characterization. Techniques: degumming, deacidification (acid refining), bleaching, deodorization, winterization. Hydrogenation of fats: production of vanaspati, margarine, ghee, and butter Specialty Oils and Products	**9**

	Conjugated linoleic acid (CLA), gamma linolenic acid (GLA), oils from microorganisms, lecithin, transgenic oils, fish oil. Production and processing of non-edible oils: castor oil, linseed oil, vegetable waxes.	
4	Applications, Quality Standards, Packaging and Storage Requirements and properties of packaging materials: tinplate, semi-rigid, glass, PET, PVC, flexible pouches. Changes during storage: causes of rancidity, atmospheric oxidation, enzyme action, free fatty acids, color changes. Industrial Applications and Standards Industrial uses: soap, candles, paints, varnishes. Quality regulations: ISI, AGMARK standards. Safety aspects and HACCP standards in oil industries.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the Characteristics and Properties of Fats and Oils	**K2**
CO2	Evaluate various mechanical extraction methods	**K3**
CO3	Explain Refining and Processing Processes of Oils	**K2**
CO4	Assess the production and processing techniques of non-edible oils and specialty oils.	**K2**
CO5	Describe quality regulations (e.g., ISI, AGMARK) and safety standards (e.g., HACCP) relevant to the oil industry.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											3
CO2	2											3
CO3	2											3
CO4	2											3
CO5	2											3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food oils and Fats - Technology, Utilization and Nutrition	Harry Lawson	CBS Publishers	1997
2	Oils and Fats in Food Industry	Gunstone	Blackwell Publishers	1ST edition, 2008

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Vegetable Oils in Food Technology: Composition, Properties and Uses	Gunstone	Wiley: Blackwell Publishers	2nd edition, 2011

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://t.ly/zFc-_
2	https://shorturl.at/CdrFb
3	https://shorturl.at/i0Jep
4	https://shorturl.at/1Zahu

SEMESTER S7

FOOD RHEOLOGY AND MICROSTRUCTURE

Course Code	**PEFTT745**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	

Course Objectives:

1. Understand the fundamental concepts of rheology and its applications in food technology.
2. Analyze the impact of food structure on the rheological properties of various food products.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Some Basic concepts of Rheology: Introduction to Rheology, Biological systems and Mechanical properties, Force deformation behaviour, Uniaxial stress, Young's modulus, Stress strain behaviour, Elastic plastic behaviour, Evaluation of Poisson's ratio, Bulk stress strain behaviour, Physical state of a material, Classical ideal materials, Time effects (viscoelasticity), Rheological equations, Rheological models	9
2	Rheological Properties: Types of Deformation: Shear Flow, Extensional (Elongational) Flow, Volumetric Flows, Viscoelastic characterization of materials- Stress-strain behaviour, Creep, Stress relaxation, Dynamic tests, Viscoelastic behaviour, Viscometry, Apparent viscosity, Intrinsic viscosity, Newtonian and Non-Newtonian fluids, Pseudoplastic flow behaviour, Thixotropic flow behaviour, Dilatant flow behaviour, Rheopectic flow behaviour, Plastic flow behaviour, Phase transition in foods.	9
3	Measurement of Rheological property and Applications: Different viscosity measurement and viscometers: Rotational viscometers, Vibrational (Oscillation) viscometer, Bostwick consistometer, Pressure-Driven Flow Viscometers, Capillary tubes and Falling sphere type, Torsion Gelometer for Solid Foods, Texture of Foods: Dough testing instruments, Application of	9

	Rheology in fluid food handling and processing	
4	Fundamentals of Food Microstructure: Introduction, Food structure, Factors affecting structure, Microstructure Elements and their Interactions: Water, Protein, Fat, Polysaccharides and other components in food, Microstructure and its relationship with quality and stability of foods: Frozen foods, confectionary and bakery products, Extruded foods, Dried foods, Non-thermal processing Technologies for Fabrication of Microstructures, Engineered food microstructure Examination of food microstructure: Light Microscopy, Fluorescence microscopy, Confocal Laser Scanning Microscopy (CLSM), Polarizing microscopy, X ray Analysis, Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), Immunolabeling Techniques	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	***Internal Ex***	***Evaluate***	***Analyse***	***Total***
5	**15**	**10**	**10**	**40**

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. **Problem Definition (2 points)**
   - Check for clarity and completeness of the problem definition.
   - Check for clear objectives and divide into sub objectives
2. **Problem Analysis (4 points)**
   - Examine the problem and identify the variables and responses
   - Divide the problem to manageable parts
3. **Implementation of software (2 points)**
   - Use suitable software for getting the solution
   - Check for the appropriateness of tools used
4. **Evaluate (4 points)**
   - Evaluate the proposed solution.
   - Assess the correctness of solution.
5. **Validation of the results (4 points)**

- Validate the results using available data.
- Do the error analysis of the results obtained.

6. **Conclusion and presentation (4 points)**
    - Summarize the results obtained
    - Do critical thinking of the results obtained and evaluate.
    - Submit a technical report

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the rheological properties of food and their significance in food processing	**K2**
CO2	Describe the various of flow behaviour	**K2**
CO3	Differentiate different types of viscometers, understand their working and applications in food industry	**K2**
CO4	Develop the conceptual knowledge of food microstructure which can be utilized at industrial levels.	**K2**
CO5	Understand the working principle and applications of different techniques used to study food microstructure	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		2	1									
CO2	3	2	2	1		2						
CO3	2				3	2						
CO4	2	1	1			2						
CO5	2				3							

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Understanding and controlling the microstructure of complex foods	D. Julian McClements	CRC Press	
2	Food Microstructure and its Relationship with Quality and Stability	Sakamon Devahastin	Woodhead Publishing	
3	Physical Properties of Plant and Animal Materials	Nuri N Mohesenin	Gordon & Breach Science Publishers	
4	Rheology of Fluid Foods and Semi Solids Principles and Applications	M. A. Rao	Springer	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Physics Physical Properties- Measurement and Applications	Ludger O Figura	Springer	
2	Food Microstructures, Microscopy, Measurement and modelling	V. J Morris and K. Groves	Woodhead Publishing	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc20_ag01/preview
2	https://onlinecourses.nptel.ac.in/noc20_ag01/preview
3	https://archive.nptel.ac.in/courses/103/106/103106131/
4	https://archive.nptel.ac.in/courses/103/106/103106131/

SEMESTER S7

ICT APPLICATIONS IN FOOD INDUSTRY

Course Code	**PEFTT751**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To understand the importance of computerization in food industry and the applications of ICT in food industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Computerization in the Food Industry Importance and benefits of ICT in food processing. Operating environments and information systems. Supervisory Control and Data Acquisition (SCADA) Systems. Overview of SCADA systems: hardware, firmware, software, and protocols. Use cases in various types of food industries. Spreadsheet Applications for Engineering Problems. Data interpretation and problem-solving using spreadsheets. Creating charts, using macros, add-ins, and Solver for engineering applications.	9
2	Web Technologies and MATLAB in Food Industry Web Hosting and Webpage Design. Basics of web hosting, FTP for file transfer. Design considerations for online food process control. MATLAB Applications in Food Industry. Introduction to MATLAB: computing basics, script files, editor/debugger. Problem-solving methodologies using MATLAB for numeric and matrix operations. Advanced Tools and Computational Fluid Dynamics (CFD)Toolboxes Useful in Food Industry Overview of MATLAB toolboxes: curve fitting, fuzzy logic, neural networks, image processing, statistical analysis.	9

3	Introduction to Computational Fluid Dynamics (CFD). Governing equations of fluid dynamics. Models of flow, boundary conditions, discretization technique. Introduction to CFD Software (GAMBIT and FLUENT) Overview of CFD software applications in food and beverage industry. Hands-on practice with GAMBIT and FLUENT.	**9**
4	LabVIEW for Data Acquisition and Control. Introduction to LabVIEW environment. Data acquisition using NI-DAQ and simulated data. Basic programming concepts in LabVIEW. ICT Applications in Food Industry. Role of ICT in food processing: robots, data collection, and wireless sensors. Data processing technologies: big data analysis, machine learning, cloud computing. Machine Learning and Image Processing. Overview of machine learning algorithms and applications in the food industry. Image processing techniques for quality control and inspection.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the importance and benefits of Information and Communication Technology (ICT) in the food industry.	**K2**
CO2	Solve numeric and matrix operations using MATLAB to address engineering challenges in food processing.	**K3**
CO3	Understand the governing equations of fluid dynamics and apply models of flow, boundary conditions, and discretization techniques in Computational Fluid Dynamics (CFD).	**K2**
CO4	Describe the role of ICT in integrating robots, wireless sensors, and data collection technologies in the food industry.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3		3							3
CO2	3	2	3		3							3
CO3	3	2	3		3							3
CO4	3	2	3		3							3
CO5	3	2	3		3							3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computer Applications in Food Technology: Use of Spreadsheets in Graphical, Statistical and Process Analysis.	R. Paul Singh	Academic Press, London	2014
2	MATLAB for Engineers	Holly Moore	Pearson	5th Edition, 2019
3	Process Control: Modeling, Design, and Simulation	B. Wayne Bequette	Prentice Hall	2nd Edition, 2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Supervisory Control and Data Acquisition (SCADA)	Stuart A. Boyer	CRC Press	1st Edition, 2006
2	Computational Fluid Dynamics: Principles and Applications	Jiri Blazek	Butterworth-Hienmann	3rd Edition, 2015
3	LabVIEW for Engineers	Ronald W. Larsen	Pearson	1st Edition, 2015

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/112105045
2	https://nptel.ac.in/courses/112105045
3	https://nptel.ac.in/courses/112105045
4	https://nptel.ac.in/courses/112105045

SEMESTER S7

ENERGY MANAGEMENT IN FOOD INSUSTRY

Course Code	**PEFTT752**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To optimize energy use in food processing operations
2. To foster an understanding of sustainable practices in food processing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Energy Management in Food Industry, Introduction to energy sources and their relevance in food processing. Basics of energy auditing and energy efficiency assessments. Energy conservation principles and techniques specific to food processing. Case studies highlighting successful energy management practices in food industry.	**9**
2	Energy Conservation Techniques- Thermal energy conservation techniques in food processing, Electrical energy conservation strategies and technologies, Renewable energy integration in food industry operations: Solar Energy: Solar PV systems: components, sizing, and installation considerations. Applications of solar energy in food processing (e.g., water heating, electricity generation). Economic feasibility analysis and incentives for solar energy adoption. Case studies of food processing facilities integrating solar energy and their experiences. Biomass Energy: Overview of biomass energy sources (biogas, biomass boilers). Utilization of biomass for heat and power generation in food processing. Environmental considerations and sustainability aspects of biomass energy.	**9**

3	Energy Management Systems (EnMS) and Standards. Overview of ISO 50001:2018 standard for energy management systems (EnMS). Implementation steps and requirements for EnMS in food processing units. Monitoring, measurement, and verification techniques for EnMS effectiveness. Case studies on EnMS implementation and certification in the food industry.	**9**
4	Socio-Economic and Environmental Aspects of Energy Management Socio-economic benefits of energy efficiency in the food industry. Environmental impacts of energy use in food processing and mitigation strategies. Greenhouse Gas Emissions Reduction: Assessment of carbon footprint in food processing operations. Strategies for mitigating greenhouse gas emissions through energy efficiency. Case studies on successful implementation of emission reduction initiatives. Role of government policies, incentives, and regulations in promoting energy efficiency. Future trends and innovations in energy management for sustainable food processing.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define and explain the basic concepts of energy management in the context of food processing.	**K2**
CO2	Apply Energy Conservation Techniques in Food Processing	**K3**
CO3	Interpret and apply the ISO 50001:2018 standard for energy management systems (EnMS) in food processing units.	**K3**
CO4	Assess the environmental impacts of energy use in food processing operations, particularly in terms of greenhouse gas emissions and resource conservation.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	2					3
CO2	3					3	2					3
CO3	3				2	3	2					3
CO4	3					3	2					3
CO5	3					3	2					3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Energy Management in the Food Industry	Abu-Mansour, Baha	CRC Press	1st Edition, 2016
2	Handbook of Energy Efficiency and Renewable Energy in Food Processing	Kutz, Myer	CRC Press	1st Edition, 2008
3	Introduction to Renewable Energy	Vaughn C. Nelson	CRC Press	1st Edition, 2011
4	Energy and the Food System	David Pimentel and Marcia H. Pimentel	CRC Press	1st Edition, 2008

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Energy Management in the Food Industry	Jochen Weiss, Jiri Klemes, and Pieternel A. Luning	CRC Press	1st Edition, 2009
2	Handbook of Energy Efficiency in Buildings: A Life Cycle Approach	Francesco Reda, Mauro Overend, and Paolo Bertoldi	Routledge	1st Edition, 2013
3	Renewable Energy Integration in Power Management	Lawrence E. Jones	Academic Press	1st Edition, 2014

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/nou23_es05/preview
2	https://onlinecourses.swayam2.ac.in/nou23_es05/preview
3	https://onlinecourses.swayam2.ac.in/nou23_es05/preview
4	https://onlinecourses.swayam2.ac.in/nou23_es05/preview

SEMESTER S7

FOOD PRODUCTS MONITORING & CONTROL

Course Code	**PEFTT753**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To gain the knowledge of instrumentation in the field of monitoring and control
2. To familiarize monitoring and control of Flavour, Colour, Structure and Chemical contamination of food.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Product monitoring and process control , the concept of Product and Process Monitoring in food industries. Process Control – Objectives, Control loop, Loop elements and their functions, Modes of process control, Control techniques, control equipment. Definition of quality - Optimization paradigm, Quality-prediction model based on quality kinetics and process state equations. Quality factors- appearance, texture and flavour. Appearance factors – size and shape, colour and gloss, consistency. Textural Factors – measuring texture, texture changes. Flavour Factors – influence of colour and texture on flavour - Taste Panels. Simulation modelling – Process and Product Optimization - Optimization procedures – Search methods, Response surface, Differentiation & Programming methods - Neural Networks - Optimization software	9
2	Instrumentation for monitoring and control Real-time Instrumentation – Sensors - Chemo sensors, biosensors, immune-sensors and DNA probes base devices with working principles. Requirements of on-line sensors. Biosensors – Construction, types, working principles, applications, merits	9

	and limitations. E-Nose & E-Tongue – Simulation of natural organs, Components & their functions, Applications. Computer based monitoring and control - Advanced instruments - microwave measurements of product variables, ultrasonic instrumentation, conductance/impedance techniques for microbial assay.	
3	Monitoring and control of Flavour & Colour Flavour analysis - Flavour bioassays – Gas Chromatography, Olfactometry techniques - Isolation, Separation and detection/Identification of flavour compounds – GC-MS, LC-MS, NMR, FTIR - Analysis of chiral compounds Formation of flavour compounds in milk and milk products during heat processing like UHT processing, caramelization and extrusion cooking. Fermentation and ripening and storage - Maillard Browning. Aroma loss or retention during the drying process - Industrial processes for extraction of desirable and undesirable volatile components from fresh and/or stored products by supercritical fluid (SCF) technique. Colour Characterization - Colour and appearance - gloss and translucence, monitoring through visual colorimeter, tri-stimulus colorimeters and reflectance spectrophotometer, CIE, Hunter- Lab, Munsel and other systems of three-dimensional expression of colour. Colour-based Sorting of foods, Computer Vision – Principles, applications and Benefits.	**9**
4	Monitoring and control of food structure Application of Thermal Analysis - Pulse Nuclear Magnetic Resonance (PNMR) spectroscopy in determination of solid-fat content (SFC) – Glass transitions in dairy products, Starch gelatinization Elucidation of crystal characteristics of milk fat in ghee and other fat-rich products by means of X ray Crystallography with reference to the impact of cooling and storage/handling conditions on the crystal nature and product texture – structure texture relationship, Emerging techniques in monitoring and control Raman Spectroscopy and ElectronSpin Spectroscopy – Working principles and applications - Monitoring of irradiated foods, detection of lipid auto- oxidation. Microwave & NIR absorption/reflection methods for Compositional analyses – Automated milk analysers; Proximate principles in cheese and milk powder	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand and Apply Principles of Product Monitoring and Process Control in Food Industries	K2
CO2	Describe the construction, types, and working principles of biosensors and their specific applications in food quality assessment.	K2
CO3	Illustrate techniques for Monitoring and Control of Flavour and Colour in Food Products	K2
CO4	Explain advanced Techniques for Monitoring and Control of Food Structure and Composition	K2

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	2										3
CO3	3	2										3
CO4	3	2										3
CO5	3	2			3							3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Process Engineering and Technology	P.J. Fellows	Woodhead Publishing	2^{nd} edition, 2009
2	Food Analysis	S. Suzanne Nielsen	Springer	5th edition, 2017
3	Food Quality: Quantitative Analysis of Physical Properties	M. Edwards, B. Kowalski, D. Armitage	Chapman and Hall	2^{nd} edition, 1993

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Biosensors in Food Processing, Safety, and Quality Control	Mehmet Mutlu, Arzum Erdem, Koray Palazoglu	CRC Press	1^{st} edition, 2010
2	Handbook of Food Analysis	Leo M.L. Nollet, Fidel Toldra	CRC Press	3^{rd} edition, 2017

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/112105045
2	https://nptel.ac.in/courses/112105045
3	https://nptel.ac.in/courses/112105045
4	https://nptel.ac.in/courses/112105045

SEMESTER S7

POST HARVEST SPOILAGE & PHYSIOLOGY OF FOODS

Course Code	**PEFTT754**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To identify the physiology changes and their reasons occurring after fruit and vegetable harvest
2. To study the methods for quality improvement of post-harvest produce

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Basic post- harvest physiology, definition, scope and importance in food production, respiration and gas exchange, Effect of respiration on physiology, factors affecting respiration, Respiration control methods, Changes during post- harvest, physical and chemical changes, transpiration, water stress. Different methods to determining maturity. Maturity indices for different cereals, pulses, oil seeds, fruits and vegetables. Ripening: Process of ripening, changes during ripening of fruits. Senescence: Physiological changes during senescence control of senescence and significance. Abscission: process of abscission, Factors affecting senescence and abscission of plant parts.	**9**
2	Factors affecting post-harvest physiology: Pre-harvest factors on post-harvest life, harvesting and handling injuries, storage conditions; temperature, RH, composition and its modification, Ethylene, Role of ethylene in ripening, Climacteric and non-climacteric fruits, ethylene biosynthesis, ethylene generators, ethylene control methods. Handling and storage: Maturity indices, Assessment of crop maturity, Changes during	**9**

	ripening, physical and chemical changes, change in colour, texture, flavour during storage, Harvesting methods, Storage conditions and types, Storage characteristics of different fruits and vegetables, MAP and CAP.	
3	Factors involved with spoilage: Biotic and abiotic factors, temperature, insect infestation, microbes-fungi, virus, bacteria etc., mechanical damage, and chilling injury, freezing injury, quality and safety factors.	**9**
4	Quality Improvement techniques: Harvesting and handling techniques, coatings and treatments, insect control and microbial control, Post- harvest treatments, GAP, GHP, GMP, HACCP, measurement of product quality methods, destructive and non- destructive tests, physical chemical, biological, visual methods.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the physiological and biochemical changes in horticultural produce after harvest and apply suitable control methods	**K2**
CO2	Describe suitable ripening techniques for the development of safe and quality ripened products for the shelf-life extension of horticultural produce	**K3**
CO3	Understand the various factors involved in spoilage and the techniques for its control	**K2**
CO4	Develop the knowledge of quality control for the quality improvement of fruits and vegetables	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										3
CO2	3	2	3									3
CO3	3											3
CO4	3	3	3		3							3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Postharvest Physiology and Pathology of Vegetables	Jerry A Bartz, Jeffrey K Brecht	CRC Press	2nd Edition, 2002
2	Postharvest Technology of Horticultural Crops	K. P. Sudheer and V. Indira	New India Publishing Agency	7th Edition, 2007

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fruit and Vegetables Harvesting, Handling and Storage	A.K Thomson	Blackwell Publishing	2nd Edition, 2014
2	Handbook of Postharvest Technology: Cereals, Fruits, Vegetables, Tea, and Spices	Amalendu Chakraverty, Arun S. Majumdar, G. S. Vijaya Raghavan and Hosahalli S. Ramaswamy	Tata McGrawhill	1st Edition, 2003

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://shorturl.at/JBLDP
2	https://shorturl.at/rwOhA
3	https://shorturl.at/sP8kx
4	https://shorturl.at/xCf2x

SEMESTER S7

SNACK FOOD TECHNOLOGY

Course Code	**PEFTT756**	**CIE Marks**	**40**
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To acquaint students with the principle involved in the manufacture of commercial snack foods
2. To familiarize students with equipment of importance in the snack food industry

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Snack foods: Types, specifications, compositions, ingredients, formulations, processing, equipment, packaging, storage and quality testing Snack food seasonings; Breakfast cereals, macaroni products and malts: Specifications, compositions, ingredients, formulations, processing, equipment, packaging, storage and quality testing.	9
2	Grain-based snacks- Technology for Whole Grains Snacks – roasted, toasted, puffed, popped, flaked, Technology for Coated Grain Snacks – salted, spiced, sweetened, Technology for Batter-Based and Dough-Based Products, Technology for Formulated Products – chips, wafers, papads, instant pre mixes	9
3	Horticulture produce-based snacks- Technology for Fruit-Based Snacks, Technology for Vegetable-Based Snacks, Technology for Coated Nuts Extruded snacks- Formulation and Processing Technology, Colouring and Flavouring, Packaging, Machinery and Equipment, Use, and Care	9
4	Pulse based snacks-technology and processing formulations, Instant Food Pre-Mixes, Shelf-Life and Quality Characteristics of Snack Foods, regulations and standards of snack foods, permitted additives in snack foods.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	The student will gain an understanding of processing techniques used to make snack foods and snack food seasonings	**K2**
CO2	Apply grain, dough and batter-based snack food formulations in real world	**K3**
CO3	Comprehend fruit & vegetable and extruded snack foods along with machinery and packaging requirements	**K2**
CO4	Analyse quality characteristics, shelf life and major regulations of snack foods	**K4**
CO5	The student will gain an understanding of processing techniques used to make snack foods and snack food seasonings	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1			1							1
CO2	2	1			3							1
CO3	2	2			3							1
CO4	2	1			3							1
CO5	2	1			2							1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Snack Foods Processing	Edmund W. Lusas, Lloyd W. Rooney	CRC Press	2001
2	The Technology of Extrusion Cooking.	Frame ND	Blackie Academic	1994
3	Snack Food	Gordon BR	AVI Publication	1st edition, 1990
4	Snack Food Technology	Samuel AM	Springer Science & Business Media	2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Extrusion -cooking techniques, Application, theory and sustainability	Leszek moscicki	Wiley-VCH	1999
2	Village scale technology for novel snack foods products based on high protein cereal legumes	H E Smith, M S Chinnan	University of Georgia	1954
3	Cereal grain for the food and beverage industries	Elke K Arendt, Emanule Zannini	Woodhead publications	1st edition, 2013
4	Food science, technology, and nutrition	Sharma A	Woodhead publishing	2nd edition, 2005

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://ugcmoocs.inflibnet.ac.in/index.php/courses/view_ug/134
2	https://onlinecourses.nptel.ac.in/noc22_ag03/preview
3	https://onlinecourses.swayam2.ac.in/cec19_ag01/preview
4	https://onlinecourses.swayam2.ac.in/nou20_ag09/preview

SEMESTER S7

CONSUMER BEHAVIOR IN FOOD MARKETING

Course Code	**PEFTT757**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To explain various theories and models related to consumer behavior and apply these concepts specifically to food marketing scenarios.
2. To identify and analyze psychological and social influences that impact consumer decision-making in food consumption.
3. To use various market research methods to gather data and apply the findings to develop effective food marketing strategies.
4. To design and assess food marketing strategies informed by consumer behavior principles, ensuring that the strategies are effective and relevant.
5. To evaluate how current trends and technological innovations influence consumer behavior and marketing strategies in the food industry.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Consumer Behaviour in Food Marketing: Definition and importance of consumer behaviour - Key theories and models of consumer behaviour - Consumer decision-making process - The role of food marketing in consumer behaviour - Trends and challenges in food marketing - Regulatory and ethical considerations	9
2	Consumer Behaviour Models and Theories: Theory of Planned Behaviour - Stimulus-Response Theory - Hedonic vs. utilitarian consumption - Cognitive Dissonance Theory - Elaboration Likelihood Model (ELM) - Expectancy-Value Theory - The impact of culture and subculture on food choices -	9

	Social class and consumer behaviour - Family and household influence on food consumption	
3	Market Research, Consumer Insights, Branding and Product Development: Qualitative vs. quantitative research methods - Surveys, focus groups, and observational studies - Demographic, psychographic, and behavioural segmentation - Identifying and analyzing target markets - Brand equity and brand loyalty - The role of branding in consumer perception - Consumer-driven product innovation - The role of sensory marketing in product development - Testing and refining food products based on consumer feedback	**9**
4	Consumer Behaviour Trends and Technology: Health and wellness trends affecting food choices- Sustainability and ethical consumption-Impact of convenience and ready-to-eat foods- Role of digital marketing in influencing food choices - Social media, influencer marketing, and e-commerce - Data privacy and consumer trust in digital platforms	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 = 24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand and apply key theories and models of consumer behaviour in the context of food marketing.	**K2**
CO2	Analyze the psychological and social factors influencing consumer choices and behaviours in the food sector.	**K3**
CO3	Conduct market research and apply insights to food marketing strategies.	**K3**
CO4	Develop and evaluate food marketing strategies based on consumer behaviour principles.	**K5**
CO5	Assess emerging trends and technological advancements affecting consumer behaviour in food marketing.	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	3	2						2
CO2	2	3	2	2		2	2					2
CO3	1	2	2	3	3	2	2					2
CO4	2	2	3	3	2	2	2	1				2
CO5	1	1	2	2	3	3	3			2		2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Understanding Consumer Behavior in Food Marketing	Barbara A. Lehnert	Springer	1st Edition, 2021
2	Consumer Behavior: Buying, Having, and Being	Michael R. Solomon	Pearson	12th Edition, 2020
3	Consumer Behavior: Insights from Indian Markets	Sujatha S. Rajan & K. M. Vasanth	Sage Publications	1st Edition, 2017
4	Food Marketing: A Global Perspective	Alex R. G. Simon & S. S. Sharma	Wiley	2nd Edition, 2018
5	Consumer Behavior in Action: Real-Life Applications for Marketing Managers	Laura J. W. Jackson & Michael A. Smith	Routledge	1st Edition, 2021

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Consumer Behavior: The Psychology of Marketing	Frank A. Kuczynski	Routledge	2nd Edition, 2020
2	Marketing Research: An Applied Orientation	Naresh K. Malhotra & David F. Birks	Pearson	7th Edition, 2021
3	Handbook of Consumer Behavior, Retailing, and Marketing	S. Ratneshwar, David G. Mick, & C. T. M. Kumar	Sage Publications	1st Edition, 2018
4	Global Perspectives on Food Marketing and Consumer Behavior	E. Shama & J. M. Kotler	Routledge	2nd Edition, 2019
5	Sustainable Food Systems: The Role of Consumers	Rachel L. S. Wright	Springer	1st Edition, 2022

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/110105029
2	https://onlinecourses.nptel.ac.in/noc22_mg47/preview
3	https://onlinecourses.swayam2.ac.in/imb21_mg31/preview
4	https://onlinecourses.swayam2.ac.in/imb24_mg34/preview

SEMESTER S7

OPTIMIZATION TECHNIQUES

Course Code	**PEFTT758**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Understand the fundamental concepts and importance of optimization in various fields
2. Develop practical skills to apply optimization techniques to solve real-world problems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Module 1: Introduction to Optimization Introduction to Optimization – Definitions – Historical development – Scope of optimization techniques. Classification of Optimization Problems – Linear and Non-linear – Static and Dynamic – Deterministic and Stochastic – Single and Multi-objective. Formulation of Optimization Problems – Objective function – Constraints – Decision variables.	**9**
2	Module 2: Classical Optimization Techniques Single Variable Optimization – Optimality criteria – Unconstrained optimization – Constrained optimization using Lagrange multipliers and Kuhn-Tucker conditions. Multi-variable Optimization – Direct search methods – Gradient-based methods – Newton's method – Conjugate gradient method. Linear Programming – Formulation – Graphical method – Simplex method – Duality theory – Sensitivity analysis.	**9**
3	Module 3: Numerical Methods and Algorithms Numerical Optimization Techniques – Introduction to numerical methods – Numerical differentiation and integration. Gradient Descent – Steepest	**9**

	descent method – Conjugate gradient method – Quasi-Newton methods. Non-linear Programming – Penalty function methods – Barrier methods – Sequential quadratic programming – Interior point methods.	
4	Module 4: Advanced Optimization Techniques Dynamic Programming – Principle of optimality – Recursive equations – Applications. Genetic Algorithms – Basics of genetic algorithms – Operators – Selection, crossover, mutation – Applications. Simulated Annealing – Principles – Algorithm – Applications. Particle Swarm Optimization – Basic concepts – Algorithm – Applications.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of optimization, including various types and formulations of optimization problems.	**K2**
CO2	Apply classical optimization techniques to both single and multi-variable problems, utilizing methods such as the simplex method and Lagrange multipliers.	**K3**
CO3	Implement numerical methods and algorithms for solving complex optimization problems, including gradient descent and interior point methods.	**K4**
CO4	Utilize advanced optimization techniques like dynamic programming, genetic algorithms, simulated annealing, and particle swarm optimization for real-world applications.	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3									
CO2	3	2										
CO3	3	2										
CO4	3		1									

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Optimization	Edwin K. P. Chong and Stanislaw H. Zak	Springer	4th Edition, 2013
2	Engineering Optimization: Theory and Practice	Singiresu S. Rao.	Wiley	4th Edition, 2009
3	Practical Optimization	Philip E. Gill, Walter Murray, and Margaret H. Wright	Academic Press	1st Edition, 1981
4	Nonlinear Programming: Theory and Algorithms	S. Bazaraa, Hanif D. Sherali, and C. M. Shetty	Springer	3rd Edition, 2006

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Optimization in Operations Research	Ronald L. Rardin	Pearson	2nd Edition, 2016
2	Linear and Nonlinear Programming	David G. Luenberger and Yinyu Ye	Springer	4th Edition, 2015
3	An Introduction to Optimization	Edwin K. P. Chong and Stanislaw H. Zak	Wiley	3rd Edition, 2008
4	Optimization: Algorithms and Applications	Rajesh Kumar Arora	CRC Press	1st Edition, 2015

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/111105039
2	https://onlinecourses.nptel.ac.in/noc21_me10/preview
3	https://nptel.ac.in/courses/106106245
4	https://nptel.ac.in/courses/108105019

SEMESTER S7

RESEARCH METHODOLOGY AND STATISTICS

Course Code	**PEFTT755**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Explain the basic concepts of research.
2. Identify various sources of information for literature review and data collection
3. Understand various research designs and techniques of data analysis, writing and evaluate its quality
4. Appreciate the components of scholarly
5. Develop an understanding of the ethical dimensions of conducting research.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Research methodology- Understanding the language of research Concepts, constructs, operational definitions, variables, propositions, hypotheses, theories, and models Research process- Literature review -Types of research- Problem identification andformulation Research question Research hypothesis - Measurement issues Methods of data collection Types of data- Primary data Scales of measurement- Sources and collection of dataObservation method Interview method- Survey- Experiments- Secondary data-Research design- Qualitative and Quantitative Research.	9
2	Processing and analysis of data- Sampling- Steps and characteristics of sampling design Sampling: concepts of Population, Sample, Sampling Frame, Sample size and its determination Types of sampling distributions	9

	Sampling error Statistics in research Descriptive statistics and inferential statistics- Measures of central tendency, dispersion, skewness, asymmetry- Measures of relationship- Correlation and regression- Simple regression analysis- Multiple regression Hypothesis Testing - parametric and non-parametric tests- Analysis of single factor experiments.	
3	Reporting and presenting research Written and oral communications - Hallmark of great scientific writing - The reading toolkit - Pre-writing considerations - Format of dissertations, research reports, and research papers - Paper title and keywords - Writing an abstract - Writing the different sections of a paper Revising a paper Responding to peer reviews Reviewing research papers Plagiarism, Conference and poster presentations.	**9**
4	Language aspects of report writing -Verb, tense and voice in scientific writing - Errors in grammar - Sentence and paragraph constructions- Paraphrasing tools and online platforms - Measures of research impact.Referencing Management Software (Mendeley, Zoteroetc) - Intellectual property rights - Copyright - Patents - The codes of ethics - Avoiding the problems of biased survey -Occupational health and safety.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	***Internal Ex***	***Evaluate***	***Analyse***	***Total***
5	**15**	**10**	**10**	**40**

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. **Problem Definition (2 points)**
    - Check for clarity and completeness of the problem definition.
    - Check for clear objectives and divide into sub objectives
2. **Problem Analysis (4 points)**
    - Examine the problem and identify the variables and responses
    - Divide the problem to manageable parts
3. **Implementation of software (2 points)**
    - Use suitable software for getting the solution

- Check for the appropriateness of tools used

4. **Evaluate (4 points)**
    - Evaluate the proposed solution.
    - Assess the correctness of solution.
5. **Validation of the results (4 points)**
    - Validate the results using available data.
    - Do the error analysis of the results obtained.
6. **Conclusion and presentation (4 points)**
    - Summarize the results obtained
    - Do critical thinking of the results obtained and evaluate.
    - Submit a technical report

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Comprehensive Understanding of Research Methodology	**K2**
CO2	Proficiency in Literature Review and Problem Formulation	**K1,K3**
CO3	Expertise in Data Collection and Analysis Methods	**K3,K4,K5**
CO4	Effective Research Communication	**K3**
CO5	Ethics and Intellectual Property Awareness	**K3,K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2								3
CO2	3	2		2								3
CO3	3	2			1							3
CO4	3									3		3
CO5	3							3				3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Business Research Methods	Cooper, D. R. and Schindler, P. S	Tata McGraw Hill,	9th edition 2006
2	Research Methods and Statistics	Jackson, S.L.	Cengage Learning India Private Limited, New Delhi	2009.
3	Scientific Writing: A Reader and Writer's Guide	Lebrun, J-L.	World Scientific Publishing Co. Pvt. Ltd., Singapore,	2007
4	Research Methods for Engineers	Thiel, D. V.	Cambridge University Press	2014

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Research methodology	CR Kothari	New Age International Publishers	2023
2	Research & publication ethics	Dr.S.B.Kishor, Dr.AjayS.Kushwaha, Dr. Gitanjali J	Das GanuPrakashan	1st edition 2023
3	Management Research Methodology	Krishnaswamy, K.N., Sivakumar, A.I., and Mathirajan, M.	Pearson Education	2006

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/121/106/121106007/
2	https://archive.nptel.ac.in/courses/121/106/121106007/
3	https://www.youtube.com/watch?v=LY1pGrkFkZg https://archive.nptel.ac.in/courses/110/105/110105091/
4	https://archive.nptel.ac.in/courses/121/106/121106007/

SEMESTER S7

FOOD ENGINEERING

Course Code	OEFTT721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To study the fundamental theories behind general process involved in commercial processing of food and major equipment types used for each process
2. To design the process parameters for thermal processing, freezing, evaporation, dehydration, separation, extraction
3. To develop skills in formulating solutions to solve problems in food industry

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Perishable, semi perishable and non-perishable foods; Non refrigerated, refrigerated transportation Precooling – room cooling, forced air cooling, package icing, hydro cooling, vacuum cooling, evaporative cooling; Storage in bins, silos, bag storage, warehouses, Cleaning of raw materials – Dry cleaning and wet cleaning; Pre-treatment – blanching, sulphur dioxide dipping; Sorting and grading according to size, shape, weight, colour Peeling - Flash steam peeling, knife peeling, abrasion peeling, caustic peeling, flame peeling Size reduction - Energy consumption laws- Rittinger's law, Kick's law, Bond's law; Size reduction equipments for solids – Impact mills, pressure mills, attrition mills, cutters, choppers, shredders, pulpers; Size reduction liquid foods - Homogenization, homogenizers.	9
2	Batch and continuous sterilization processes (including steps and various machineries involved) used in canning of foods; Commercial sterilization, Constructional and operational features of pasteurizer; homogenizer;	9

	Constructional features and principles of single effect evaporators (including mass and energy balances) used for concentration of liquid foods.	
3	Constructional features of cold storage and basic design approach; Different types of freezers including plate contact freezer, air blast freezer; Cryogenic freezing; Refrigerated mobile vans. Theory of drying and mechanism of moisture transfer in drying; Drying kinetics and constant & falling rate periods in drying; Constructional & operational features of various types of cross-flow, through flow and recirculatory dryers – Tray dryer, roller dryer, drum dryer, spray dryer, fluidized bed dryer, freeze dryer and solar dryer, rotary dryer, tunnel dryer, other grain dryers (LSU-type).	**9**
4	Heat exchangers (Co-current and counter-current heat exchanger); Constructional features of various types of heat exchangers – DPHE, Shell& tube heat exchanger, Plate heat exchanger, extended surface heat exchangers; Theory and operation of extrusion systems used in food industry; Cold extrusion and Extrusion cooking systems; Single and twin-screw extruders – constructional and operational features including advantages/disadvantages with case studies.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the mechanism of transportation, storage, cleaning, peeling and size reduction of food materials	**K2**
CO2	Summarize the food processing techniques of thermal processing, eg, sterilization, evaporation, separation, extraction.	**K2**
CO3	Explain the knowledge of different freezers and dryers	**K2**
CO4	Explain basic principles of different types of heat exchangers and extruders	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	3	3		3							3
CO3	3				3							3
CO4	3				3							3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals of Food Process Engineering	R. T. Toledo	Springer	4th Edition, 2007
2	Food Process Engineering & Technology	Z. Berk	Academic Press	3rd Edition, 2018
3	Food Process Engineering Operations	G. D. Saravacos& Z. B. Maroulis	CRC Press	1st Edition, 2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Food Process Engineering	A. Ibraz& G. V. Barbosa-Canovas	CRC Press	1st Edition, 2014
2	Postharvest Technology and Food Process Engineering	AmalenduChakraverty& R. Paul Singh	CRC Press	1st Edition, 2014

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://acesse.dev/EgwPu
2	https://encr.pw/8H4vq
3	https://acesse.dev/EbSWf
4	https://l1nq.com/MaB5x

SEMESTER S7

FOOD PROCESS AND EQUIPMENT DESIGN

Course Code	**OEFTT722**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To acquire basic understanding of design parameters and knowledge of design procedures for commonly used food processing equipment

SYLLABUS

Module No.	**Syllabus Description**	**Contact Hours**
1	Introduction and Design of vessels: Nature of equipment, general design procedure, equipment classification. Properties of materials -Types of stresses and strains - Poisson's ratio- Hooke's Law – stress strain diagram-factor of safety, Materials of construction, Corrosion Pressure vessel- design of cylindrical and spherical shell subject to internal and external pressure, Silo design- Rankine Theory-Airy equation-Janssen equation	**9**
2	Design of heat exchangers, evaporators and dryers: Design of heat exchangers –Double pipe heat exchangers, plate heat exchanger, shell and tube heat exchangers, design of evaporators-mass and energy balance-single effect and multiple effect evaporators, Design of dryers-mass and energy balance, Psychrometrics-drying time calculation.	**9**
3	Design of supplemental processes: Design of food extruders – power requirements, Design of cold storage – factors to be considered –estimation of cooling load – construction and operation. Design of Agitation Vessels-power requirements, Design of centrifugal separator- rate of separation,	**9**

4	Design of Material handling, Cleaning and Separation devices: Material Handling devices-Design -belt conveyor, bucket elevator, screw conveyor, chain conveyor,pneumatic conveyor. Cleaning and grading-screens-effectiveness of screen-Air screen cleaners - design considerations, Disc separator-Spiral Separator-Specific gravity Separator, Pneumatic Separator-Inclined draper-Velvet roll Separator, Design of Cyclone separator-Magnetic Separator -Colour Separator	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the basics concepts related to design of equipments	**K2**
CO2	Select the design parameters of heat exchangers, evaporators and dryers	**K3**
CO3	Choose the design parameters of extruders, cold storage, agitation vessel and centrifugal separator	**K3**
CO4	Explain the design considerations of cleaning, separation and material handling devices	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2								3	
CO2	3	3	3				2			1	3	2
CO3	3	3	3				1			1	3	1
CO4	3	3	1				2			1	3	2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Process Equipment Design	Joshi, M.V and V.V.Mahajani	New India Publishing Agency, New Delhi.	3rd Edition, 2004
2	Introduction to Food Engineering	R. Paul Singh and Dennis R. Heldman	Elsevier, Amsterdam, Netherlands	5th Edition, 2014
3	Unit Operations of Chemical Engineering,.	McCabe, w.l. Julian Smith, Peter Harriott	McGraw-Hill, Inc., NY, USA.	7th Edition, 2004

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of Food Processing Equipment	George D. Saravacos, Athanasios E. Kostaropoulos,	Springer	2012
2	Food Process Design,	Zacharias B. Maroulis, George D. Saravacos,	CRC Press	2003
3	Postharvest Technology and Food Process Engineering	AmalenduChakraverty, R. Paul Singh	CRC Press	2016

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc21_ch18/preview
2	https://archive.nptel.ac.in/courses/112/105/112105248/
3	https://nptel.ac.in/courses/103105060
4	https://archive.nptel.ac.in/courses/113/105/113105104/

SEMESTER S7

FOOD LAWS AND LEGISLATION

Course Code	**OEFTT723**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To provide a general overview of food laws and regulations
2. To assist students in finding sources of information pertaining to food laws and regulations

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Concept of food safety, food security and nutrition security, Indian food regulatory regime (existing & old), FSSAI, PFA Act and Rules, food licensing and registration system, AGMARK, Legal metrology act 2009**,** Export (Quality Control and Inspection) Amendment Act, 1984, Voluntary Standards and Certification, BIS, food licensing and registration system, food import clearance system, essential commodity act, 1995.	**9**
2	Food hazard contamination-biological (bacteria, viruses and parasites), chemical (toxic constituents / hazardous materials) pesticides residues / environmental pollution / chemicals) and physical factors, Risk assessment studies: risk management, risk characterisation and communication, concept and implementation of HAACP in food premises, use of permitted additives like colours, preservatives, emulsifiers, stabilisers, antioxidants, etc. and their regulations.	**9**
3	Regulations of nutrition labelling, nutrient level claims, and health claims, Regulations of standard of identity and quality, Introduction to the Food, Drug, and Cosmetic Act, Introduction to OIE & IPPC, Other International	**9**

	Food Standards (e.g. European Commission, USFDA etc), Introduction to WTO Agreements: SPS and TBT Agreement, Export & Import Laws and Regulations, WHO/FAO Expert Bodies (JECFA/JEMRA/JMPR)	
4	Voluntary quality standards and certifications-GMP, GHP, GAP, Good Animal Husbandry Practices, ISO 9000, ISO 22000, 1SO 14000, ISO 17025, PAS 22000, FSSC 22000, BRC, Halal & Kosher standards, concept of six sigma, TQC, NABL, NABH, NBQP	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concept of food laws and regulations and major regulatory bodies in India	**K2**
CO2	Apply the concept of food contamination, risk assessment and HACCP application in food	**K3**
CO3	Summarize regulations related to labelling and various international food safety standards and agreements	**K2**
CO4	Analyse sanitary regulations and ISO specifications	**K4**
CO5	Explain the basic concept of food laws and regulations and major regulatory bodies in India	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	2	3				1
CO2	3					3	3	3				1
CO3	3					2	2	3				1
CO4	3					3	2	1				1
CO5	3					1	2	1				1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of indices of food quality and authenticity	Singal R S	Woodhead publications, Cambridge, UK	1st edition, 1997
2	Food safety and standard act, 2006, Rules ang regulations, Volume 1, 2 & 3	Rajan Nijhawan	ILBCO	25th edition, 2024
3	Principles and practices of safe processing of foods	Shapton DA	Butterworth Publication	1st edition, 2013
4	Techniques of food analysis	Winton AL	Allied Science Publications, New Delhi	1999

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The chemical analysis of food and food products	Jacob MB	CBS Publications, New Delhi	1999
2	Food Safety Handbooks	Food Safety Handbooks	Wiley Interscience, UK	2005
3	Food Safety Regulations, Concerns and Trade: The Developing Country Perspective	Mehta, Rajesh and J. George	Macmillan	2005
4	Guide to US Food Laws and Regulations	Patricia A. Curtis	Wiley Online Library Online Books	2^{nd} edition, 2022

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/cec20_ag06/preview
2	https://onlinecourses.swayam2.ac.in/cec22_ag01/preview
3	https://onlinecourses.swayam2.ac.in/nou22_ag11/preview
4	https://onlinecourses.nptel.ac.in/noc24_lw03/preview

SEMESTER S7

FOOD INDUSTRY MANAGEMENT

Course Code	OEFTT724	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To study the basic concepts of management and various forms of business organization and trade unions function in professional organizations
2. To study the planning, organizing and staffing functions of management in professional organization
3. To study the leading, controlling and decision making functions of management in professional organization
4. To learn the organizational theory in professional organization
5. To learn the principles of productivity and modern concepts in management in professional organization.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Management: Introduction; Definition and Functions – Approaches to the study of Management – Mintzberg's Ten Managerial Roles – Principles of Taylor; Fayol; Weber; Parker – Forms of Organization: Sole Proprietorship; Partnership; Company (Private and Public); Cooperative – Public Sector Vs Private Sector Organization – Business Environment: Economic; Social; Political; Legal – Trade Union: Definition; Functions; Merits & Demerits.	9
2	Planning: Characteristics, Importance, Steps, Limitation, Planning Premises, Strategic Planning. Organizing: Organizing Theory, Principles, Types-Departmentalization, Centralization and Decentralization. Controlling: Authority & Responsibility – Staffing, Recruiting and Selection Process,	9

	Human Resource Development (HRD) Concept. Decision Making: Leadership Traits. Organizational Conflict: Positive Aspects, Individual Role, Interpersonal, Intra Group, Inter Group Conflict Management, Guidelines to managing Conflict.	
3	Finance: Assessing, acquiring and allocating funds, cash flow statement-balance sheet, financial ratio, break even analysis, concept, application in food industry, project appraisal, techno- economic feasibility report on an identified opportunity - any food processing; bakery and confectionary , fruit and vegetable processing, oil and fat ,milk and milk products etc. Market management: concepts- consumer market, business market, marketing environment- market segmentation- market measurement and forecasting-advertisement - publicity market information system- market researchmanagement of distribution channel, Consumer buying behaviour, factors influencing consumer buying behaviour. Export trade- Government regulations, GAIT, WTO regulation.	**9**
4	Productivity: Concept; Measurements; Affecting Factors; Methods to Improve – Modern Topics (concept, feature/characteristics, procedure, merits and demerits): Business Process Reengineering (BPR); Benchmarking; SWOT/SWOC Analysis; Total Productive Maintenance; Enterprise Resource Planning (ERP); Management of Information Systems (MIS).	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Summarize the basic concept of management	**K2**
CO2	Explain appropriate requirement of planning, organizing, controlling and decision making functions of management in professional organization	**K2**
CO3	Explain the function of financial, marketing management techniques	**K2**
CO4	Develop the principle of productivity and modern concepts in management in professional organization	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	3	3									3
CO3	3	2										3
CO4	3	3	3		3							3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Principles of Management	M. Govindarajan and S. Natarajan	Prentice Hall of India	1st Edition, 2009
2	Fundamentals of Financial Mangement	Brigham and Eugene F	The Dryden Press	14th Edition, 1989
3	Marketing management	Philip Kotler	Prentice Hall of India	15th Edition, 1985
4	Organization Theory and Design	Richard L. Daft	South Western College Publishing	12th Edition, 2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Principles of Management: A Modern Approach	Saxena and P. K.	Global India Publications	1st Edition, 2009
2	Essentials of Management: An International Perspective	Koontz. H. and Weihrich. H.	Tata McGrawhill	11th Edition, 2010

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://encr.pw/7pYoI
2	https://l1nq.com/WAOyz
3	https://acesse.dev/oSNld
4	https://acesse.dev/FZttF

SEMESTER S7

CONSUMER AND CONVENIENCE FOODS

Course Code	**OEFTT725**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1.To gain knowledge on convenience foods

2.To acquire knowledge on food processing techniques

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Food product development –Development of new product, need for developing new products, Developing marketing strategy for new product, Strategies in product development, success and failure factors for new products.	**9**
2	Snack foods: Snack foods Popped snacks – Popcorn –popping procedures, loss during popping, measurement of expansion, factors affecting quality of popcorn, storage. Puffed snacks – Puffable materials, extrusion methods ,drying, Addition of flavours and colours, Simulated popcorn. Baked snacks – Proportion and role of ingredients; Sweet based –plain cookies, wire cut cookies; Salt based – soda crackers and cheese crackers.	**9**
3	Convenience Foods For Defense Service: Convenience foods for defense services –IMF and Hurdle Technology-Principles. Processing of dehydrated vegetables, vegetable powder, IMF fruit slices, IMF fruit bars, fruit milk, soup powder. Foods designed by DRDO for defense services – list and principle of processing applied.	**9**

4	Ready to eat and extruded foods foods: Ready to eat foods– principle of retort processing, technique, production, advantages and disadvantages. Ready to eat foods available in India. Marketing and future prospects. Extruded foods – Principle of extruders, Production of pasta- noodle and macaroni products, Common extruders used in food industry,Merits and demerits of extruder technology, Uses of extruded foods, Factors affecting extrusion performance.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Summarize the concept of preparing convenience foods.	**K2**
CO2	Explain the knowledge in production of high nutritive value convenience foods.	**K2**
CO3	Identify the quality of the foods.	**K3**
CO4	Identify the shelf life and acceptability of the foods among the consumers.	**K3**
CO5	Select new products imbibed with probiotics, prebiotics and antioxidants.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2					1						3
CO2			2				3					
CO3	2											
CO4		3	3		3	3						
CO5	3				3	3	2					

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Reframing Convenience Food	Peter JacksonHelene Brembeck, Jonathan Everts	Palgrave Macmillan	2018
2	Convenience Foods and the Consumer — Current Questions and Controversies	JC. Somogyi, E.H. Koskinen	Forum of nutrition	Volume 45 1990
3	Convenience Foods Technology	AlagiParvarti	Self publisher	2023
4	Food Analysis	S. Suzanne Nielsen	Springer	4^{th} edition, 2020

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Hurdle Technologies: Combination Treatments for Food Stability, Safety and Quality: Combination Treatment for Food Stability, Safety and Quality	Lother Leistner	Kluwer Academic/Plenum Publishers	2^{nd}edition, 2002
2	The Technology of Extrusion Cooking	N.D. Frame	Springer	2019
3	Advances in Food Extrusion Technology	MedeniMaskan, AylinAltan	CRC Press	1^{st} edition, 2016
4	In Defence of Food: An Eaters Manifest	Michael Pollan	PENGUIN UK	2009

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_ag03/preview
2	https://onlinecourses.nptel.ac.in/noc24_ag07/preview
3	https://onlinecourses.swayam2.ac.in/nou22_hc01/preview
4	https://onlinecourses.nptel.ac.in/noc22_mg47/preview

SEMESTER S7

BIOPROCESS ENGINEERING

Course Code	**OEFTT726**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To understand the basic principles of various biochemical processes and realize the importance of different design parameters in bioreactor operation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	A historical overview of industrial fermentation process – traditional and modern biotechnology - industrially useful microorganisms . Process flow sheeting – block diagrams with industrial pictorial representation for various equipments, Isolation, preservation and improvement of industrial microorganisms for overproduction of primary and secondary metabolites, medium requirements for fermentation process, Basic design of the fermenter	**9**
2	Production and purification of primary metabolites: Industrial processes for the manufacture of the following products: Organic acids-citric acid, lactic acid itaconic acid and acetic acid, Production of amino acids - commercially important amino acids; alcohols: ethanol, acetone	**9**
3	Production of secondary metabolites: - Industrial production processes for various classes of secondary metabolites: antibiotics: beta-lactams- penicillin and cephalosporin, aminoglycosidesstreptomycin, quinines, commercially important vitamins and steroids. Production and purification of enzymes and other by-products: Microbial production of industrial enzymes: proteases, amylases, lipases and cellulases, Production of biofertilizers- manufacture,	**9**

	formulation and utilization, biopesticides:-Important biopesticides- Bt-toxin, Kasugamycin, Beauverin, Devine.	
4	Biopreservatives, biopolymers- Xanthan gum and PHB, Beverages:-production of beverages, production of baker's yeast. Bioremediation-microbes in mining, ore leaching, oil recovery, waste water treatment, biodegradation of non cellulose and cellulosic wastes for environmental conservation, production of vaccines, production of monoclonal antibodies. Extraction and preparation of crude enzymes, purification and characterization of enzymes from plant, application of enzymes in industry, analytical purposes and medical therapy	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain industrial fermentation and design of industrial fermenter	**K2**
CO2	Explain primary and secondary metabolites	**K2**
CO3	Explain the process technologies for commercial production of products by identifying engineering problems associated with bioreactors	**K2**
CO4	Summarize the application of enzymes in industries and to explain the role of microorganisms in bioremediation and biopesticides	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2		3	1								
CO2	2	3										
CO3	2	2	2	2							2	
CO4	2	1									1	

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Biochemical Engineering Fundamentals	James Bailey and David Ollis	McGraw-Hill, Inc., NY, USA.	2nd Edition, 2017
2	Bioprocess Engineering	Michael L. Shuler and Fikret Kargi	Pearson Education India	2nd Edition, 2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Bioprocess Engineering Principle	Pauline M. Doran	Academic Press	2nd Edition, 2012
2	A fist Course in Bioprocess Engineering	Kaushik Nath	PHI Learning Pvt. Ltd.	1st Edition, 2024

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_bt19/preview
2	https://onlinecourses.swayam2.ac.in/nou24_bt11/preview
3	https://onlinecourses.nptel.ac.in/noc21_bt28/preview
4	https://onlinecourses.nptel.ac.in/noc23_bt61/preview

SEMESTER S7

FERMENTATION & ENZYME TECHNOLOGY

Course Code	**OEFTT727**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To study fundamentals of enzyme structure, functions, kinetics of soluble and immobilized enzymes
2. To learn about various designs of fermenter and fermentation processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	INTRODUCTION TO ENZYMES: Classification of enzymes, Mechanisms of enzyme action; concept of active site and energetics of enzyme-substrate complex formation; specificity of enzyme action; principles of catalysis – collision theory, transition state theory, Coenzymes and Cofactors- Prosthetic group, coenzymes involved in different metabolic pathways. Classification of coenzymes. Isozymes, Abzymes, Synzyme. Kinetics of single substrate reactions; estimation of Michaelis- Menten parameters, multisubstrate reaction mechanisms and kinetics	**9**
2	ENZYME IMMOBILIZATION & APPLIED ENZYMOLOGY: Enzyme inhibition - types of inhibitors - competitive, non-competitive and uncompetitive, their mode of action, turnover number; ph and temperature effect on enzymes. Physical and chemical techniques for enzyme immobilization – adsorption, matrix entrapment, encapsulation, cross-linking, covalent binding etc., - examples, advantages and disadvantages. Industrial Enzymes- Thermophilic enzymes, amylases, lipases, proteolytic enzymes in meat industry, enzymes used in various fermentation processes.	**9**

3	INTRODUCTION TO FERMENTATION PROCESSES: Text Books: General requirements and Microbial culture selection for fermentation processes, Media formulation, inoculum development and process optimization- medium composition, energy, CO_2, nitrogen and other growth factors, buffering and foam agents. Types of fermentation- ethanolic fermentation – mixed alcoholic and acid fermentation – lactic acid fermentation.	**9**
4	OVERVIEW OF FERMENTATION TECHNOLOGY & INDUSTRIAL IMPORTANCE: Design of fermenter - types of fermenters -different parts- agitator, impellers, aerator, baffles, mode of configurations, process control, functions of various parts of fermenter. Recovery and purifications of food products. Selection of industrially important microorganisms; Fermentation in foods; application in baking, sauces, sauerkraut, sausages, pickle, beer and wine fermentation.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the classification of enzymes and their mechanism of action	**K2**
CO2	Develop kinetics of enzyme catalysed reactions, enzyme inhibitory and regulatory process.	**K3**
CO3	Outline different types of enzyme immobilization and applications of enzymes and their future potential.	**K2**
CO4	Summarize the design, parts of fermenters and its applications in various food industries.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	2										3
CO3	3	2										3
CO4	3	2										3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Biochemical Engineering and Biotechnology	GhasemNajafpour	Elsevier	2nd Edition, 2015
2	Principles of Fermentation Technology	Peter F. Stanbury, Allan Whitaker, Stephen J. Hall	Butterworth-Heinemann	3rd Edition, 2016
3	Fermentation Processes Engineering in the Food Industry	J.A. Charles, R. V. Chinnayakanahalli, N. Niranjan	CRC Press	1st Edition, 2013

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Enzyme Technology	Martin F. Chaplin, Christopher Bucke	Cambridge University Press	3rd Edition, 1990
2	Principles of Enzymology for the Food Sciences	John R. Whitaker, Alphons G. J. Voragen, Dominic W. S. Wong	Marcel Dekker	2ndEdition, 1993
3	Industrial Enzymology: The Application of Enzymes in Industry	Julian E. Smith, Bernard E. Smith	Nature Publishing Group	2nd Edition, 1989

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc23_bt05/preview
2	https://onlinecourses.swayam2.ac.in/cec20_bt20/preview
3	https://archive.nptel.ac.in/courses/102/102/102102033/
4	https://onlinecourses.nptel.ac.in/noc24_bt17/preview

SEMESTER 8

FOOD TECHNOLOGY

SEMESTER S8

FOOD PLANT LAYOUT AND DESIGN

Course Code	PEFTT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To develop the knowledge of plant design and layout of specific food industries
2. To enable the students to learn about the regulations governing food plant design
3. To enable the students to learn how to scale up plant designs

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic concepts of plant layout and design including a basic understanding of equipment layout, ventilation; Design consideration for the location of food plants. The material of Construction for food processing equipment; Specifications of processing equipment and accessories; provision for waste disposal, and safety arrangements aspects of plant layout and design; Symbols used for food plant design and layout	9
2	Introduction to feasibility study and analysis Layout and design aspects of pilot and semi-commercial food processing plants; Reference to the design of bakery and biscuit, fruits, vegetable, and beverage processing, and dairy industries; Hygienic Design of Food Plants; specific FSSAI requirements in food plant layout and design	9
3	Application of Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM) in project planning and monitoring; application of ISO, HACCP in food plant, Experimentation in Pilot Plant	9

4	Introduction to project engineering; Process design development: Optimum economic design, Feasibility Survey; Preparation of flow sheets for material movement and utility consumption in food plants; Cost estimation for a Food Plant; Scale-up, Food processing enterprise, Plant operation	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the concept and different aspect required for food plant design	**K2**
CO2	Select appropriate requirement of design and layout for the specific food industry with compliance with FSSAI	**K3**
CO3	Illustrate the function of PERT, CPM, ISO and HACCP in food plant design	**K2**
CO4	Develop a food plant layout with taken into account different legal, economic, social aspects	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	3	3				2					3
CO3	3	2			3		2					3
CO4	3	3	3		3		3					3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Plant Layout and Design	James M Moore	Mcmillan& Co.	1959
2	Plant Layout and Material Handling	J M Apple	John Willey & Sons	3rd Edition, 1977

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Textbook of Dairy Plant Layout and Design	Prof.LalatChander	Indian Council of Agriculture Research	3rd Edition, 2021
2	Food Processing Plant	Slade, F. H.	Leonardhill Books, London	1967

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://encr.pw/C87wJ
2	https://l1nq.com/SUdsz
3	https://encr.pw/YfAPt
4	https://acesse.dev/N9lvX

SEMESTER S8

ENTREPRENEURSHIP DEVELOPMENT IN FOOD TECHNOLOGY

Course Code	**PEFTT862**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Acquire knowledge in Entrepreneurship Development
2. Able to study and prepare the business plan for any organization
3. Classify and study the organizational structure between small, medium, and large scale manufacturing industries

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Entrepreneurship concept- Entrepreneurship as a Career- Entrepreneur Personality Characteristics Knowledge- Skills- Attitude Requirement; Business Environment- Role of Family and Society Entrepreneurship Development Training and Other Support Organizational Services- Central and State Government Industrial Policies and Regulations, MoFPI scheme and support to budding food entrepreneurs, Skill Development by Central Government, International Business.	**9**
2	Sources of Product for Business- Prefeasibility Study- Criteria for Selection of Product- Ownership Capital- Budgeting Project Profile Preparation- Matching Entrepreneur with the Project- Feasibility Report Preparation and Evaluation Criteria; legal aspect; Selection of land and factory sheds; Swot Analysis of Food Industry- opportunities for entrepreneurs in food industry	**9**
3	Finance and Human Resource Mobilization- Financial Analysis – Fund management; Operations Planning- Market and Channel election- Growth	**9**

	Strategies- Product Launching – Conflict and Change management in new industrial enterprise – Management Reporting – Information system Monitoring.	
4	Monitoring and Evaluation of Business- Preventing Sickness and Rehabilitation of Business Units; Effective Management of Small Business, Overview of Start-up food business and challenges, Trade License and Registration, Managing Innovation and sustaining operation services in competitive environment.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain opportunities to set-up Food Processing Industries	**K2**
CO2	Identify the market competitors and conduct and prepare survey reports accordingly	**K3**
CO3	Illustrate the finance, human resource and operation strategy for the effective market growth	**K3**
CO4	Describe characteristics of the effective and sustainable business ecosystem	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										3
CO2	3	3										3
CO3	3	2										3
CO4	3	2	3									3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Entrepreneurship	Hisrich	Tata McGraw Hill, New Delhi	11th Edition, 2005
2	Entrepreneurship Development in Food Industry	Dr. Sandip T. Gaikwad	Blue Leaf publication	2021
3	Entrepreneurship Development in Food Processing	K P Sudheer	New India Publishing Agency	2017

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook for New Enterpreneurs	Khanka, S.S.	S. Chand and Co. Limited	2001
2	Enterpreneurship Development and Communication Skills	R.R. Chole, P.S. Kapse & P.R. Deshmukh	Scientific Publishers	2017

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://acesse.dev/QrYTU
2	https://encr.pw/GzszN
3	https://acesse.dev/WAOyz
4	https://l1nq.com/iaKBH

SEMESTER S8

NUTRACEUTICALS AND FUNCTIONAL FOODS

Course Code	**PEFTT863**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To understand the historical development, definitions, and classifications of nutraceuticals and functional foods.
2. To analyze the role of bioavailability and synergy of functional components in managing diseases such as cardiovascular diseases, diabetes, cancer, hypertension, and obesity.
3. To evaluate different food sources and their benefits as functional foods.
4. To examine the marketing strategies, ethical considerations, and consumer behaviour related to functional foods and nutraceuticals.
5. To familiarize and compare the regulatory frameworks and guidelines for functional foods and nutraceuticals.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to nutraceuticals and functional foods, historical perspective of nutraceuticals, definition, nature, nutraceutical compounds, applications, classification based on chemical/biochemical nature with descriptions, scope and future prospects, Overview of functional foods, definition, classification functional food science and food technology, Impact of food technology on functional food development, Key issues in the Indian functional food industry and nutraceuticals.	9
2	Bioavailability and synergy of functional components in food matrices, antioxidants and disease management, concept of free radicals and antioxidants, role of antioxidants as nutraceuticals and functional foods, disease management, management of cardiovascular diseases, diabetes,	9

	cancer, hypertension, and obesity by nutraceutical compounds, mechanisms of action of these compounds.	
3	Food sources and their benefits, different foods as functional foods, cereal products (oats, wheat bran, rice bran, etc.) fruits and vegetables, milk and milk products, legumes, nuts, oil seeds, and sea foods, herbs, spices, and medicinal plants, coffee, tea, and other beverages as functional foods/drinks and their protective effects,Organosulphur Compounds, pigments: Carotenoids, Lycopene, curcumin, Non nutrient effect of specific nutrients: proteins and peptides and nucleotides, vitamins, minerals.	**9**
4	Marketing and regulatory issues for functional foods and nutraceuticals, CODEX guidelines, EU guidelines, FSSAI guidelines, regulations for nutraceuticals and functional foods, recent updates and amendments, comparative analysis of global regulatory frameworks, challenges in regulatory compliance and future directions.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the historical context, definitions, nature, and classifications of nutraceuticals and functional foods.	**K2**
CO2	Explain the mechanisms by which antioxidants and nutraceutical compounds manage diseases.	**K2**
CO3	Identify various functional foods and assess their health benefits.	**K2**
CO4	Develop insights into marketing strategies, ethical issues, and consumer behaviours related to functional foods and nutraceuticals.	**K3**
CO5	Compare global regulatory guidelines for functional foods and nutraceuticals.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											3
CO2	2											3
CO3	3					2						3
CO4	3	3	2	3							2	3
CO5	2	2		2								3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Functional Foods	John Shi, G. Mazza, and Marc Le Maguer	CRC Press Inc	1st edition, 3 June 2015
2	Bioprocesses and Biotechnology for Functional Foods and Nutraceuticals.	FereidoonShahidi, J. Bruce German , Jean-Richard Neeser	CRC Press Inc	2004
3	Functional Foods and Nutraceuticals	Rotimi E Aluko	Springer-Verlag New York Inc.	2012
4	Handbook of Nutraceuticals and Functional Foods	Wildman, R. E.	CRC Press Inc	2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Nutraceuticals Functional Foods	Gurpreet Singh	Nova Science Publishers Inc	2023
2	Handbook of Nutraceuticals Volume I Ingredients	Yashwant V. Pathak	CRS Press	2009

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/ugc19_hs33/preview
2	https://nptel.ac.in/courses/126105023
3	https://onlinecourses.swayam2.ac.in/ugc19_hs33/preview
4	https://ugcmoocs.inflibnet.ac.in/index.php/courses/view_ug/290

SEMESTER S8

AUTOMATION IN FOOD INDUSTRY

Course Code	**PEFTT864**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Understand the role and importance of automation in the food industry
2. Identify different areas where automation can be applied in food processing and production

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Module 1:Introduction to Automation in the Food Industry Overview of Automation – Definition and scope of automation – History and evolution of automation in the food industry – Types of Automation Systems – Hard and soft automation – Fixed and flexible automation – Benefits and Challenges – Advantages of automation in food production – Common challenges and limitations – Applications of Automation – Case studies and examples of automated systems in food processing – Ethical and Social Considerations – Impact of automation on workforce and society	**9**
2	Module 2: Automation Technologies and Tools Sensors and Actuators – Types of sensors used in food industry automation – Actuators and their role in automation – Control Systems – Introduction to PLC (Programmable Logic Controllers) – SCADA (Supervisory Control and Data Acquisition) systems – Robotics in Food Processing – Types of robots used in the food industry – Applications of robotics in food packaging and handling – Computer Vision Systems – Basics of computer vision – Applications in quality control and inspection – Emerging Technologies – AI	**9**

	and machine learning in food automation – Internet of Things (IoT) in food processing	
3	Module 3: Design and Implementation of Automated Systems System Design Principles – Key considerations in designing automated systems – Integration of various components – Automation Project Management – Planning and execution of automation projects – Cost-benefit analysis – Case Studies – Analysis of successful automation projects in the food industry – Safety and Compliance – Safety standards and regulations – Ensuring compliance with food safety standards – Maintenance and Troubleshooting – Strategies for maintaining automated systems – Common troubleshooting techniques	**9**
4	Module 4: Future Trends and Innovations in Food Industry Automation Sustainable Automation – Green automation technologies – Energy-efficient systems – Advanced Robotics – Collaborative robots (cobots) – Autonomous systems – Smart Manufacturing – Industry 4.0 and its impact on the food industry – Smart factories and digital twins – Predictive Maintenance – Use of predictive analytics in maintenance – Benefits and implementation strategies – Research and Development – Current trends in R&D – Potential future innovations in food automation	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
1. 2 Questions from each module. **2.** Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of optimization, including various types and formulations of optimization problems.	**K2**
CO2	Apply classical optimization techniques to both single and multi-variable problems, utilizing methods such as the simplex method and Lagrange multipliers.	**K3**
CO3	Develop numerical methods and algorithms for solving complex optimization problems, including gradient descent and interior point methods.	**K3**
CO4	Utilize advanced optimization techniques like dynamic programming, genetic algorithms, simulated annealing, and particle swarm optimization for real-world applications.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											1
CO2	3		3									1
CO3	3	2										1
CO4	3	2	1									1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Process Automation: Principles and Applications	B. W. Gould and G. R. Stewart	Springer	1st Edition, 1991
2	Automation in the Food Industry: Current and Future Technologies	Nicholas M. Holden and Ricardo Simpson	Woodhead Publishing	1st Edition, 2012
3	Food Process Engineering and Technology	Zeki Berk	Academic Press	2nd Edition, 2013
4	Fundamentals of Food Process Engineering	Romeo T. Toledo	Springer	3rd Edition, 2007

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of Food Process Design	Ahmed, Mohammad ShafiurRahman	Wiley-Blackwell	1st Edition, 2012
2	Food Engineering Handbook: Food Process Engineering	TheodorosVarzakas, ConstantinaTzia	CRC Press	1st Edition, 2014
3	Introduction to Food Engineering	R. Paul Singh, Dennis R. Heldman	Academic Press	5th Edition, 2013
4	Food Process Engineering Operations	George D. Saravacos, Zacharias B. Maroulis	CRC Press	1st Edition, 2011

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_me139/preview
2	https://onlinecourses.nptel.ac.in/noc24_ag15/preview
3	https://onlinecourses.nptel.ac.in/noc24_ag16/preview
4	https://onlinecourses.nptel.ac.in/noc24_me139/preview

SEMESTER S8

APPLICATIONS OF RENEWABLE ENERGY IN FOOD PROCESSING

Course Code	**PEFTT866**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

- To study fundamentals of enzyme structure, functions, kinetics of soluble and immobilized enzymes
- To learn about various designs of fermenter and fermentation processes

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Renewable Energy Sources Solar energy: principles, technologies (solar thermal, photovoltaic), applications in food processing. Wind energy: turbine technology, wind farms, integration with food processing operations. Biomass energy: types of biomasses, conversion technologies (combustion, gasification, anaerobic digestion), heating, electricity generation. Hydroelectric and geothermal energy: principles, utilization in food processing (heating, cooling, electricity). Environmental and Economic Benefits, Importance of renewable energy in sustainable food processing. Environmental impacts and benefits: reduction of carbon footprint, energy efficiency. Economic feasibility: cost-benefit analysis, return on investment (ROI), incentives and subsidies.	**9**
2	Solar Energy Applications in Food Processing: Solar Thermal Systems, Design considerations: collectors, storage systems. Applications in food processing: heating, drying, thermal processes. Case studies and examples of successful implementations. Solar Photovoltaic Systems PV technology:	**9**

	components, PV arrays, grid-tied vs off-grid systems. Integration with food processing facilities: electricity generation, energy management. Case studies and best practices.	
3	Biomass and Wind Energy Integration: Biomass Energy for Food Processing Types of biomass feedstocks: agricultural residues, forestry waste, energy crops. Bioenergy conversion technologies: combustion, gasification, anaerobic digestion. Application in food processing: steam generation, combined heat and power (CHP), biogas production. Wind Energy Integration Wind turbine technology: types (horizontal axis, vertical axis), selection criteria. Wind farm design considerations: site assessment, grid connection, operational challenges. Case studies of wind energy applications in food processing.	**9**
4	Regulatory Framework and Policies: Introduction to Regulatory Landscape. Overview of national and international policies supporting renewable energy adoption in food processing. Key regulatory bodies and their roles: environmental agencies, energy ministries, regulatory commissions. Compliance Requirements. Permitting processes: environmental impact assessments (EIAs), licensing for renewable energy installations. Standards and codes: adherence to safety, efficiency, and environmental performance standards. Incentives and Financial Mechanisms. Government incentives: tax credits, grants, subsidies for renewable energy projects. Feed-in tariffs (FITs) and power purchase agreements (PPAs) for renewable energy generation. Case studies illustrating successful utilization of incentives in food processing industries.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
1. 2 Questions from each module. **2.** Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the types of renewable energy systems	**K2**
CO2	Summarize application of solar energy in food processing	**K2**
CO3	Outline different types biomass and wind energy systems	**K2**
CO4	Explain the regulatory framework and policies related to renewable energy	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2					3					3
CO2	3	2					3					3
CO3	3	2					3					3
CO4	3	2					3					3
CO5	3	2					3					3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Renewable Energy: Power for a Sustainable Future	Godfrey Boyle	Oxford University Press	4th Edition, 2012
2	Renewable Energy Resources	John Twidell, Tony Weir	Routledge	3rd Edition, 2015
3	Renewable Energy Systems: A Smart Energy Systems Approach to the Choice and Modeling of 100% Renewable Solutions	Henrik Lund	Academic Press	2nd Edition, 2020

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Biomass for Renewable Energy, Fuels, and Chemicals	Donald L. Klass	Academic Press	2nd Edition, 1998
2	Wind Energy Explained: Theory, Design and Application	James F. Manwell, Jon G. McGowan, Anthony L. Rogers	Wiley-Blackwell	2ndEdition, 2010
3	Solar Energy Engineering: Processes and Systems	Soteris A. Kalogirou	Academic Press	2nd Edition, 2013

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/103/107/103107157/
2	https://onlinecourses.nptel.ac.in/noc20_ph14/preview
3	https://onlinecourses.nptel.ac.in/noc22_ch27/preview
4	https://onlinecourses.swayam2.ac.in/nou23_es03/preview

SEMESTER S8

BY PRODUCT UTILIZATION IN FOOD INDUSTRY

Course Code	**PEFTT 867**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To understand different food industry wastes and recovery methods of beneficial components
2. To valorisation of food industry waste products and its utilisation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	By-product Utilization from Cereals :Importance of by-product utilization - Different sources of wastes from food industries- Nature of different waste - Thermal and biotechnological uses of rice husk. Pyrolysis and gasification of rice husk - Cement preparation and different thermal applications - Utilisation of rice bran - stabilization - Defatted bran utilisation.By products of wheat milling - germs and bran - By products of pulse milling - husk - germs and broken. Extraction of rice bran - refining - uses of bran.	9
2	Fruit Industry Waste Utilization: Different sources of wastes from fruit and vegetable industries, Status and types of waste available by products and utilization - mango - citrus - apple - guava - grape waste, Status and types of waste available by products and utilization - vinegar production, SCP production - organic acid production from vegetable waste - utilization of moringa - potato - leafy vegetable waste, Distillation for production of alcohol - oil extraction from waste - waste management in sugar mills, Citric acid production from fruit waste.	9
3	Fish, Meat and Poultry Waste Utilization: Fish industry by products and waste utilisation - fish by-products - production of fish meal, Fish protein	9

	concentrate - fish protein hydrolysate fish liver oil and fish silage - Production of chitin – chitosan, Production of non-food items from fish processing wastes - meat and poultry waste recycling.	
4	Waste from tuber crops - effluent safe disposal - effluent treatment plant - waste recycling plant, Feasibility report for food industries using food waste and by products, Waste from Coconuts - uses of coir pith - biogas production - particle board - utilization of coconut husk - coir fibre, Coconut shell utilization - methods for production of shell charcoal, Fuel briquette - machineries used. Effluent safe disposal - effluent treatment plant Marks Waste water treatment - physical and chemical methods, Sludge disposal	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify and distinguish types of waste generated from food industries and find suitable application based on its properties.	**K2**
CO2	Identify the effective methods for energy recovery from waste products and thereby to improve economic efficiency.	**K2**
CO3	Develop various value-added products from food industry waste	**K3**
CO4	Identify the control environmental impacts associated with food industry waste disposal by using appropriate waste management techniques.	**K3**
CO5	Apply the principles of waste water recycling to restore water supply and to control pollution.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2					1						
CO2			2				3					
CO3	2											
CO4		3	3		3	3						
CO5	3				3	3	2					

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Utilization of By-Products and Treatment of Waste in the Food Industry	VassoOreopoulou, Winfried Russ	Springer	Vol 3, 2007
2	Food processing by-products and their utilization	A K Anal	Wiley Blackwell	1st Edition , 2007
3	Animal By-Product Processing & Utilization	Herbert W. Ockerman, Conly L. Hansen	CRC Press,	1999
4	Roots, Tubers, and Bulb Crop Wastes: Management by Biorefinery Approaches	Ramesh C Ray	Springer	2024

5	Waste Water Recycling and Management	Sadhan Kumar Gosh	Springer	Vol 3, 2017
6	Cereals and Their By-Products	JagbirRehal, KulwinderKaur, Preetinder Kaur	CRC Press	1st Edition, 2023

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Activated Sludge Technologies for Treating Industrial Wastewaters	WJoseph G. Cleary, P.E., BCEE	DES Tech Publications	2014
2	Manufacture, Composition, and Utilization of Dairy Byproducts for Feed	Mayne Reid Coe	Classic Reprint	2019
3	Handbook of Fruit Wastes and By-Products, Chemistry, Processing Technology, and Utilization	Khalid Muzaffar, Sajad Ahmad Sofi, Shabir Ahmad Mir	CRC Press	1st Edition, 2022
4	Essential Oils: Extraction, Characterization and Applications	Gulzar Ahmad, Mohammad Javed Ansari	Atlantic Publishers	2022

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/103/107/103107088/
2	https://archive.nptel.ac.in/courses/127/105/127105018/
3	https://archive.nptel.ac.in/content/storage2/courses/105105048/M1L1.pdf
4	https://archive.nptel.ac.in/content/storage2/courses/105104102/Lecture%2039.htm

SEMESTER S8

HACCP PLANNING AND IMPLEMENTATION

Course Code	PEFTT868	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To develop the knowledge of students regarding hazard control and management principles, tools and their application
2. To enable the students to be aware of the voluntary and mandatory food standards and certifications in place- globally and nationally

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Food Safety; History of HACCP, HACCP Principles, HACCP Plan, Role of Pre requisite programmes (PRP), Operational Pre-REQUISITE Programmes (OPRP) and GMP in Food Industry, Documentation required for implementation of HACCP. HACCP levels, CP's, CCP Controls, Audit: First, second- & third-party audits, CAPA report. Comprehensive global perspective of HACCP (USA, Canada, UK, EU, Africa, Japan). HACCP as a part of ISO 22000/FSSC 22000 8.	9
2	Introduction to Food Processing, Food processing and its types, Microbial, Chemical, Physical Hazards, HACCP generic Model, Importance of Equipment/ Process Selection, Advantages in Implementing HACCP, Risks at different stages of Food Chain. HACCP Implementation in Storage and Transport, Retail and Distribution. Example of Implementation of HACCP in different Food Industry. Implementation of HACCP in a food industry/retail food establishment/catering industry. Case Studies.	9

3	Food Plant Design, layout and Food Logistics, Food Packaging Technology and Labelling Food Microbiology, food borne illness and hazards, Food Sensory Evaluation, Entrepreneurship Development in Food Processing, Case studies, Quality Risk Assessment, Quality Risk Management: Ins and Outs, Deviation Management, CAPA, and Change control, Case Study.	**9**
4	HACCP based approach towards Food Safety;Introduction of VACCP, TACCP, principles, contribution towards Food Preservation, Processing and Packaging, Food Processing Operations, Good Manufacturing Practices, principles including novel and emerging methods. Hazards Associated with bakery and dairy foods, production of safe bakery and dairy foods – Pre requisite programmes and HACCP, Risk assessment at different stages of bakery and dairy food process, Application of VACCP and TACCP system in Sea food Processing.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the HACCP and its principles	**K2**
CO2	Make use of HACCP in food industries	**K3**
CO3	Summarize the overall protection and handling of food materials	**K2**
CO4	Develop HACCP plan in the Food Plant Design, layout, Food Logistics, Food Packaging Technology and Labelling	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	3										3
CO3	3											3
CO4	3	3	3		3							3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	HACCP HACCP System in the Food Industry	Marta Leticia Almengor Hecht	Our Knowledge Publishing	2023
2	The HACCP Food Safety: Training Manual	Tara Paster	John Wiley & Sons	2006

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	HACCP: A Food Industry Briefing	Sara E. Mortimore and Carol A. Wallace	Willey-Blackwell	2nd Edition, 2015

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://shorturl.at/sfHvO
2	https://rb.gy/cj7177
3	https://acesse.dev/8DRLR
4	https://l1nq.com/EgwPu

SEMESTER S8

AGRO-INDUSTRIAL PROJECT PLANNING AND MANAGEMENT

Course Code	**PEFTT865**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To provide insight to students and encouragement to become entrepreneur in food processing sector

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Production and Inventory Management Introduction to Food and agri business Management- nature of processing industry, production, planning – Batch and continuous production, process planning, definition and concepts, inventory control- classification, economic ordering, inventory models, ABC Analysis	9
2	Financial Management Assessing, acquiring and allocating funds, cash flow statement- balance sheet, financial ratio, break even analysis, concept, application in food industry – project appraisal	9
3	Market Management Concepts- consumer market, business market, marketing environment-market segmentation market measurement and forecasting- advertisement-publicity market information system market research-management of distribution channel, Consumer buying behaviour, factors influencing consumer buying behaviour. Export trade- Government regulations, GAIT, WTO regulation	9

4	Business Plan Reports Preparation of project report; Market feasibility reports; Break even Analysis Techno- economic feasibility report on an identified opportunity-any food processing; bakery and confectionary , fruit and vegetable processing, oil and fat ,milk and milk products etc	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	***Internal Ex***	***Evaluate***	***Analyse***	***Total***
5	**15**	**10**	**10**	**40**

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. **Problem Definition (2 points)**
   - Check for clarity and completeness of the problem definition.
   - Check for clear objectives and divide into sub objectives
2. **Problem Analysis (4 points)**
   - Examine the problem and identify the variables and responses
   - Divide the problem to manageable parts
3. **Implementation of software (2 points)**
   - Use suitable software for getting the solution
   - Check for the appropriateness of tools used
4. **Evaluate (4 points)**
   - Evaluate the proposed solution.
   - Assess the correctness of solution.
5. **Validation of the results (4 points)**
   - Validate the results using available data.
   - Do the error analysis of the results obtained.
6. **Conclusion and presentation (4 points)**
   - Summarize the results obtained
   - Do critical thinking of the results obtained and evaluate.
   - Submit a technical report

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	**Part B**	**Total**
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	**Bloom's Knowledge Level (KL)**
CO1	Explain production and inventory management concepts	**K2**
CO2	Explain financial and marketing management techniques	**K2**
CO3	Illustrate about the market strategy, selection of market for product launching, consumer testing by market survey	**K2**
CO4	Apply knowledge on project report preparation and its significance	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	2					2		2				2
CO2	2											2
CO3	2	2	2				2	2	2		2	2
CO4	3							2				2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Managing Agricultural Finance	A. S. Kahlon and Karam Singh	Allied	1st Edition, 2007
2	Marketing Management	Kevin Lane Keller and Philip Kotler	Pearson	15th Edition, 2014
3	Entrepreneurship and Skill Development in Horticultural Processing	K. P. Sudheer and V. Indira	New India Publishing Agency	2018

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Management in Agriculture Finance	S.C. Jain	Vora and Company Publishers Pvt. Ltd.	1st Edition, 1970
2	Marketing Management	Sherilaker	Himalya Publishing Company	1989

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc23_mg33/preview
2	https://onlinecourses.nptel.ac.in/noc20_mg65/preview
3	https://onlinecourses.swayam2.ac.in/nou19_ag08/preview
4	https://onlinecourses.nptel.ac.in/noc24_hs59/preview

SEMESTER S8

FUNDAMENTALS OF FOOD PROCESSING

Course Code	**OEFTT831**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To understand the fundamental principles and concepts of food preservation and processing.
2. To introduce the role and significance of regulatory bodies in maintaining food standards, and emphasizes the essential principles of food safety and quality control.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Food Basics** Food definition, Food constituents- Moisture, Carbohydrates, Proteins, Lipids, Vitamins and minerals, Food and its functions, Food groups, Major properties of food, Compositional and nutritional aspects- fruits and vegetables, dairy, cereals, pulses, oilseeds, fisheries, meat and poultry, consumer foods etc.	**9**
2	**Food spoilage and preservation** Perishable, semi perishable and non-perishable foods, Food deterioration-types and causes, Major microbes causing food spoilage, Factors affecting microbial growth, Principles and methods of food preservation- preservation by low temperature, high temperature, preservatives, dehydration and food irradiation	**9**
3	**Food Processing** Need of Food processing, Primary, Secondary and tertiary processing, Processed food products types and method of preparation- cereals and pulses, extruded products, fruits and vegetables, minimally processed	**9**

	products, dairy, flesh foods, confectionary and fermented products. Status of food processing sector in India- fruits and vegetables, dairy, cereals, pulses, fisheries, meat and poultry, consumer foods, Major challenges and growth potential.	
4	**Food processing regulation bodies** The role of Codex alimentarius commission, FSSAI, MOFPI, KINFRA, BIS, APEDA, MPEDA and AGMARK, Mega food parks and packaging Centers, Role of food technologist in society. **Food Safety** Introduction to food safety and food quality importance, Quality attributes of food, factors affecting quality, Principles of HACCP, ISO 22000, TQM, GHP, GMP, SOP and SSOP	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Provide a comprehensive understanding of food definition, its constituents, the functions of food and food groups.	**K1**
CO2	Understand the types and causes of food deterioration, and identify the major microbes responsible for food spoilage.	**K3**
CO3	Make use of the need for food processing and methods of preparation of processed food products.	**K3**
CO4	Explore the roles of key regulatory bodies in ensuring food safety and standards	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	2			1							
CO3	3	1	1		1							
CO4	3	2	1		1	1		1				

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Processing Technology: Principles and Practice	P. Fellows	CRC Press, Boca Raton, FL, USA	2000, 2nd Ed.
2	Food Microbiology	William C. Frazier and & Dennis C. Westfoff	Tata McGraw-Hill Education, New Delhi.	4th Ed. 1987
3	Fundamentals of Food Processing Operation	Heid, J.L. and Joslyn, M.A.	The AVI Publishing Co; Westport	1967.
4	Fundamentals Of Food Engineering	Stanley E. Charm	Medtech Science Press	2019

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food processing Handbook Ed.	James G Brennan	WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim	2006
2	Fundamentals Of Food Process Engineering,	Toledo R T	Springer, ISBN: 9783319900971	Fourth Edition 2018
3	Food process engineering and technology	Zeki Berk	ELSEVIER	2013
4	Elements of Food Technology	Norman W. Desrosier	Avi Pub Co	1977

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/126/105/126105011/
2	https://onlinecourses.nptel.ac.in/noc22_ag03/preview
3	https://onlinecourses.nptel.ac.in/noc22_ag03/preview
4	https://archive.nptel.ac.in/courses/126/105/126105011/

SEMESTER S8

FOOD PLANT LAYOUT AND DESIGN

Course Code	OEFTT832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To familiarize basic concepts of plant layout and design with special reference to food process industries
2. To select a location for the food industry and develop a plant layout using the systematic layout planning procedure

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Basics Concepts, Location and Layout of Food plant:**Basic concepts of plant layout and design with special reference to food process industries, manufacturing processes-types, Application of HACCP concept, ISO, FPO & MPO requirements in food plant layout and design.Plant location, location theory and models, Plant location factors-plant site selection, Economic plant size-plant layout objectives-classical and practical layout, Characteristics of an efficient layout.	9
2	**Plant layout procedure:** Preparation of flow sheets for material movement and utility consumption in food plants, SLP and its application in development of plant layout, computerised layout planning-types-application in development of plant layout	9
3	**Food plant layout considerations:** Building planning, flooring and types, Basic understanding of equipment layout and ventilation in food process plants, miscellaneous aspects of plant layout and design like provision for packaging and labelling, waste disposal, safety arrangements, Development of general food plant layout and design.	9

4	**Development of food plant layout:** Plant layout and design of Baking oven and frying plant, chocolate and ice-cream plant, fruits and vegetables processing industries including beverages, milk and milk products, fat and oil processing, meat and poultry processing,	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the design considerations and food safety standards in development of food plant layout types along with site selection	**K3**
CO2	Apply the systematic layout planning in development of food plant layout.	**K3**
CO3	Construct flow sheet for the material movement and utility consumption in food industry.	**K3**
CO4	Design and develop plant layout and plant design for food processing industries.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2			2						
CO2	3	2	2		2					2	2	
CO3	3	2										
CO4	2	3	3								2	

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Plant layout and design	James M Moore	Macmillan, New York	1963
2	Food Plant Layout andSanitation	J. A. Awan	Unitech Communications	2010
3	Food Plant Design	Antonio Lopez-Gomez, Gustavo V. Barbosa-Canovas	CRC Press, Boca Raton, FL, USA	2005

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Plant layout and Material Handling	J M Apple	John Willey &Sons	1977
2	Food processing plant	Slade, F.H,	Leonardhill Books,London	1967
3	Milk plant layout	Hall,H.S and Y.Rosen,	F.A.O. Publication	1976

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://drmcet.digimat.in/nptel/courses/video/112107292/L22.html
2	http://acl.digimat.in/nptel/courses/video/112102106/L32.html
3	http://acl.digimat.in/nptel/courses/video/112102106/L33.html
4	https://onlinecourses.nptel.ac.in/noc24_ag15/preview

SEMESTER S8

NUTRACEUTICALS AND FUNCTIONAL FOODS

Course Code	**OEFTT833**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3 :0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. This course will examine bioactive components of functional foods and nutraceuticals, their sources, chemistry, process technology, efficacy, safety and regulation.
2. Thorough knowledge and understanding of functional foods and nutraceuticals help the students to design innovative functional food products. An understanding on the health benefits of various nutraceuticals is emphasized.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Nutraceuticals as Science** Historical overview to functional foods and nutraceuticals. Classification and future prospects of nutraceuticals. Sources of nutraceuticals, dietary supplements, fortified foods, phytochemicals. Zoo chemicals and microbes in food. Scope of nutraceuticals in food industry, national and global scenario. **Marketing, Safety, and Regulatory aspects of nutraceuticals and Functional Foods** Health claims, regulations, and safety issues – National and International. Research frontiers in functional foods, delivery of immunomodulators through functional foods. Concept of personalized medicine.	9
2	**Functional Foods** Conventional foods vs. Functional foods as well as nutraceuticals vs. pharmaceuticals. Introduction to biomarkers and food technology. Functional Lipids, Functional Carbohydrates from cereals, Functional Proteins.	9

	Nutrigenetics and Nutrigenomics. Probiotics, prebiotics and postbiotics, Functional beverages. Design a health drink suitable for diabetic patient.	
3	**Properties, structure, and function of various nutraceuticals.** Natural occurrence of certain phytochemicals - Dietary anti-oxidants and flavonoids, Glucosamine, Octacosanol, Lycopene, Solasodine, Capsaicin, Carnitine, Melatonin and Ornithine alpha ketoglutarate. Carotenoids, phytoestrogens, omega – 3 fatty acids. Role of anti-oxidants in cancer prevention.	**9**
4	**Food as medicines** Nutraceuticals bridging the gap between food and medicine - Nutraceuticals in treatment for cognitive decline, Arthritis, Bronchitis, circulatory problems, hypoglycemia, Nephrological disorders, Liver disorders, Osteoporosis, Psoriasis and Ulcers. Brief idea about some Nutraceutical rich supplements such as Bee pollen, Caffeine, Green tea, Curcumin, Lecithin, Mushroom extract, Cordyceps, Chlorophyll, Phycocyanin, Kelp and single cell protein from Spirulina.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall general knowledge on development of functional foods and nutraceuticals	**K1**
CO2	Select conventional foods and functional foods as well as nutraceuticals vs. pharmaceuticals	**K3**
CO3	Identify several natural ingredients used in the food preparation are important sources of nutraceuticals.	**K3**
CO4	Utilize potential health benefits of functional foods and know the basic physicochemical properties of various nutraceuticals.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1		2		1	1					
CO2	2	2	2	2		2	2					
CO3	3	2	1			1	1					
CO4	2	3	3	1		1						

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of Nutraceuticals and Functional Foods	Robert E C Wildman, Robert Wildman,	Taylor C Wallace	Second edition (2006)
2	Functional Foods–Concept to Product	Gibson GR &William CM	Woodh ead Publishing	(2000).CRC Press.
3	Functional Foods: designer Foods, Pharma Foods, nutraceuticals	Goldberg Israel	An Aspan publication	Chapman&Hall,NewYork,Lond (1994)

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Functional Foods and Nutraceuticals	Aluko,Rotimi.	Springer-VerlagNew York Inc	(2012)
2	Natural Products: The Secondary Metabolites	Hanson, JamesR	Royal Society of Chemistry	(2003),
3	Functional Foods, Nutraceuticals and Natural Products: ConceptsandApplications	Vattem D. A, Maitin V.	DEStechPublications, Inc., PA, USA.	(2016),

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/Cnq0JFID7sA
2	https://youtu.be/R7BonXAiOE4
3	https://shorturl.at/MLGSK
4	https://rb.gy/nldgp9

SEMESTER S8

FOOD INDUSTRY WASTE MANAGEMENT

Course Code	**OEFTT834**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To provide knowledge of Food Processing Industry Waste Management.
2. To provide knowledge of the various aspects and methods adopted for Waste Management.
3. To provide knowledge regarding the environmental considerations and Laws pertaining to these.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Types of waste and magnitude of waste generation in different food processing industries, concept, scope and importance of waste management and effluent treatment, Standards for emission or discharge of environmental pollutants from food processing Industries as per the updated provision of Environment (Protection) Act, 1986.	9
2	Characterization and Effective Utilization of by-products from Cereals, Pulses and Oilseeds Industries. Fruits, Vegetables and Plantation products, Fermented foods, Dairy Industry and Milk Products Industry, Fish, Meat, Egg and Poultry processing industries.	9
3	Characterization of food Industry effluents, Physical and Chemical parameters, Oxygen demands and their interrelationships, Residues (Solids), Fats, Oils and Grease, Forms of Nitrogen, Sulphur and Phosphorus, Anions and Cations, Surfactants, Colour, Odour, Taste,	9

	Toxicity. Unit concept of treatment of Food Industry Effluent. Screening, Sedimentation and Floatation as pre and primary treatments.	
4	Food effluent treatment: Biological oxidations: Objects, Organisms, Reactions involved, Oxygen requirements, Aeration devices / Equipment used. Systems: Lagoons, activated sludge process, oxidation ditches. Bio Remediation. Advanced wastewater treatment systems. Physical separations, Micro-strainers, Filters, Ultra filtration and reverse osmosis. Physico-chemical separations: Activated Carbon Adsorption, Ion Exchange, Electro-dialysis and Magnetic Separation.	**9**

Course Assessment Method

(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Recall the laws and regulations pertaining to Waste Management and Disposal.	**K1**
CO2	Develop the byproducts associated with various types of food processing industries and its utilization.	**K3**
CO3	Utilize the different parameters associated with the characterization of food industry waste.	**K3**
CO4	Demonstrate various food industry effluent treatment operations.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3	2										
CO4	3		1				1					

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Food Processing Waste Management	J.H. Green, A.Kramer	AVI Publishing Company	1979/1981
2	Handbook of Waste Management and Co-Product Recovery in Food Processing	K. Waldron	Elsevier Science	Volume 1 2007
3	Food Processing Waste Management: Treatment and Utilization Technology	V K Joshi & S K Sharma	New India Publishing Agency	2011
4	Environmental Pollution Control Engineering	C S Rao	New Age International Publishers	2006
5	Waste Management for the Food Industries	Ioannis Arvanitoyannis	Elsevier Science	2010
6	The Environment (Protection) Act 1986.		(Act No.29 of 1986)	1986, amended 1991

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1.	Food Processing Waste and Utilization: Tackling Pollution and Enhancing Product Recovery	Ajay Singh, Pradyuman Kumar, Sanju Bala Dhull	CRC Press/Taylor & Francis Group	2022
2.	Food Waste Recovery: Processing Technologies, Industrial Techniques, and Applications	Charis M. Galanakis	Elsevier Science	2020
3.	The Complete Book on Managing Food Processing Industry Waste	Dr. H. Panda	Asia Pacific Business Press Inc.	2011

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/126105023
2	https://www.youtube.com/watch?v=rOOQB4hZHC4
3	https://nptel.ac.in/courses/126105023
4	https://www.youtube.com/watch?v=rOOQB4hZHC4

SEMESTER S8

FOOD PRODUCT DESIGN AND DEVELOPMENT

Course Code	**OEFTT835**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

The main objective is the acquisition by the students of strategic and technological criteria for the design and development of sustainable food products inorder to prepare a placement of the graduate in the R&D team of food companies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**New product development**: Introduction- new products, customers and consumers, value addition, and market. Marketing characteristics of new products-product life cycle and profit picture. Corporate avenues for growth and profitability, opportunities in the marketplace for new product development, technological advances driving new product development, government's role in new product development.	**9**
2	**Designing new products**: New Food Product Development (NPD) process and activities, NPD success factors, design thinking process, new product design, food innovation case studies, market-oriented NPD methodologies, organization for successful NPD; Recipe development; use of traditional recipe and modification; involvement of consumers, chefs and recipe experts; selection of materials/ingredients for specific purposes; modifications for production on large scale, cost effectiveness, nutritional needs or uniqueness; use of novel food ingredients and novel processing technologies.	**9**
3	**Standardization & Large scale production**: Process and equipment	**9**

	design; manufacturing protocol, establishing process parameters for optimum quality; sensory evaluation; food testing lab requirements; different techniques and tests; statistical quality control; comparison of market samples; stages of the integration of market and sensory analysis.	
4	**Quality, Safety & Regulatory aspects**: Product stability; evaluation of shelf life; changes in sensory attributes and effects of environmental conditions; accelerated shelf life determination; developing packaging systems for maximum stability and cost effectiveness; regulatory aspects; approval for proprietary product, food safety management system and quality audits for a food product, regulatory aspects of FSSAI for a food product.	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	**3.** Each question carries 9 marks. **4.** Two questions will be given from each module, out of which 1 question should be answered. **5.** Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the areas of possible innovations based on the life cycle of the food product.	**K3**
CO2	Select advanced formulation technology principles for the design and development of novel food products or the repositioning of products already on the market.	**K3**
CO3	Organize a team, verified through the realization of a product design.	**K3**
CO4	Make use of the concepts and terminology utilized in R&D and food design for repositioning or product innovation.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3				3	3	3				
CO2	3	3				3	3	3				
CO3	3	3				3	3	3				
CO4	3	3				3	3	3				

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Developing New Food Products for a changing market place	Brody, A. L., and John B. L.	Taylor and Francis Group, UK	2nd Edition, CRC press, 2008
2	Food Product Development: Based on Experience	Catherine Side	Lowa State Press, Blackwell publications	2nd Edition, 2008
3	Food Process Engineering and Technology.	Zeki Berk.	Academic Press. 360 Park Avenue South, New York, USA.	2009.
4	New Food Product Development: From Concept to Marketplace	Gordon W Fuller	CRC press, Taylor and Francis Group, UK	3rd Edition, 2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Food Engineering	R. Paul Singh and Dennis R. Heldman	Elsevier, Amsterdam, Netherlands	2014, 5th Edition.
2	Fundamentals of Food Process Engineering,	Earle, R.L.	Pergamon Press. Oxford. U.K.	2003
3	Food Process Engineering Operations,	Saravacos G D	Taylor and Francis, 2011,ISBN: 9781420083538	2012
4	Consumer-led Food Product Development	Macfie, H	CRC press ,Wood Head publications	1st Edition, 2007

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	Lec 01 Introduction to Product Design and Development (youtube.com)
2	Lec 02 Product Life-Cycle (youtube.com)
3	Lec 03 Product Policy of An Organization and Selection of Profitable Products (youtube.com)
4	Lec 04 Product Design (youtube.com)
5	Lec 05 Product Design Steps and Product Analysis (youtube.com)

SEMESTER S8

CEREALS AND LEGUME TECHNOLOGY

Course Code	OEFTT836	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

Aim of this course is create the basic knowledge on different types of Cereals & Legumes and its processing methodologies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Rice processing** Status of grain processing industries in India–rice-varieties-nutrient composition. Parboiling – physico chemical changes during parboiling – soaking – steaming – drying. Modern methods of parboiling-Rice milling. Equipments involved in the rice milling processing–Cleaner–Stoner–Husker-Separator-Whitening machine. By product utilization–processing of rice bran oil	9
2	**Wheat and Barley Processing** Wheat- nutrient composition – classification of wheat. Milling of wheat-equipments used in the milling – break roll reduction roll- purifier – plan sifter – scalping - scratch system. Types of wheat flour .Barley–Nutrient composition –Commercial production of malt	9
3	**Corn, oats and millet processing** Milling of corn -Dry milling and wet milling- Gluten and starch separation – enzyme hydrolysis – acid hydrolysis- HFCS. Processing of oats. Millet –	9

	major millet – minor millet - Processing of pearl and finger millets.	
4	**Value added products from Cereals and Processing of Legumes** Processing of breakfast cereals – processing of extruded products (noodles and pasta)-bread - Processing of protein isolates, flaked products- Processing of popcorn, puffed snacks, -Processing of special dietary foods. Legumes Types – nutrient composition – anti nutritional factors present in the pulses – Commercial milling of pulse – premilling techniques- Processing of soya bean.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the processing of rice and utilization of by-products.	**K2**
CO2	Explain the processing of barley and wheat	**K2**
CO3	Explain the processing of corn, oat and millets	**K2**
CO4	Identify the value added products and their processing	**K3**
CO5	Select the processing of various legumes and their products.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										1
CO2	3	2	2									1
CO3	2	2	3									1
CO4	2	2										1
CO5	2	3										1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The Chemistry and Technology of Cereals as Food and Feed	Matz, Samuel A.	CBS	2nd Edition, 2014.
2	Post-Harvest Technology of Cereals, Pulses and Oil Seeds	Chakraverty, A.	Oxford & IBH	Third Edition, 2008
3	Technology of Cereals	N.L.Kent	Elsevier, 1994.	Fourth Edition

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Principles of Cereal Science and Technology	Delcour, Jan A. and R. Carl Hoseney.	American Association of Cereal Chemists	3rdEdition 2010
2	Handbook of Cereal Science and Technology	Karl Kulp	CRC Press	2nd Rev. Edition2000.
3	Millets Nutritional Value And Processing Technology ,	Chavan, U D & J V Patil,	Astrall Publications	1st edition, , 2016
4	Cereals Processing Technology, Bio-Green	Gavin ed Owens	Bio-Green Elsevier	1st Edition (2015), 978-935107329,2015
5	Advances in Cereal Science: Implications to Food Processing and Health Promotion,	Joseph Awika (Editor), VienoPiironen (Editor), Scott Bean (Editor	OUP USA	ISBN13: 9780841226364 eISBN: 9780841226418 (7June2012),

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.youtube.com/watch?v=pY-W5HLf3R4
2	https://www.youtube.com/watch?v=9I6-eVtcpao
3	https://www.youtube.com/watch?v=S3uetP9g72Y
4	https://www.youtube.com/watch?v=dB7Tb-WvOXs

SEMESTER S8

ENTREPRENEURSHIP DEVELOPMENT IN FOOD INDUSTRY

Course Code	OEFTT837	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. The objective of the course is to provide insight to students and encouragement to become entrepreneur in food processing sector

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Classification of small, medium and largescale manufacturing industries; Opportunities of food processing industries in India and agencies for promotion of food processing industries; Source of machine and equipment. Entrepreneurship Development Training and Other Support Organizational Services - Central and State Government Industrial Policies and Regulations	9
2	Fundamentals of marketing principles and marketing mix, Sales and distribution management, Costing and cost management, pricing methods, fundamentals of operations and supply chain management, organization structure and human resource management, Trade license and registration marks. Systems approach and consideration in an entrepreneurial venture. Management reporting and information system for monitoring and control of the new enterprise.	9
3	Opportunity identification and feasibility studies, financial analysis, technical entrepreneurship, Problem-solving, decision-making processes and tools, conflict and change management in a new industrial enterprise, Systems approach and consideration in an entrepreneurial venture. Management reporting and information system for monitoring and control	9

	of the new enterprise, managing Innovation.	
4	Preparation of project report; Market feasibility reports; Break even Analysis Techno- economic feasibility report on an identified opportunity-any food processing; bakery and confectionary , fruit and vegetable processing, oil and fat ,milk and milk products etc. Marketing challenges and approaches for new products and services sustaining in competitive environment	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain motivation for entrepreneurial career	**K2**
CO2	Explain the ability to discern distinct entrepreneurial traits	**K2**
CO3	Make use of conducting feasibility analysis	**K3**
CO4	Organize the preparation of business plan	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2					2	2	2				
CO2	2											
CO3	2	2	2	2			2	2		2	2	2
CO4	3	2					2	2	2	2	2	

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Entrepreneurship development.	C.B. Gupta and N.P. Srinivasan	S. Chand & Sons, New Delhi.	2012
2	Entrepreneurship Development. New Age	Anil Kumar, S., Poornima, S.C., Mini, K., Abraham and Jayashree, K	International Publishers, New Delhi.	2003
3	Management: Theory and Practice	Gupta, C.B.	Sultan Chand & Sons, New Delh	2001.
4	Dynamics of Entrepreneurial Development and Management	Vasant Desai	.. Himalaya Publishing House, New Delhi	2000
5	repreneurship and skill development in Horticultural Processing	K.P. Sudheer and V.Indira	New India Publishing Agency, New Delhi.	2008

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Creativity and Innovation	Couger C	IPP	1990
2	Understanding Enterprise : Entrepreneurship and Small Business	Creativity and Innovation	Palgrave	2000
3	The Future of Food Business The Facts, the Impacts, the Acts	Marcos Fava Neves	World Scientific	2011
4	Innovation in Food Ecosystems Entrepreneurship for a Sustainable Future	Paola De Bernardi, Danny Azucar	Springer International Publishing	2019

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/cec19_mg39/preview
2	_https://onlinecourses.nptel.ac.in/noc24_mg128/preview
3	https://www.youtube.com/watch?v=BepKLu1mx4U
4	https://onlinecourses.nptel.ac.in/noc22_mg71/preview